Introspective Imitation Dynamics and the Limits of Extrinsic Update Rules in Heterogeneous Public Goods Games

Harry Foster, Vince Knight, Sebastian Krapohl

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1 Introduction

Reducing global carbon emissions is one of the defining collective action problems of our time. Countries differ substantially in their economic capacity, historical emissions, and the costs they face when decarbonising, yet they share a mutual interest in the global public good of a stable climate. This structure maps onto a *public goods game* [26, 8]: each actor decides whether to incur a private cost to contribute to a shared resource whose total value is multiplied by some factor r and distributed equally. The temptation to free-ride means that purely selfish reasoning leads to a tragedy of the commons [6], even when collective cooperation would benefit everyone. Both theory [24] and controlled experiments [15] show that framing cooperation as a collective-risk problem, where failure to meet a cooperation threshold incurs a shared loss, can substantially raise cooperation rates. The challenge of overcoming free-riding is not merely theoretical. Nordhaus [17] proposed voluntary *climate clubs* in which member nations adopt a common carbon price and levy tariffs on non-members, turning the international climate problem into a coordination game with enforcement. The European Union's Carbon Border Adjustment Mechanism [4], which requires importers to purchase certificates reflecting the carbon content of goods produced under weaker environmental regulation, represents a real-world instance of this idea.

Evolutionary game theory offers a framework for studying how cooperation can emerge among self-interested actors without assuming rational foresight [9, 18]. In this approach, strategies that perform well spread through a population via one of several update rules, while poorly performing strategies decline. The conditions under which cooperative strategies can invade and stabilise have been extensively studied in two-player settings such as the iterated prisoner's dilemma [11, 1] and the snowdrift game [33]. The public goods game generalises these pairwise dilemmas to N players [7], making it well suited to studying multi-lateral cooperation such as climate agreements. In finite populations, evolutionary outcomes are stochastic: a rare mutant strategy spreads or vanishes with a probability that depends on the population size, payoff structure, and update rule [20, 32].

A feature of real-world collective action that standard models typically set aside is that actors are not identical. Players differ in wealth, productive capacity, and the costs they face, and this inequality can reshape the dynamics of cooperation [8]. In spatial and graphical settings, where players sit on a network and participate simultaneously in multiple games with their neighbours [19, 29, 21], heterogeneity generally promotes cooperation. In [25] it is shown that social diversity in endowments promotes the emergence of cooperation in the public goods game directly: heterogeneous populations outperform their homogeneous counterparts over a wide range of parameters. This echoes earlier work showing that scale-free network topology, a form of structural heterogeneity, provides a unifying framework for the emergence of cooperation [23]. When contributions depend on dynamically accumulated reputation [14], high-reputation defectors receive reduced transfers, weakening defection's advantage. When heterogeneity arises from differences in network degree so that players allocate contributions according to the strength of their social ties [13], the payoff structure penalises high-degree defectors and protects cooperating clusters from invasion. The characteristic spatial mechanism is the formation of cooperating clusters that are more profitable than their defecting neighbours; provided r is sufficiently large, these clusters resist invasion and eventually spread. Heterogeneity does not, however, unconditionally favour cooperation. Flores et al. [5] show that when the contribution level is itself a strategic variable rather than a fixed attribute, low-value contributions act as a form of defection against the most collectively beneficial strategies, so that the emergence of cooperation does not guarantee the emergence of high-value cooperation.

The studies above are almost exclusively concerned with spatially structured populations, where network topology is central. Much less is known about fully heterogeneous, *well-mixed* populations, in which all individuals are distinguished by a fixed attribute (such as their contribution capacity) independently of any graph structure. Furthermore, most existing analyses rely on simulation or mean-field approximations; exact treatments of heterogeneous evolutionary dynamics remain rare. How the choice of update rule interacts with population heterogeneity is also poorly understood. The Moran process [16, 10], Fermi imitation dynamics [30], aspiration dynamics [**?**], and introspection dynamics [2, 3] all model strategy revision in distinct ways, and each may respond differently to player-specific payoffs. In particular, *intrinsic* dynamics, in which a player evaluates a prospective strategy change based on their own counterfactual payoff, are better suited to heterogeneous populations than *extrinsic* dynamics, in which a player may copy a high-fitness neighbour without considering whether the same strategy would serve them as well. The introspection dynamics framework developed in [2] and extended to multiplayer asymmetric games in [3] formalises this distinction for homogeneous populations; extending it to the fully heterogeneous setting motivates the present work.

We make the following contributions. First, we establish a general framework for evolutionary dynamics on fully heterogeneous, well-mixed populations of N ordered individuals, in which each player may have a distinct payoff function and contribution level. Second, for $N = 2$, we derive exact transition matrices and, where applicable, exact absorption and steady-state distributions for four established dynamics (the Moran process, Fermi imitation dynamics, aspiration dynamics, and introspection dynamics) together with their mutation-augmented variants, applied to a heterogeneous public goods game. Third, we introduce a novel update rule, *Introspective Imitation Dynamics*, in which players select a role model proportionally to fitness (as in the Moran process) but evaluate whether to adopt that strategy by comparing their own current and counterfactual payoffs (as in introspection dynamics). This rule is well suited to heterogeneous populations, where copying a high-fitness individual is likely although irrational if the same strategy yields a worse payoff for the copying player. Fourth, we show that purely extrinsic dynamics are subject to a ceiling on the probability of cooperation that intrinsic dynamics can exceed, and we characterise the conditions under which this advantage arises, with the ratio of the public goods multiplier r to population size N as the central governing parameter. We study two heterogeneous population structures: a *linear* population, in which players' capacities are graded uniformly across the group, and a *binomial* population, consisting of a low-capacity majority and a high-capacity minority.

The remainder of the paper is organised as follows. Section 2 introduces the population model, defines all five dynamics and their mutation variants, and specifies the heterogeneous public goods game. Section 3 presents our results, and supplementary material containing extended parameter sweeps is provided in Section 4.

2 The General Heterogeneous Population Dynamic Model

Following [2, 3], we define a general framework for evolutionary dynamics on a fully heterogeneous, well-mixed population.

2.1 Population and State Space

The population consists of N **ordered** individuals. Each individual selects an action from a common action set $A = (a_1, a_2, \ldots, a_k)$; we assume throughout that all individuals share the same action set of size k although this is not strictly required for the introspective and aspiration dynamics of [2, **?**]. A *state* of the population is an ordered N-tuple $\mathbf{a} = (a_1, a_2, \ldots, a_N) \in A^N$, and the *state space* S is the set of all such tuples. A payoff function $\pi\colon S \to \mathbb{R}^N$ assigns to each state a vector of individual payoffs; we write $\pi_i(\mathbf{a})$ for the payoff of individual i in state $\mathbf{a}$.

The population evolves as a Markov chain [27] on S. Transitions can only occur between states that differ in exactly one coordinate, that is, states $\mathbf{a}$ and $\mathbf{b}$ with Hamming distance $h(\mathbf{a}, \mathbf{b}) = 1$ [12]. We call the unique coordinate at which $\mathbf{a}$ and $\mathbf{b}$ differ the *index of difference*, denoted $I(\mathbf{a}, \mathbf{b})$ [3]. The set of states reachable from $\mathbf{a}$ in one step is $\mathrm{Neb}(\mathbf{a}) = \{\mathbf{b} \in S : h(\mathbf{a}, \mathbf{b}) = 1\}$ [3].

2.2 Population Dynamics

We adapt four established dynamics to the heterogeneous setting: the Moran process [16], Fermi imitation dynamics [30, 14, 13], aspiration dynamics [**?**], and introspection dynamics [2, 3]. We also introduce a novel fifth dynamic, *Introspective Imitation Dynamics*, which combines the fitness-proportionate role-model selection of the Moran process with the self-referential evaluation step of introspection dynamics.

Choice and Selection intensity

In Moran-process-type dynamics, which model biological evolution, we introduce a parameter ϵ referred to as *selection intensity* via the mapping

$$f_i(\mathbf{a}) = 1 + \epsilon \pi_i(\mathbf{a}).$$

Appropriate bounds on ϵ ensure that the fitness values f_i remain non-negative, which is necessary for the reproduction probabilities in the Moran process to be well-defined. Smaller values of ϵ correspond to weaker selection, placing less emphasis on payoff differences in the evolutionary update. In the limiting case $\epsilon = 0$, the process reduces to neutral drift, in which reproduction is independent of payoff. In contrast, population dynamics based on the Fermi update rule [30] employ a parameter β, which we refer to as the *choice intensity*. This parameter determines how sensitively individuals respond to payoff differences when making decisions. Depending on the specific dynamic under consideration, these payoff differences may arise from comparing an individual's current payoff with that of another individual in the population, or from comparing their current payoff with the payoff they would obtain by switching strategy.

The Moran Process

The Moran process [16] is defined by the following algorithm:

1. A player is chosen to be copied with a probability proportional to their fitness in the population
2. A player is chosen with probability $\frac{1}{N}$ to change their strategy

The process goes from state to state according to the entry $T_{\mathbf{ab}}$ of a transition matrix T in which each row and column correspond to a state in S is defined as the probability of a transition $\mathbf{a} = (a_1, a_2, \ldots, a_N) \to \mathbf{b} = (b_1, b_2, \ldots, b_N)$. For the Moran process, this is as follows:

$$T_{\mathbf{ab}} = \begin{cases} \frac{\sum_{a_i = b_{I(\mathbf{a},\mathbf{b})}} f(a_i)}{N \sum_{a_j} f(a_j)} & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \text{, differing at position } I(\mathbf{a}, \mathbf{b}) \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b} \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \tag{1}$$

An important case which proceeds from (1) is that of a transition $\mathbf{a} \to \mathbf{b}$ where $\mathbf{b}$ contains an individual of a *type* not found in $\mathbf{a}$. For example, the transition $(0, 1) \to (0, 2)$. This would be forbidden by the intuition of a standard Moran process, as the new individual is the duplication of another individual in $\mathbf{a}$. However, we can see that the standard formula yields $T_{ab} = 0$ because there does not exist $a_i \in \mathbf{a} : a_i = b_{i^*}$.

We can now discuss one of the key ideas in the Moran process. As no new action types can be introduced to the population once they are fully removed, we have a set of absorbing states $\mathbf{a}$ such that $a_1 = a_2 = ...$, for which we have $p_{\mathbf{a} \to \mathbf{a}} = 1$. We denote the set of absorbing states S^{ω}. We therefore have that the Moran process forms an absorbing Markov chain [27], and using standard results can investigate the probability of the population being absorbed into each absorbing state.

Consider a game with two players, each playing one of two actions, giving the state space $\mathbf{a} = (0, 0)$, $\mathbf{b} = (0, 1)$, $\mathbf{c} = (1, 0)$, $\mathbf{d} = (1, 1)$. The explicit transition matrix for this case is given in equation (6) in the supplementary material.

Fermi Imitation Dynamics

The Fermi Imitation Dynamic is another method of modelling evolution in a population. It models the following process:

- Two players, i and j are selected at random. Unlike the original formulation [30], player j need not be a neighbour of player i, reflecting the well-mixed setting considered here.
- Player i copies the action type of the player j with a probability according to the Fermi function $\phi(\pi_i(\mathbf{a}) - \pi_j(\mathbf{a})) = \frac{1}{1+e^{(\beta(\pi_i(\mathbf{a}) - \pi_j(\mathbf{a})))}}$

Where β is the choice intensity of the system, representing how sensitive our players' choice is to the difference between their current payoff, and the payoff which they are considering. This method has been commonly used in studies of evolutionary games, for example [14, 13, 9, 30].

Unlike a Moran Process, we can see that the players are not selected with a probability proportional to their fitness. However, Fermi Imitation Dynamics compared the fitness after the selection of players. This means that in Fermi Imitation Dynamics, there is not necessarily a player changing action in each timestep, whereas in a Moran process, there is always a player changing action type (although it may be a change to the same action type that they are currently performing) We now define our transition matrix according to the Fermi function:

$$T_{\mathbf{ab}} = \begin{cases} \left(\frac{1}{N(N-1)} \sum_{a_j = b_{I(\mathbf{a},\mathbf{b})}} \phi(\pi_{\mathrm{I}(\mathbf{a},\ \mathbf{b})}(a) - \pi_j(\mathbf{a}))\right) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b} \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \tag{2}$$

Here we see how the Fermi function can be used to define the transition probability between states in a Markov chain. Importantly, we see that the first case of equation 2 is always in the interval $(0, 1)$, as the Fermi function always takes a value on $(0, 1)$, and the summation term is at most $N - 1$ values of said function, and thus takes a value $< N - 1$.

Aspiration Dynamics

Aspiration dynamics models an update rule based on how far away a player's performance in a game is from an aspirational constant fitness value:

1. A player i is chosen with probability $\frac{1}{N}$ to reconsider their strategy $a_{i,k}$
2. This player changes strategy with a probability $\phi(\pi_i(\mathbf{a}) - A_i)$

Here, A_i is player i's *aspiration*, meaning the payoff which they desire to make from a given game. A player will accept a new strategy with a higher than random probability if their current payoff is lower than their aspired payoff. This dynamic is almost always defined for two actions, traditionally cooperation and defection [**?**].

Introspection Dynamics

In introspection dynamics [2], individuals are selected to consider if a change of their action choice would lead to an increase in payoff. The new action is selected with a probability that depends on the potential increase and the selection intensity β:

1. A player i is chosen with probability $\frac{1}{N}$ to reconsider their strategy $a_{i,k}$
2. Player i chooses another possible strategy, $a_{i,l}$, from the set of all possible strategies.
3. The chosen player replaces their strategy with probability $p_{a_{I(\mathbf{a},\mathbf{b})} \to b_{I(\mathbf{a},\mathbf{b})}}(a_{-I(\mathbf{a},\mathbf{b})}) = \phi(\Delta\pi_{I(\mathbf{a},\mathbf{b})})$, where $\Delta\pi_{I(\mathbf{a},\mathbf{b})} = \pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a}) - \pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{b})$ is the difference between the fitness of the player's current strategy and the possible fitness of the new strategy.

This process is well suited to a heterogeneous population. This is because in some cases using imitation dynamics, a player may copy a strategy due to its fitness when played by another individual in the population, despite the fact that such a strategy actually has a lower payoff for the focal player than their current strategy. In introspection dynamics, players do not consider the fitness of other individuals, and so cannot be fooled by high fitness strategies with attributes different to that of the focal player.

Introspective Imitation Dynamics

Introspection dynamics allows players to learn based on their own fitness, resulting in a reduced probability of state changes that worsen the focal individual's fitness in a heterogeneous population. Now we consider a population dynamic that utilises this property of introspection dynamics but within the context of an imitative process. Consider a population of N players with the same action set. We define the introspective imitation process as follows:

1. A focal player is chosen at random to reconsider their strategy
2. Another player a_j in the population is chosen with a probability proportional to their fitness within the population to have their strategy considered
3. The focal player changes their strategy with probability $\phi(\Delta(f_{I(\mathbf{a},\mathbf{b})}))$

We define the transition matrix T of an introspective imitation process as follows:

$$T_{\mathbf{ab}} = \begin{cases} \frac{1}{N} \frac{\sum_{a_j = b_{I(\mathbf{a},\mathbf{b})}} f_j(\mathbf{a})}{\sum_k f_k(\mathbf{a})} \phi(\Delta(f_{I(\mathbf{a},\mathbf{b})})) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b} \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \tag{3}$$

We see that players can only copy the actions present in the state, however they determine whether or not to copy a player based on their own fitness rather than by considering the fitness of another player. This models a behaviour where individuals learn from each other, but do not blindly trust that the strategy will work for themself just because it worked for another player. Instead, players look to others to see what action they *could* play, but judge based on their own fitness to decide what action they *should* play.

Classification of dynamics

Figure 1, shows a diagrammatic representation of the population dynamics shown in this section. It also proposes a classification our population dynamics into one of two categories. The first are population dynamics which accept strategies from the population based on the fitness of said strategies when played by other individuals. These dynamics are poorly suited to a heterogeneous population, as individuals do not consider how the change will affect them. However, these do offer a good model for biological processes such as the evolution of traits in nature, as in these cases the fitness of an individual causes them to pass on their traits. The second are population dynamics which consider their own change in payoff in the case that they accept a new strategy. These dynamics are more well suited to a heterogeneous population, modelling state changes where individuals make an active decision to change state in order to improve their own fitness.

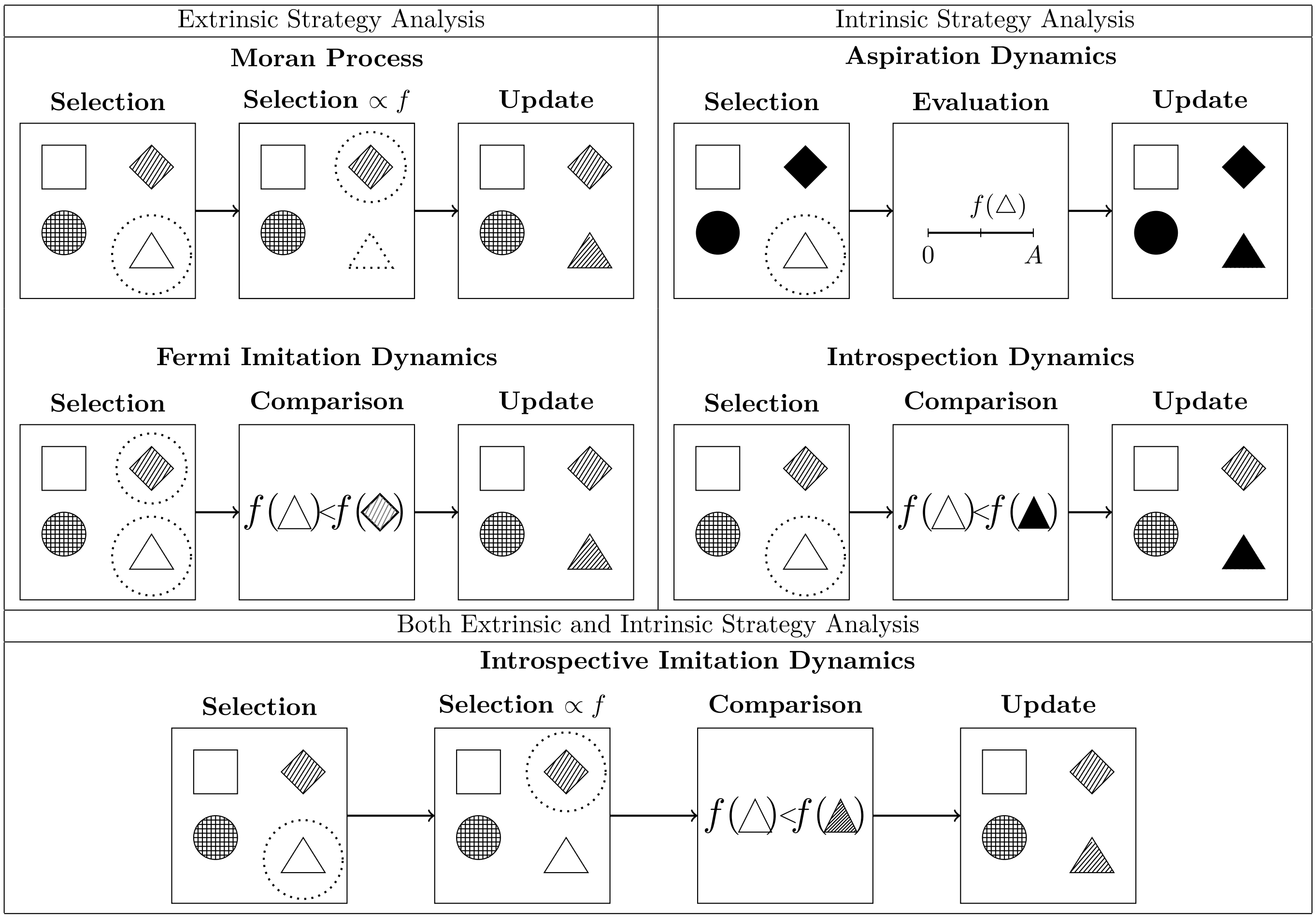


Figure 1: A comparison of different population dynamics. We assume in all cases that the triangle is chosen for replacement with a probability $\frac{1}{N}$ and the diamond is chosen with a varying probability depending on the process. In the case of Introspection dynamics, we assume that the strategy represented by the shape being filled in is chosen with probability $\frac{1}{k-1}$, and for aspiration dynamics there is only one other strategy to accept.

2.3 Mutation

Three of the dynamics in our model correspond to absorbing Markov chains, where the only relevant outcome is the probability of absorption into a given absorbing state. The remaining two dynamics are ergodic chains, for which we can compute the long-run probability of cooperation. To enable meaningful comparison across all models, we introduce mutation into our framework.

Mutation ensures that all processes become ergodic, while also introducing a controlled amount of noise: at any moment, individuals may switch to a different strategy. assumption.

Given an original process and two states $\mathbf{a}$ and $\mathbf{b}$, the probability of the original process *with mutation* transitioning from $\mathbf{a}$ to $\mathbf{b}$ is given by:

$$P(\mathbf{a} \to \mathbf{b}) = P(\text{pick individual } I(\mathbf{a}, \mathbf{b})) \cdot \big(P(\mathbf{a} \to \mathbf{b}) \cdot P(\text{no mutation}) + P(\mathbf{a}_{I(\mathbf{a},\mathbf{b})} \text{ mutates} \to \mathbf{b}_{I(\mathbf{a},\mathbf{b})})\big)$$

We assume a mutation parameter $\mu \in [0,1]_{\mathbb{R}}^{N\times k}$, where μ_{ij} denotes the probability that individual i adopts strategy j via mutation. For a population process with transition matrix T, the mutated process is described by $T^{(\mu)}$. In this process, a transition from state $\mathbf{a}$ to $\mathbf{b}$ occurs with probability $T_{\mathbf{ab}}$; mutation then acts on the resulting state with probability dictated by μ.

For a focal individual i, the probability that no mutation occurs is $1 - \sum_{j=1}^{k} \mu_{ij}$. The transition matrix $T^{(\mu)}$ for the mutated process is:

$$T^{(\mu)}_{\mathbf{ab}} = \begin{cases} T_{\mathbf{ab}} \left(1 - \sum_{j=1}^{k} \mu_{I(\mathbf{a},\mathbf{b}),j}\right) + \frac{\mu_{I(\mathbf{a},\mathbf{b}),\mathbf{b}_{I(\mathbf{a},\mathbf{b})}}}{N} & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}), \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b}\in S\setminus\{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b}. \end{cases} \tag{4}$$

Note that if $\sum_{j=1}^{k} \mu_{ij} = 1$, mutation fully determines the behavior of individual i.

This construction ensures that the process is ergodic, rather than absorbing. Consequently, we no longer obtain absorption matrices for models with mutation; instead, we derive a unique steady state that is independent of the initial state.

Detailed transition matrices for each of the previously studied processes, now incorporating mutation, are provided in the supplementary materials.

2.4 The Public Goods Game

To illustrate all of the dynamics of Section 2 we will use the Public Goods game.

We apply the dynamics above to the public goods game [26, 9]. In this game, individuals choose whether or not to contribute to a common resource pool. The total contribution is multiplied by a factor $r > 1$ and distributed equally among all N players. The model captures the temptation to free-ride [6]: a player who withholds their contribution still receives an equal share of the multiplied pool, so individual incentives push against the collectively optimal outcome of universal cooperation.

We can model a public goods game by providing the following payoff function:

$$\pi_i(\mathbf{a}) = \frac{r \sum_{j=1}^{N} \alpha_j}{N} - \alpha_i \tag{5}$$

Where α_j is the contribution by individual j. This is a similar heterogeneous public good game as the one described in [8] although in that model the return r is also a heterogeneous property of each player.

3 Results

3.1 Analytic Results

3.2 Numeric Results

For the purposes of this section, a large scale parameter sweep is carried out. To the authors knowledge this is the largest such parameter sweep. It is done to the highest standards of reproducibility [34, 22, 35, 28, 31] and all data has been archived and is a further contribution of the work. For the purpose of this research, the authors developed a software library `ludics` (written in python) which is installable. The archive of the software is at [**?**] and the documentation can be found at . All the development of the software can be found at .

Parameter Space

- Number of experiments.
- Size of population (simple 1 to 9 right?)
- r
- Total contribution of the population M (I don't think this technically exist for the binomial population right? But include the range.

- Two population structures.
- Mutation values.

What we measure

Figure 2 shows the marginal distribution of the cooperation probability p_C across all parameter combinations and dynamics. Most configurations yield low cooperation, but the right tail is non-trivial: meaningful cooperation is achievable under a subset of conditions.

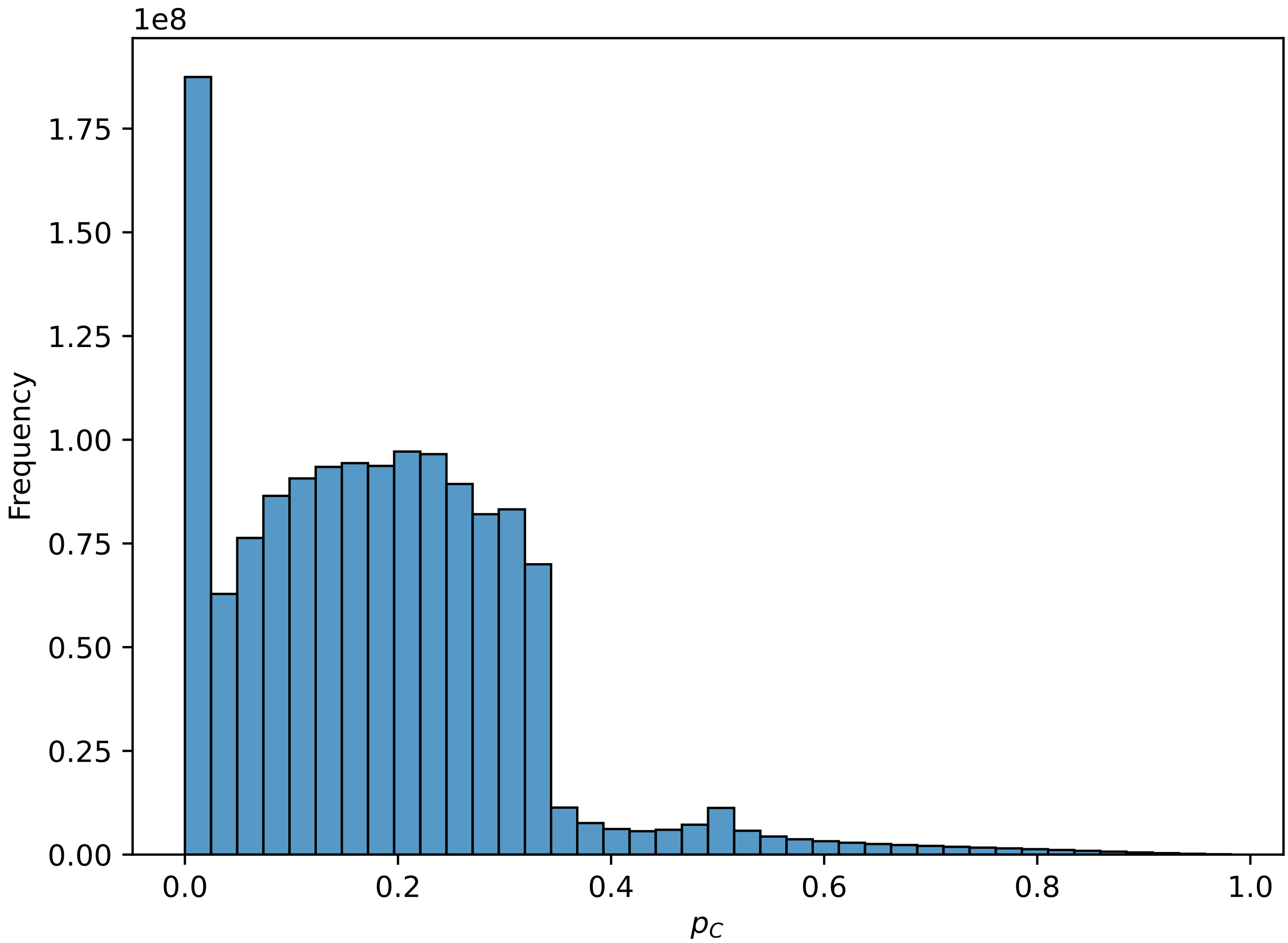


Figure 2: Distribution of p_C across all dynamics and parameter settings.

Separating by dynamic (Figure 3) reveals a striking difference: Fermi imitation dynamics consistently produces the lowest p_C values, while Introspective Imitation Dynamics achieves the highest among all absorbing-chain dynamics. This suggests that the choice of update rule is at least as consequential as the payoff parameters.

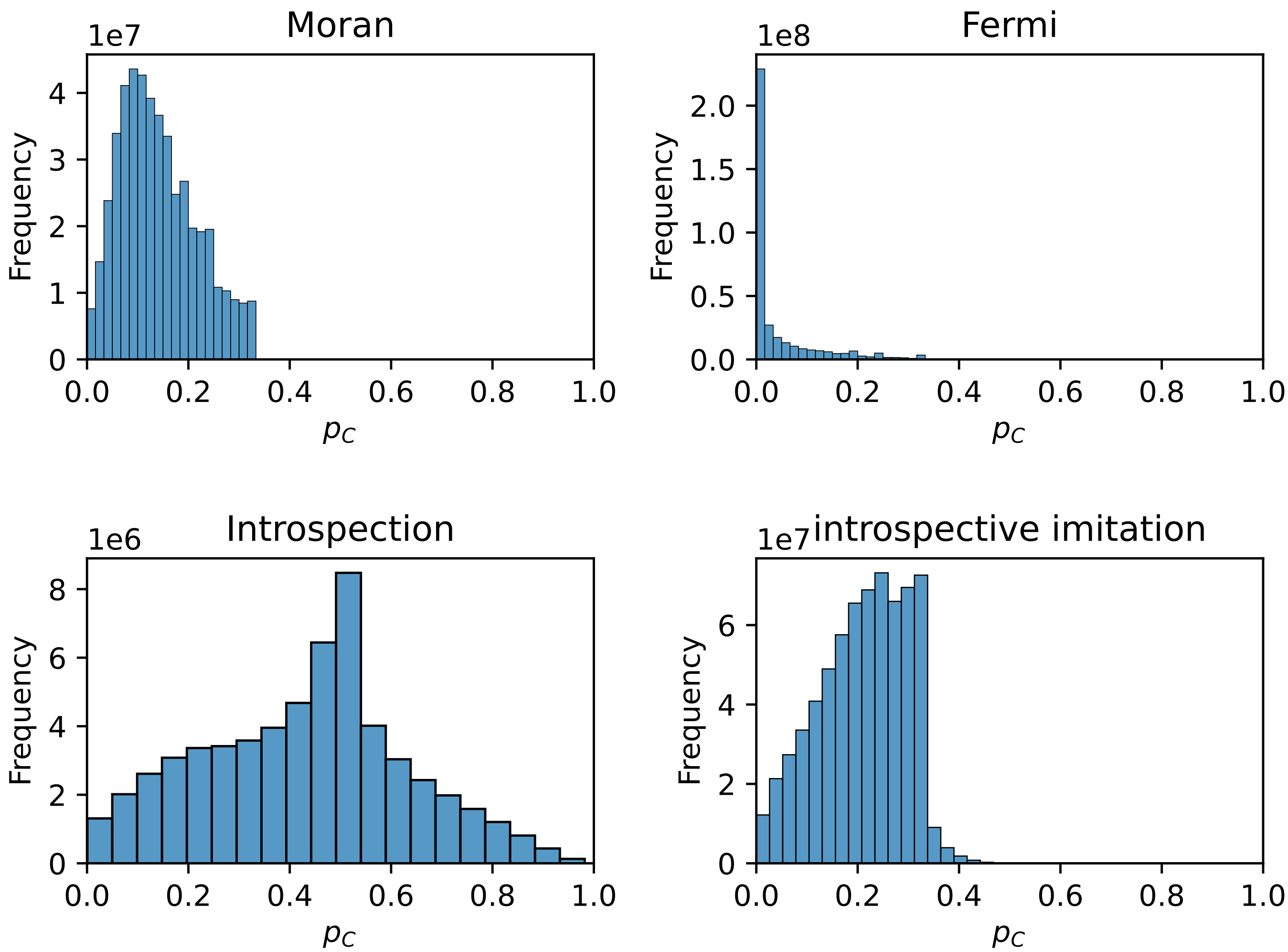


Figure 3: Distribution of p_C broken down by population dynamic.

3.3 Effect of population heterogeneity on extrinsic dynamics

We examine how varying the degree of contribution heterogeneity (the spread of α_i values in the linear population) affects p_C for a low-contributing mutant invading an $N = 3$ cooperating population.

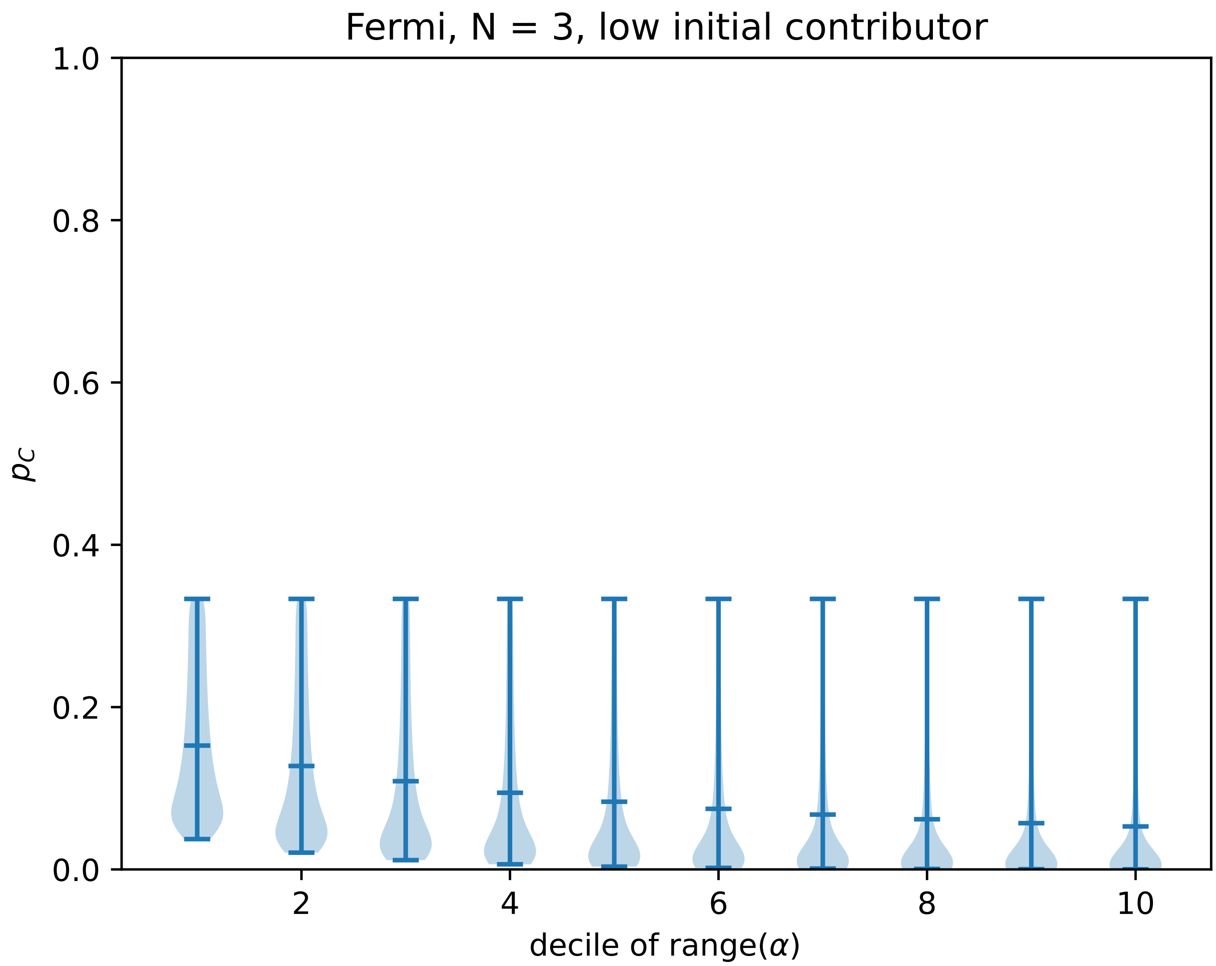


Figure 4: p_C under Fermi imitation dynamics as heterogeneity varies ($N = 3$, linear population, low mutant).

Under Fermi imitation dynamics, greater heterogeneity decreases p_C. Higher values of α_i increase the payoff gap between cooperators and defectors within each state, making it more likely that a defector will be copied and less likely that a high contributor will persist.

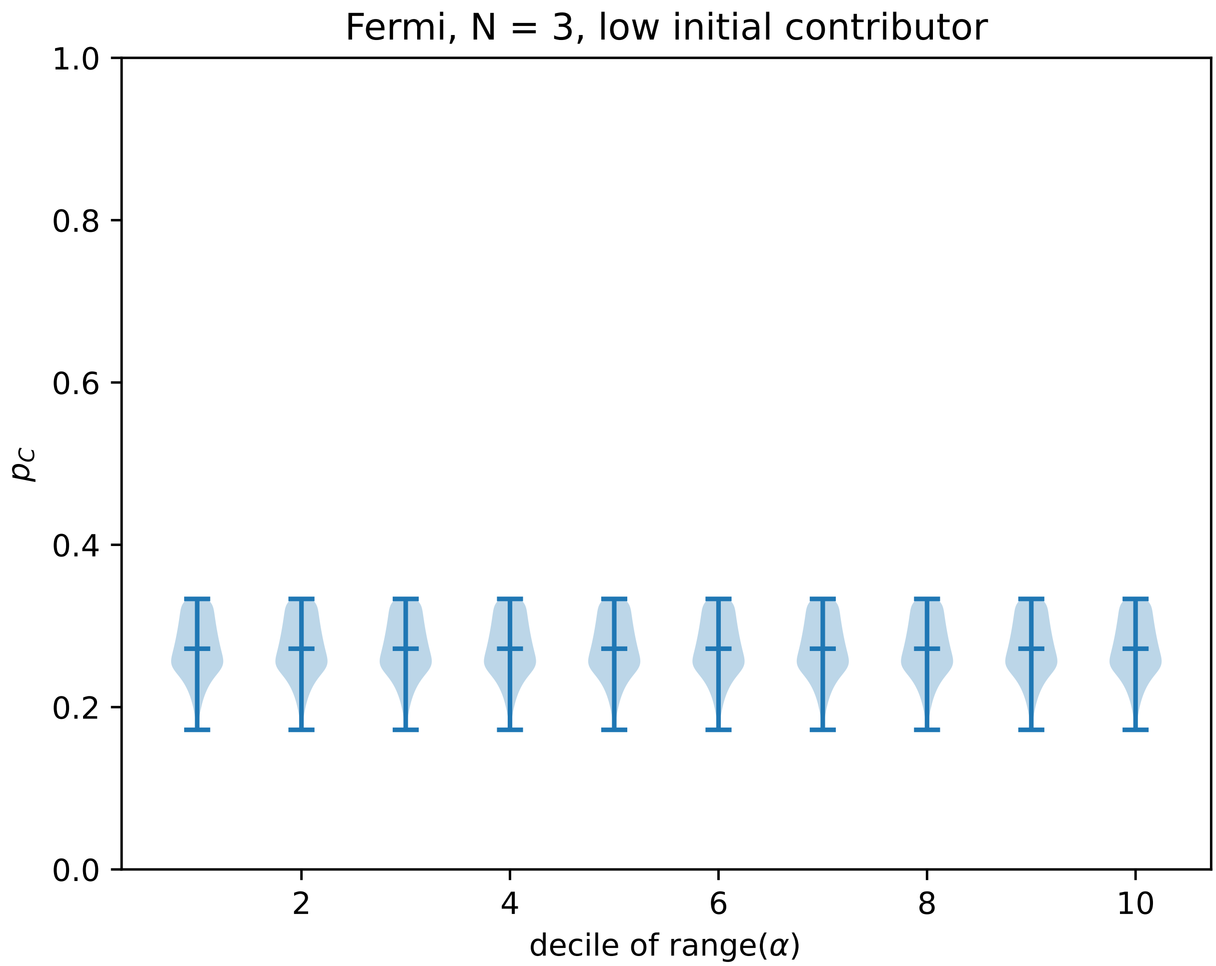


Figure 5: p_C under the Moran process as heterogeneity varies ($N = 3$, linear population, low mutant).

Under the Moran process, the degree of linear heterogeneity has no discernible effect on p_C (Figure 5): every violin across the heterogeneity deciles is centred at the same median and has the same spread. The same invariance holds for Introspective Imitation Dynamics under linear population structures (see supplementary material). However, both dynamics are sensitive to the *binomial* population structure.

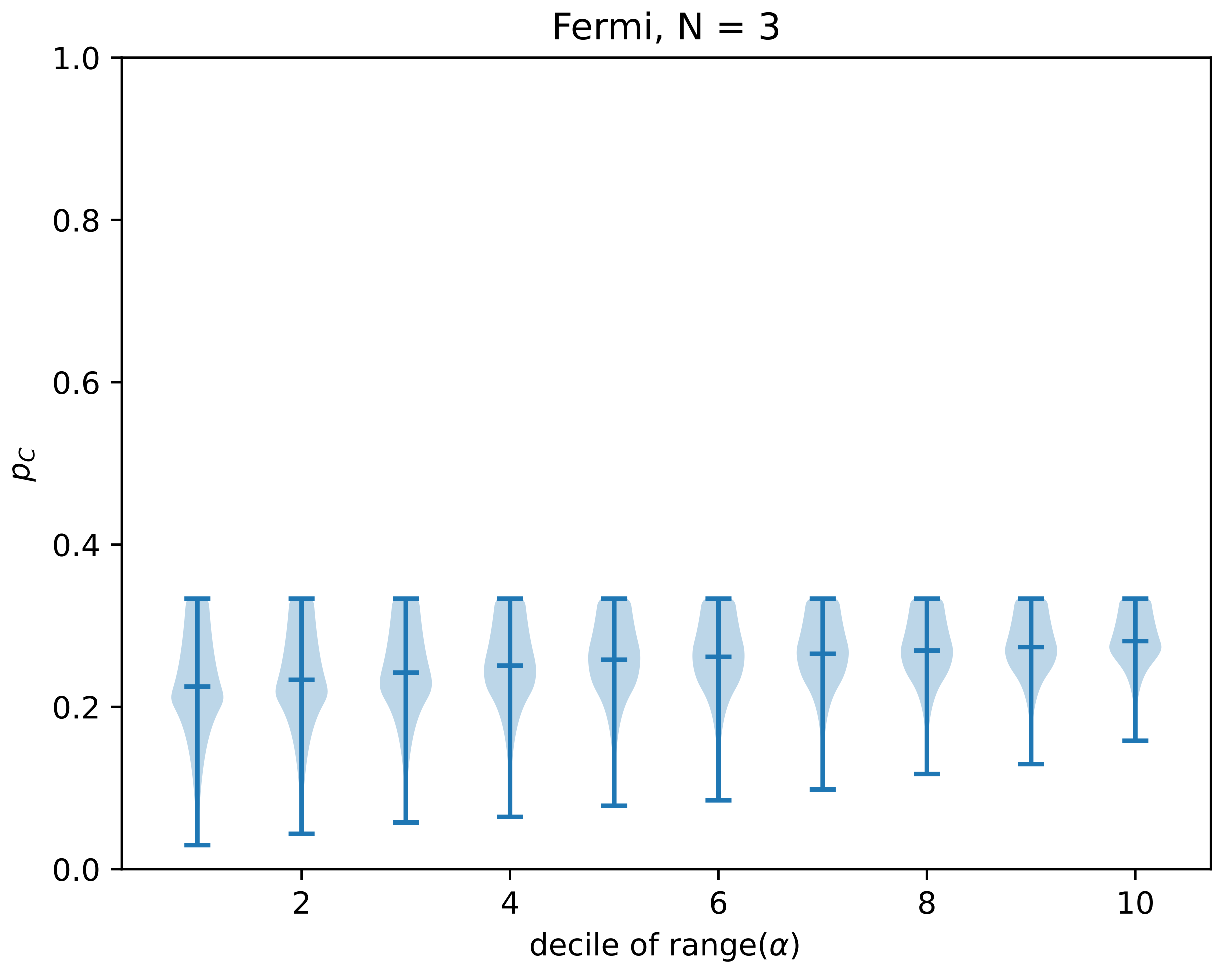


Figure 6: p_C under the Moran process as binomial heterogeneity varies ($N = 3$, binomial population, low mutant). Unlike the linear case, a clear upward trend in the median is visible.

Under the binomial population structure (Figure 6), both the Moran process and Introspective Imitation Dynamics exhibit a slight but consistent increase in p_C as the contribution gap between low and high contributors widens. This suggests that the invariance observed under linear heterogeneity is specific to that population structure: it does not reflect a general insensitivity of these dynamics to heterogeneity.

3.4 The cooperation ceiling under purely extrinsic dynamics

Both extrinsic dynamics exhibit an upper bound on cooperation that cannot be exceeded by any parameter choice. When $\alpha_i > 0$ for all i, a contributing individual always earns less than a defector in the same state: the contributor pays α_i while receiving only $r\alpha_i/N$ of it back (since the multiplied pool is split equally), so a defector strictly benefits from the contributor's effort without bearing its cost. In a purely extrinsic dynamic, where strategy revision compares players' current payoffs directly, the probability of adopting a cooperating strategy is therefore always lower than the probability of adopting a defecting one. Cooperation cannot exceed the neutral-drift baseline:

Proposition 1 (Extrinsic ceiling)**.** *Under any purely extrinsic dynamic, the stationary probability of full cooperation satisfies $p_C \leq \frac{1}{N}$ for all $\alpha_i > 0$.*

For the $N = 3$ case shown, this gives a ceiling of $p_C \leq \frac{1}{3}$, visible as a hard upper boundary in both Figures 4 and 5.

Although Introspective Imitation Dynamics breaks through this ceiling (see Section 3.5), it does not escape it entirely: across all parameter combinations examined, p_C under Introspective Imitation Dynamics has a hard empirical upper bound near $p_C \approx 0.35$ to 0.40 for $N = 3$. This is only modestly above the $1/3$ extrinsic ceiling, and well below the values achievable by introspection dynamics, which can approach $p_C = 1$.

3.5 Intrinsic dynamics exceed the extrinsic ceiling

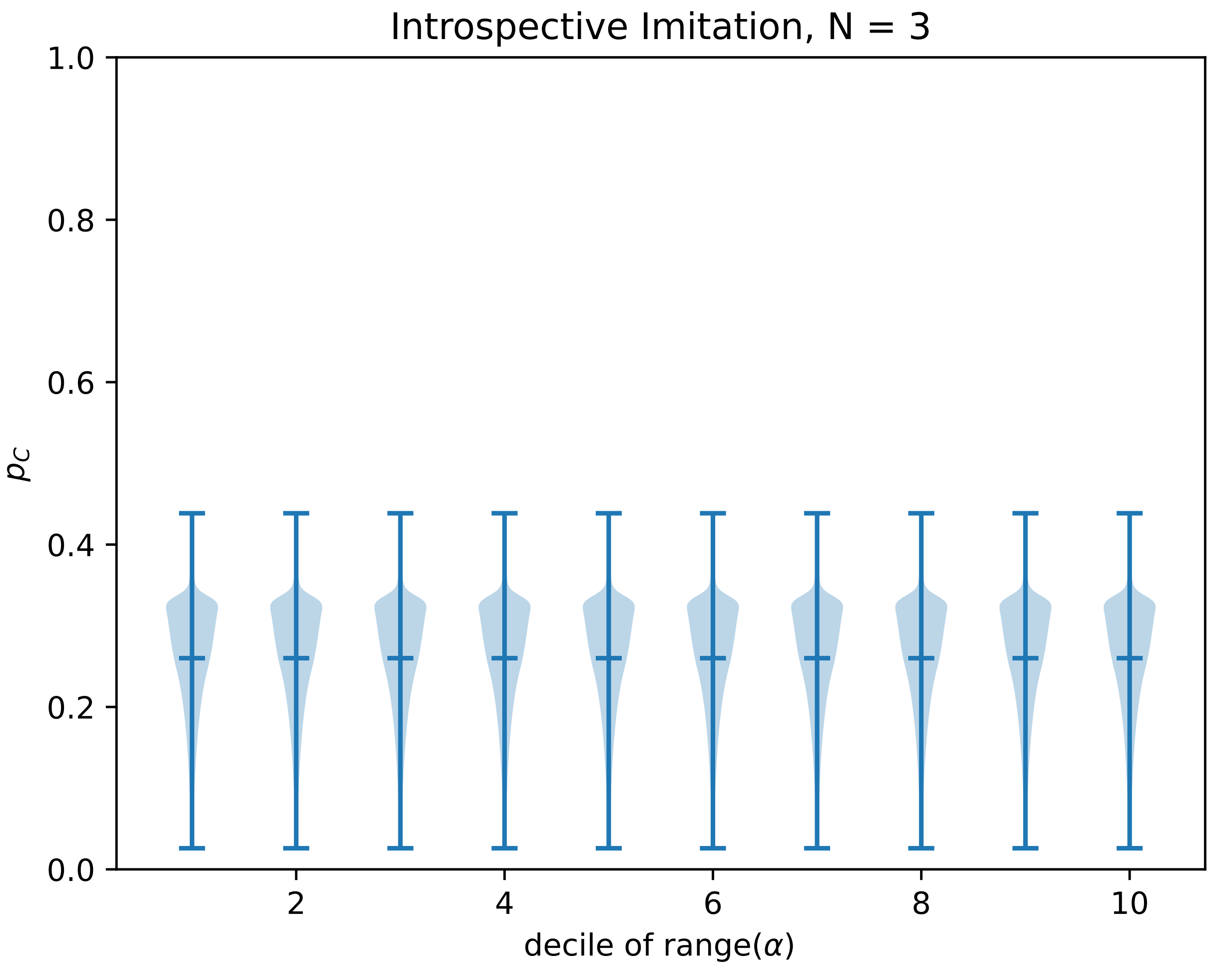


Figure 7: p_C under Introspective Imitation Dynamics ($N = 3$, linear population, low mutant).

Introspective Imitation Dynamics breaks through the extrinsic ceiling. The introspective step evaluates whether the focal player's own payoff would improve under the candidate strategy, independently of whether a defecting neighbour currently outperforms cooperators. When cooperation is individually beneficial for the focal player, this step assigns a probability greater than $\frac{1}{2}$ to accepting it, enabling $p_C > \frac{1}{N}$.

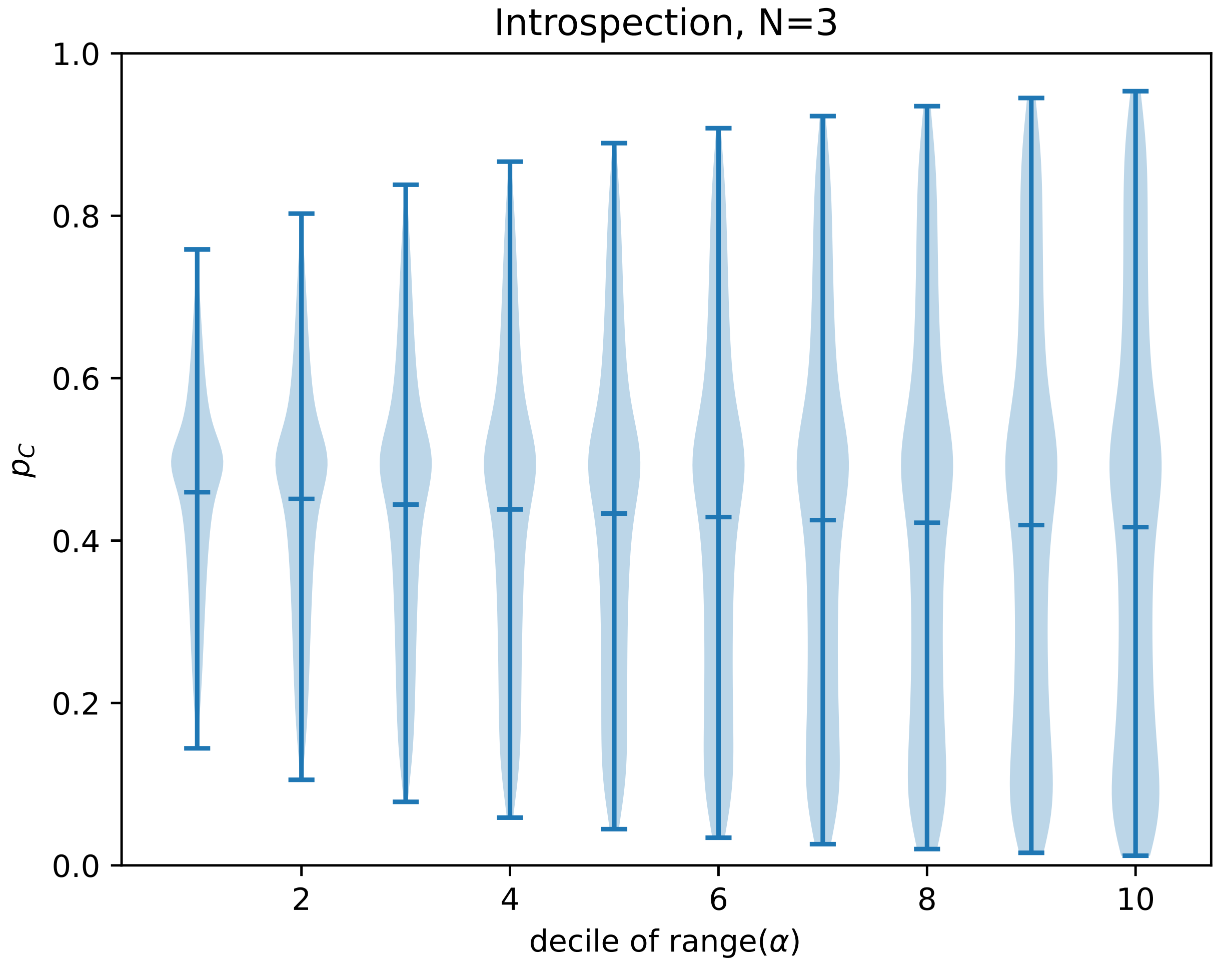


Figure 8: p_C under introspection dynamics ($N = 3$, linear population).

The same qualitative result holds for introspection dynamics, though p_C there measures a steady-state probability rather than an absorption probability, so comparisons across dynamics should be made with care.

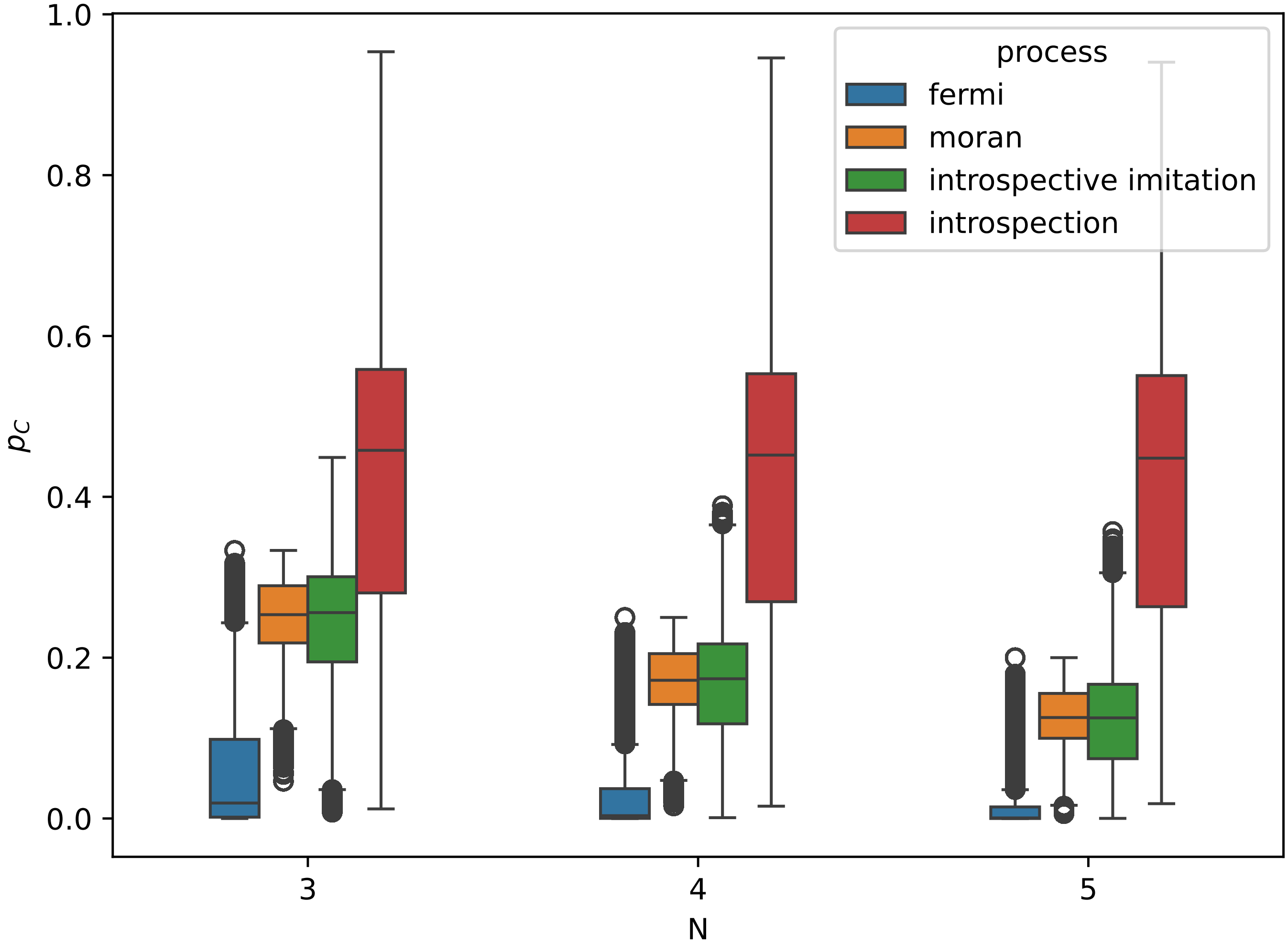


Figure 9: Maximum p_C achieved by each dynamic across all parameter settings, for multiple values of N.

The advantage of intrinsic dynamics over extrinsic dynamics holds across all values of N and both population structures examined, as Figure 9 and the extended parameter sweeps in the supplementary material confirm.

3.6 The $r = N$ threshold

To identify when intrinsic dynamics begin to outperform extrinsic ones, we compare p_C values from the Moran process and Introspective Imitation Dynamics directly; the latter is the Moran process augmented with a single introspective step, making the comparison clean.

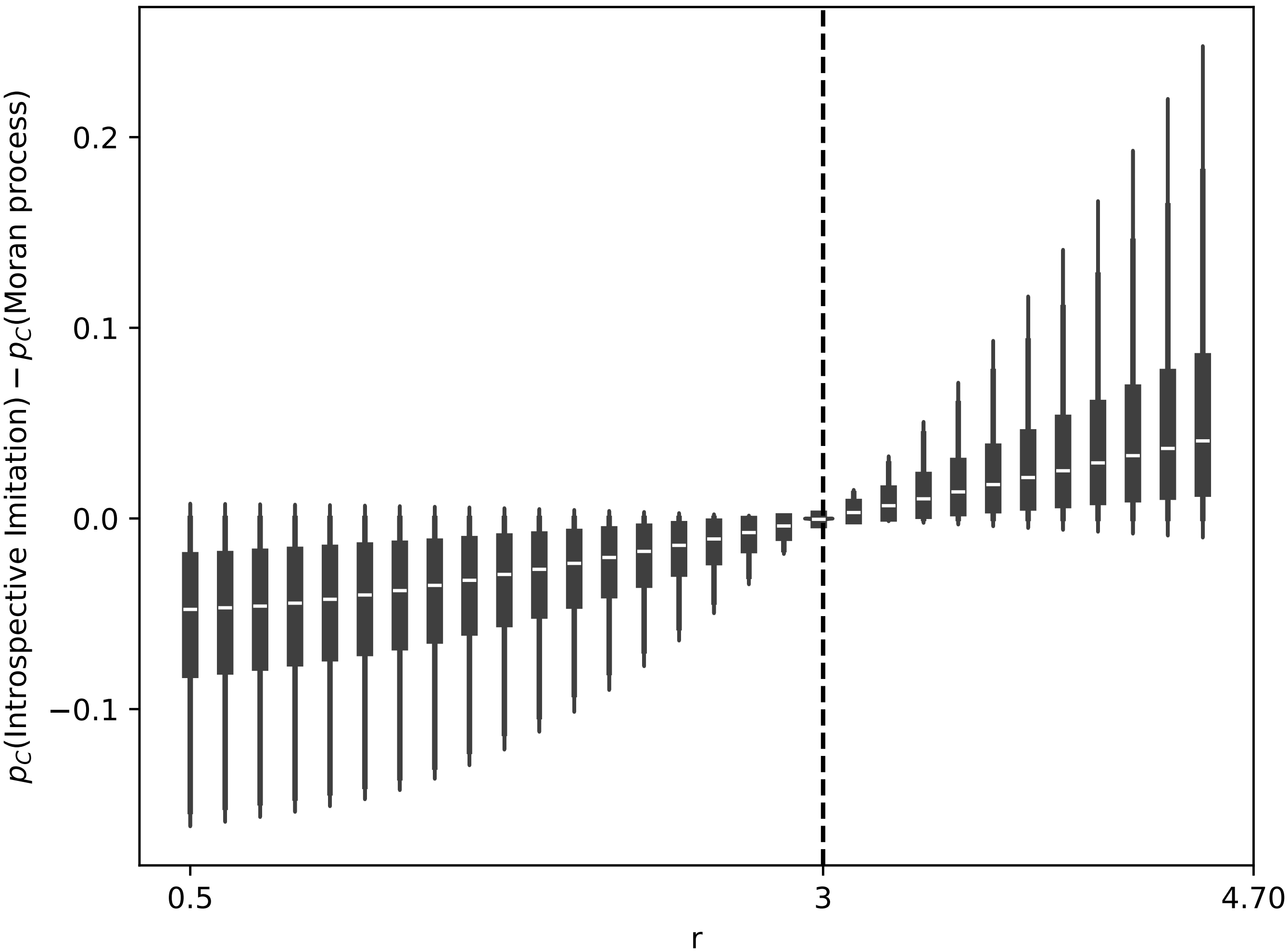


Figure 10: Comparison of p_C under the Moran process and Introspective Imitation Dynamics for $N = 3$.

The parameter $r = N$ marks the precise threshold at which intrinsic dynamics begin to favour cooperation over defection.

Proposition 2 ($r = N$ threshold)**.** *For any heterogeneous population with $\alpha_i > 0$ for all i, the introspective evaluation step of Introspective Imitation Dynamics favours cooperation if and only if $r > N$.*

Proof. Individual i evaluates switching from defection to cooperation by comparing their payoff $\pi_i(\mathbf{a})$ in the current state to the payoff $\pi_i(\mathbf{b})$ they would receive in the state where they cooperate and all others remain unchanged. Using the convention $\Delta\pi_i = \pi_i(\mathbf{a}) - \pi_i(\mathbf{b})$,

$$\Delta\pi_i = \alpha_i - \frac{r\alpha_i}{N} = \alpha_i\Big(1 - \frac{r}{N}\Big).$$

Since $\alpha_i > 0$, this is negative if and only if $r > N$. Since ϕ is a decreasing function, $\phi(\Delta\pi_i) > \frac{1}{2}$ precisely when $\Delta\pi_i < 0$, i.e. when $r > N$, and $\phi(\Delta\pi_i) < \frac{1}{2}$ when $r < N$. □

When $r < N$, the introspective step actively suppresses cooperation relative to the extrinsic baseline. This is visible in Figure 10 as a region where Introspective Imitation Dynamics underperforms the Moran process. The pattern holds across all values of N (see supplementary material).

A further empirical signature of the $r = N$ threshold is a sharp collapse of variance. In Figure 11, which shows $p_C(\text{IID}) - p_C(\text{Moran})$ across the full r range, the interquartile range narrows to nearly zero precisely at $r = N$, before widening again for larger r. This variance minimum coincides exactly with the theoretical threshold, providing an unusually clean empirical marker for the analytical result.

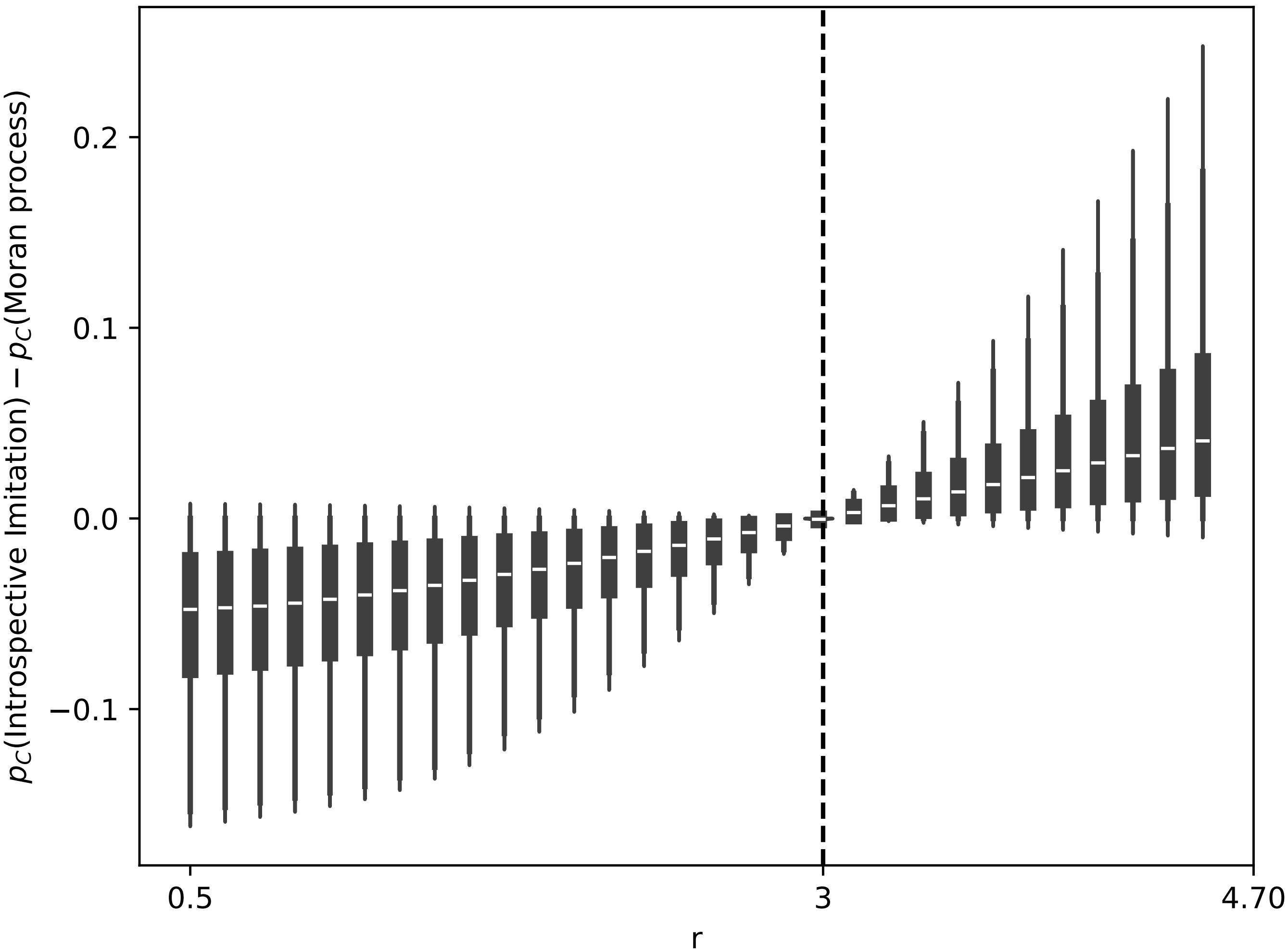


Figure 11: $p_C(\mathrm{IID}) - p_C(\mathrm{Moran})$ as a function of r for $N = 3$ (linear population). The dashed vertical line marks $r = N = 3$. The IQR collapses to near zero at the threshold before expanding for $r > N$.

3.7 Choice intensity and its interaction with dynamics type

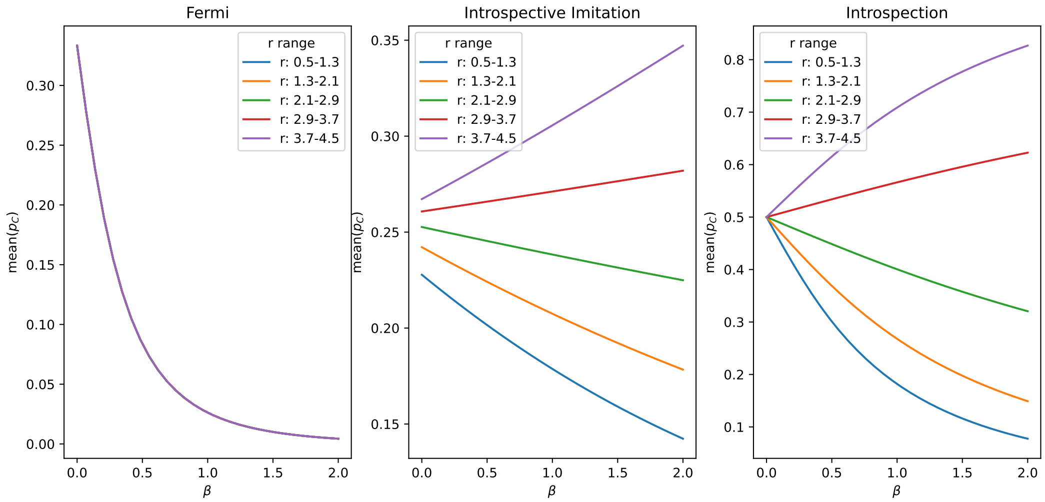


Figure 12: p_C as a function of choice intensity β and multiplier r, comparing extrinsic and intrinsic dynamics.

Higher choice intensity β consistently reduces p_C under purely extrinsic dynamics, regardless of r. A high β causes players to respond sharply to payoff differences, and since defectors always have higher payoffs than cooperators under extrinsic comparison, stronger selection accelerates the collapse of cooperation.

Under intrinsic dynamics the relationship reverses when $r > N$. The introspective step compares a player's current payoff against their counterfactual payoff under cooperation; when $r > N$ this comparison favours cooperation, and higher β amplifies that advantage, raising p_C.

Introspection dynamics has a further notable property at $\beta = 0$: all lines in Figure 12 converge to $p_C = 0.5$, independently of r or population structure. At zero choice intensity players update their strategy uniformly at random, so the stationary distribution places equal weight on all states; since exactly half of all states have the focal player cooperating, $p_C = 0.5$ follows immediately. This neutral-drift baseline therefore serves as an anchor: introspection raises p_C above 0.5 when $r > N$ and lowers it below 0.5 when $r < N$, always converging to 0.5 at $\beta = 0$.

3.8 Aspiration dynamics

3.9 Dominance relations between dynamics

Having examined each dynamic in isolation, we now compare them directly to establish a dominance ordering in terms of p_C.

Introspection always outperforms Fermi. Figure 13 shows p_C(Introspection)$-p_C$(Fermi) across the full parameter sweep. The lower tail of every violin sits at or above zero: introspection dynamics *uniformly dominates* Fermi imitation dynamics for all parameter combinations examined. The gap grows monotonically with r, reaching a median difference of approximately 0.7 at the upper end of the sweep.

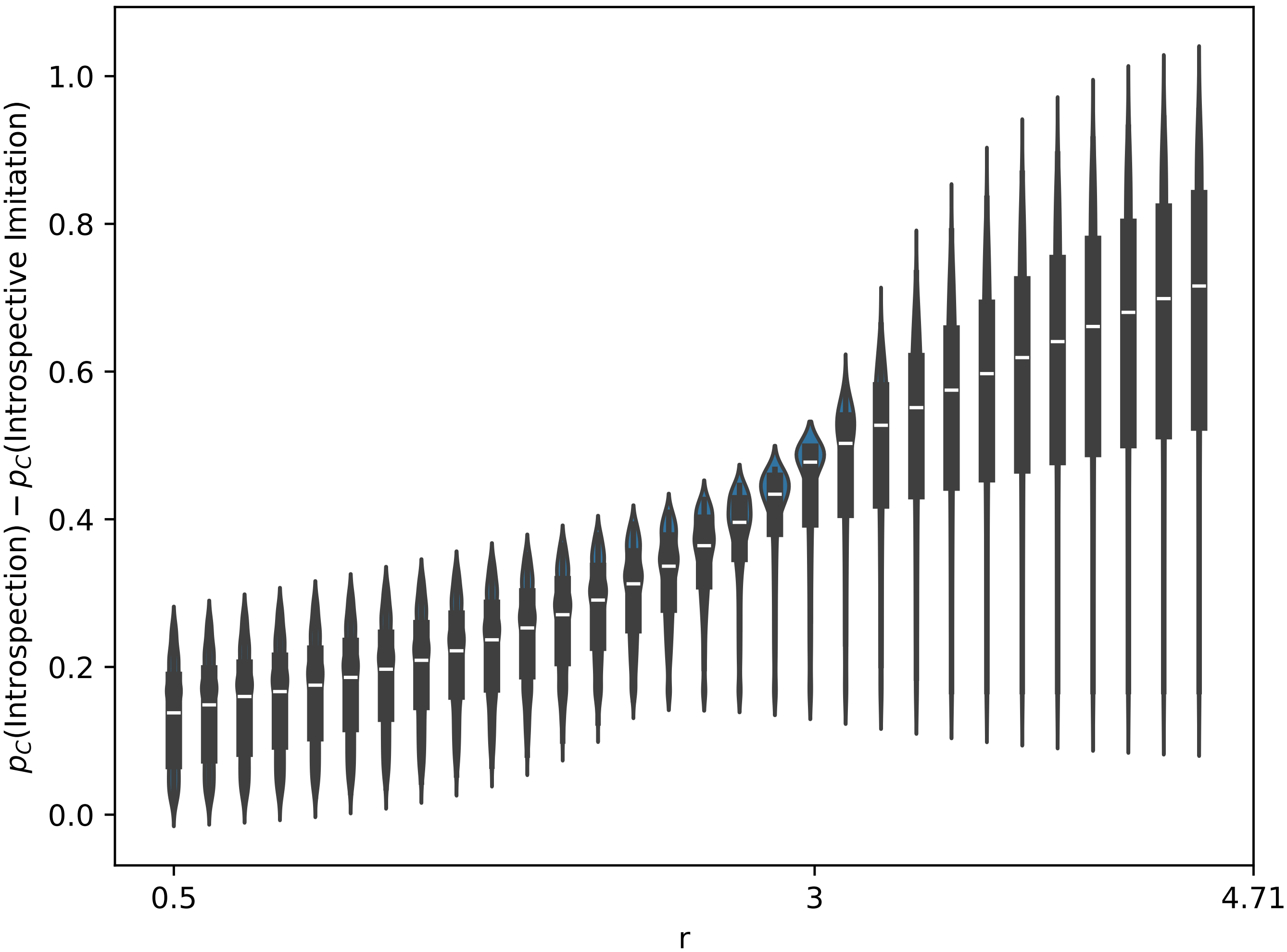


Figure 13: p_C(Introspection) $-$ p_C(Fermi) as a function of r for $N = 3$ (linear population). The difference is non-negative for all parameter combinations, establishing a uniform dominance of introspection over Fermi dynamics.

The Moran–Fermi ordering is r-invariant. Unlike every other pairwise comparison, Figure 14 shows that p_C(Moran) $-$ p_C(Fermi) has nearly identical violin shape, spread, and median across the entire r range. There is no threshold effect at $r = N$: the comparison between these two dynamics is effectively insensitive to r. This parallels the finding that Fermi dynamics is itself r-insensitive (Section on choice intensity). On average the Moran process outperforms Fermi imitation, though for some parameter configurations Fermi achieves a higher p_C.

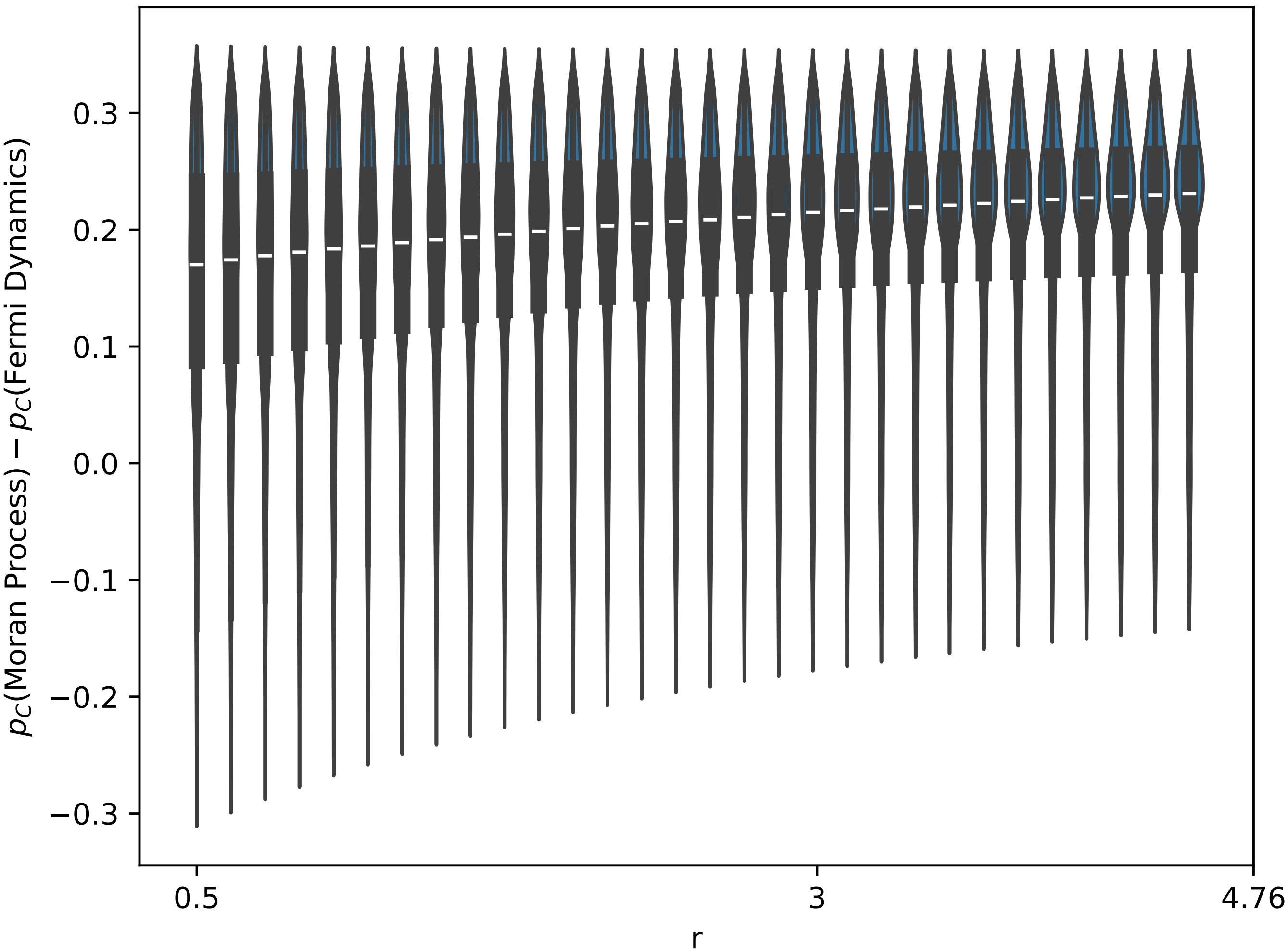


Figure 14: $p_C(\text{Moran}) - p_C(\text{Fermi})$ as a function of r for $N = 3$ (linear population). Unlike all other pairwise comparisons, the violin shape and median are approximately constant across the entire r range.

Introspection's advantage over Moran grows with N**.** At $N = 3$, $p_C(\text{Introspection}) - p_C(\text{Moran})$ can be negative for $r < N$: there exist parameter configurations under which the Moran process outperforms introspection dynamics. However, at $N = 9$ (Figure 15), the same difference is *always non-negative* across the entire r range, including values well below $r = N = 9$. This transition from a sign-changing difference to a uniformly positive one reflects the shrinking of the Moran ceiling $1/N$ as N increases: with a lower ceiling, the Moran process is more easily dominated by introspection dynamics across all multipliers.

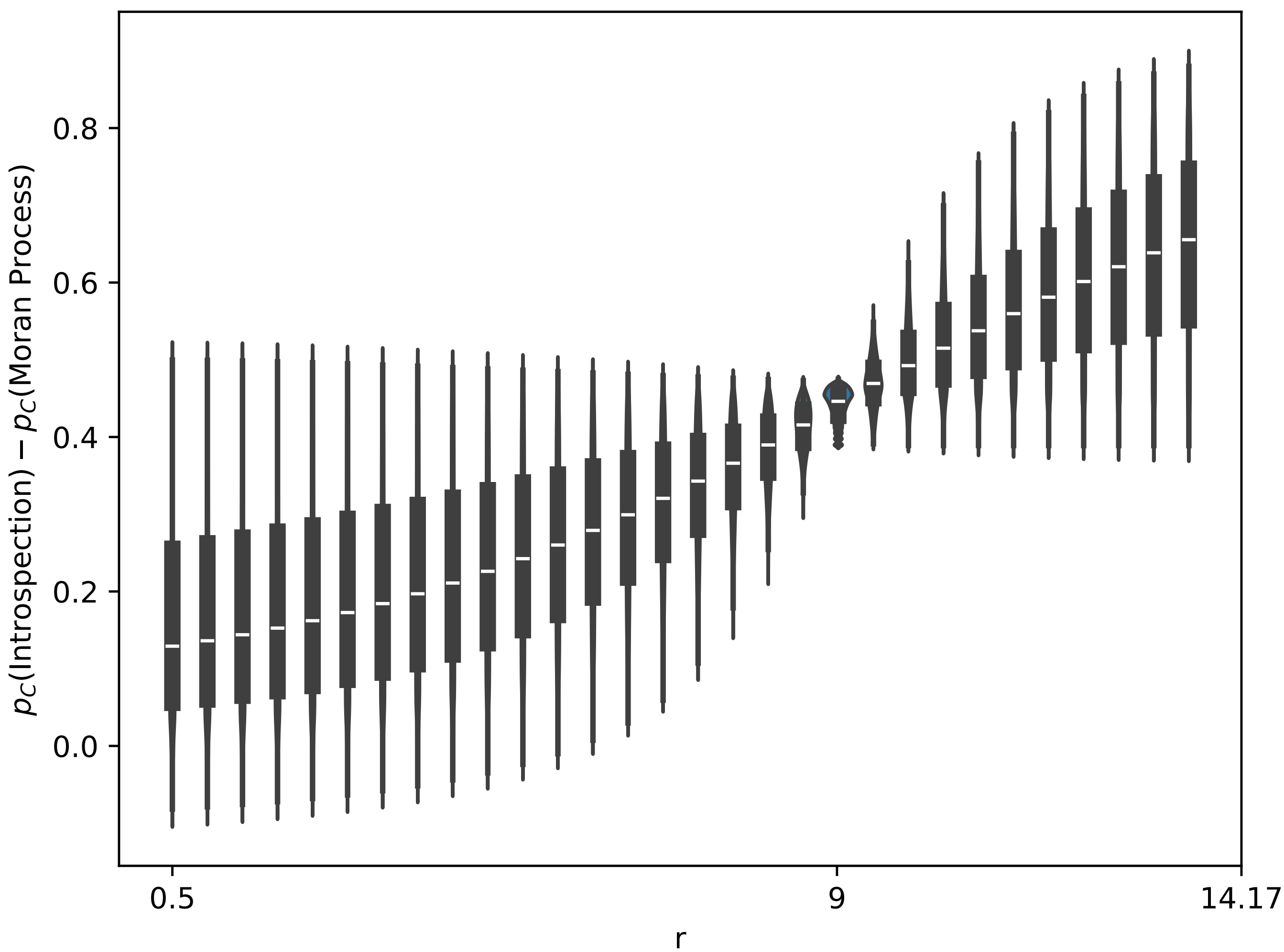


Figure 15: p_C(Introspection) $-$ p_C(Moran) as a function of r for $N = 9$ (linear population). At this population size the difference is non-negative for all r, in contrast to the $N = 3$ case where the difference can be negative for $r < N$.

4 Conclusion

We have introduced and analysed a framework for heterogeneous public goods games under five distinct population dynamics, two purely extrinsic (Moran process and Fermi imitation) and three intrinsic or mixed (aspiration, introspection, and Introspective Imitation Dynamics). Exact results are obtained by computing transition matrices and stationary distributions analytically for the two-player case, with the general-N case studied numerically.

The central theoretical finding is a dichotomy between extrinsic and intrinsic dynamics. Under any purely extrinsic dynamic, cooperation is bounded above by $p_C \leq 1/N$ (Proposition 1): payoff comparisons between players always disfavour the cooperating strategy, so the neutral-drift baseline cannot be exceeded. Intrinsic dynamics, which evaluate payoffs from the focal player's own perspective, break through this ceiling whenever $r > N$ (Proposition 2). The $r = N$ threshold is sharp: it is not only the point at which the expected difference in p_C changes sign, but also the point at which the variance in that difference collapses to a minimum.

The empirical results reveal a rich structure within and across dynamics. Introspection dynamics uniformly dominates Fermi imitation over all parameter combinations, and its advantage over the Moran process grows with population size N, becoming unconditional for large enough N. The Moran–Fermi comparison, by contrast, is approximately r-invariant: neither dynamics has a meaningful threshold response to r. Population heterogeneity affects the two extrinsic dynamics differently depending on structure: linear heterogeneity is neutral for both Moran and IID, but binomial heterogeneity produces a consistent positive effect on p_C for both.

Introspective Imitation Dynamics is introduced here as a novel hybrid dynamic, combining the imitative learning structure of the Moran process with the payoff self-evaluation of introspection dynamics. It is the only absorbing dynamic that exceeds the $1/N$ ceiling, and its behaviour changes qualitatively at $r = N$. Its hard empirical upper bound near $p_C \approx 0.35$–0.40 for $N = 3$, modestly above the ceiling, suggests that introspective self-evaluation alone cannot fully substitute for the unconstrained exploration of strategy space available to introspection dynamics.

The framework developed here, however, provides a complete and exact treatment of the two-player heterogeneous public goods game under each dynamic, and establishes the key qualitative distinctions that persist across population sizes and structures.

5 Supplementary Information

5.1 Explicit transition matrices for the two-player game

In each subsection below, states are ordered $\mathbf{a} = (0,0)$, $\mathbf{b} = (0,1)$, $\mathbf{c} = (1,0)$, $\mathbf{d} = (1,1)$; rows and columns of each matrix correspond to these states in the same order.

Moran process

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{f_1(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} & 1 - \frac{f_1(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} - \frac{f_2(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} & 0 & \frac{f_2(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} \\ \frac{f_2(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} & 0 & 1 - \frac{f_1(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} - \frac{f_2(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} & \frac{f_1(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{6}$$

Now, by taking the fundamental matrix of T and performing a right multiplication by R, the matrix with entries which represent transitions from transitive states to absorbing states[27], we can obtain the absorption matrix for this system.

For the above game, we find the following equation:

$$\left(I - \begin{bmatrix} 1 - \frac{f_1(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} - \frac{f_2(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} & 0 \\ 0 & 1 - \frac{f_1(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} - \frac{f_2(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} \end{bmatrix} \right)^{-1} \begin{bmatrix} \frac{f_1(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} & \frac{f_2(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} \\ \frac{f_2(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} & \frac{f_1(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} \end{bmatrix} \tag{7}$$

And solve to find our absorption matrix:

$$\begin{bmatrix} \frac{f_1(\mathbf{b})}{\left(\frac{f_1(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} + \frac{f_2(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})}\right)(2f_1(\mathbf{b})+2f_2(\mathbf{b}))} & \frac{f_2(\mathbf{b})}{\left(\frac{f_1(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} + \frac{f_2(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})}\right)(2f_1(\mathbf{b})+2f_2(\mathbf{b}))} \\ \frac{f_2(\mathbf{c})}{\left(\frac{f_1(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} + \frac{f_2(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})}\right)(2f_1(\mathbf{c})+2f_2(\mathbf{c}))} & \frac{f_1(\mathbf{c})}{\left(\frac{f_1(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} + \frac{f_2(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})}\right)(2f_1(\mathbf{c})+2f_2(\mathbf{c}))} \end{bmatrix} \tag{8}$$

In this matrix, the rows represent our initial transitive state (in our system, row 1 represents $\mathbf{b}$ and row 2 represents $\mathbf{c}$), and our columns represent which state the system is absorbed into (with $\mathbf{a}$ on the left, and $\mathbf{d}$ on the right). It is immediately clear from this result that the probability of the system ending up in a particular absorbing state does not rely on the fitness of individuals in the absorbing state, but rather their fitness in the transitive states. Thus, players can remain in a state which could be bettered by a single change, however they will not make such a change if no individual of another type exists in the population.

Fermi imitation dynamics

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{2}\phi(\pi_2(\mathbf{b}) - \pi_1(\mathbf{b})) & \frac{1}{2} & 0 & \frac{1}{2}\phi(\pi_1(\mathbf{b}) - \pi_2(\mathbf{b})) \\ \frac{1}{2}\phi(\pi_1(\mathbf{c}) - \pi_2(\mathbf{c})) & 0 & \frac{1}{2} & \frac{1}{2}\phi(\pi_2(\mathbf{c}) - \pi_1(\mathbf{c})) \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{9}$$

For the two-player state space $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$, the explicit transition matrix is given in equation (9) in the supplementary material. Applying the same absorption matrix calculation as for the Moran process, we obtain:

$$\begin{bmatrix} \phi(\pi_2(\mathbf{b}) - \pi_1(\mathbf{b})) & \phi(\pi_1(\mathbf{b}) - \pi_2(\mathbf{b})) \\ \phi(\pi_1(\mathbf{c}) - \pi_2(\mathbf{c})) & \phi(\pi_2(\mathbf{c}) - \pi_1(\mathbf{c})) \end{bmatrix} \tag{10}$$

This shows that, similarly to the Moran process, our absorption probabilities rely on the fitness of players in transitive states and not those of players in the absorbing states. Fermi imitation dynamics is therefore poorly suited to heterogeneous populations: a player may copy a high-fitness strategy without considering whether that strategy would benefit them, and may even enter an absorbing state that worsens their own payoff.

Aspiration Dynamics

The related transition matrix is:

$$T_{\mathbf{ab}} = \begin{cases} \dfrac{1}{N}\,\phi(\pi_{I(\mathbf{a},\mathbf{b})}(\mathbf{a}) - A_{I(\mathbf{a},\mathbf{b})}) & \text{if } \mathbf{b} \in \mathrm{Neb}(\mathbf{a}) \text{ and } j = I(\mathbf{a},\mathbf{b}), \\ 0 & \text{if } \mathbf{b} \notin \mathrm{Neb}(\mathbf{a})\text{and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \tag{11}$$

Now, consider the transition matrix for our general four-state system:

$$\begin{bmatrix} 1 - \frac{1}{2(1+e^{-\beta(A_2-f_2(a))})} - \frac{1}{2(1+e^{-\beta(A_1-f_1(a))})} & \frac{e^{A_2\beta}}{2(e^{A_2\beta}+e^{\beta f_2(a)})} & \frac{e^{A_1\beta}}{2(e^{A_1\beta}+e^{\beta f_1(a)})} & 0 \\ \frac{e^{A_2\beta}}{2(e^{A_2\beta}+e^{\beta f_2(b)})} & 1 - \frac{1}{2(1+e^{-\beta(A_2-f_2(b))})} - \frac{1}{2(1+e^{-\beta(A_1-f_1(b))})} & 0 & \frac{e^{A_1\beta}}{2(e^{A_1\beta}+e^{\beta f_1(b)})} \\ \frac{e^{A_1\beta}}{2(e^{A_1\beta}+e^{\beta f_1(c)})} & 0 & 1 - \frac{1}{2(1+e^{-\beta(A_2-f_2(c))})} - \frac{1}{2(1+e^{-\beta(A_1-f_1(c))})} & \frac{e^{A_2\beta}}{2(e^{A_2\beta}+e^{\beta f_2(c)})} \\ 0 & \frac{e^{A_1\beta}}{2(e^{A_1\beta}+e^{\beta f_1(d)})} & \frac{e^{A_2\beta}}{2(e^{A_2\beta}+e^{\beta f_2(d)})} & 1 - \frac{1}{2(1+e^{-\beta(A_2-f_2(d))})} - \frac{1}{2(1+e^{-\beta(A_1-f_1(d))})} \end{bmatrix} \tag{12}$$

We can see that as our players no longer consider the actions of the rest of the population when considering their decision, we no longer have an absorbing Markov chain. Thus, we no longer consider an absorption matrix, but instead we consider the steady state of our Markov chain [27] This is the proportion of time spent in each state when the Markov chain reaches a stable distribution across the states. Any irreducible, aperiodic, finite Markov chain has a unique stationary distribution, and the chain converges to it from any starting state [**?**].. For our four state system, we acquire the following result for a stationary distribution:

Introspective Dynamics

The transition matrix for the introspection dynamics process [3] is given by:

$$T_{\mathbf{ab}} = \begin{cases} \dfrac{1}{N(m_j-1)}\,\phi(\Delta\pi_{I(\mathbf{a},\mathbf{b})}) & \text{if } \mathbf{b} \in \mathrm{Neb}(\mathbf{a}) \text{ and } j = I(\mathbf{a},\mathbf{b}), \\ 0 & \text{if } \mathbf{b} \notin \mathrm{Neb}(\mathbf{a})\text{and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \tag{13}$$

where m_j is the number of actions available to individual j. Since we assume that all individuals have the same action set with size k, we can replace $m_j - 1$ with $k - 1$.

Now, Couto also calculated in [2] the explicit strategy distribution for an asymmetric interaction between two players. We can adapt these results to define our transition matrix for our example state space **a,b,c,d**:

$$\begin{bmatrix} 1 - \frac{1}{2\left(e^{\beta(\pi_2(a)-\pi_2(b))}+1\right)} - \frac{1}{2\left(e^{\beta(\pi_1(a)-\pi_1(c))}+1\right)} & \frac{1}{2\left(e^{\beta(\pi_2(a)-\pi_2(b))}+1\right)} & \frac{1}{2\left(e^{\beta(\pi_1(a)-\pi_1(c))}+1\right)} & 0 \\ \frac{1}{2\left(e^{\beta(-\pi_2(a)+\pi_2(b))}+1\right)} & 1 - \frac{1}{2\left(e^{\beta(\pi_1(b)-\pi_1(d))}+1\right)} - \frac{1}{2\left(1+e^{-\beta(\pi_2(a)-\pi_2(b))}\right)} & 0 & \frac{1}{2\left(e^{\beta(\pi_1(b)-\pi_1(d))}+1\right)} \\ \frac{1}{2\left(e^{\beta(-\pi_1(a)+\pi_1(c))}+1\right)} & 0 & 1 - \frac{1}{2\left(e^{\beta(\pi_2(c)-\pi_2(d))}+1\right)} - \frac{1}{2\left(1+e^{-\beta(\pi_1(a)-\pi_1(c))}\right)} & \frac{1}{2\left(e^{\beta(\pi_2(c)-\pi_2(d))}+1\right)} \\ 0 & \frac{1}{2\left(e^{\beta(-\pi_1(b)+\pi_1(d))}+1\right)} & \frac{1}{2\left(e^{\beta(-\pi_2(c)+\pi_2(d))}+1\right)} & 1 - \frac{1}{2\left(1+e^{-\beta(\pi_2(c)-\pi_2(d))}\right)} - \frac{1}{2\left(1+e^{-\beta(\pi_1(b)-\pi_1(d))}\right)} \end{bmatrix} \tag{14}$$

We then consider the steady state vectors for this system. For this, we take the left eigenvector of the transition matrix, which is given by:

Unlike in the case of imitation dynamics, there are no absorbing states for a Markov chain transitioning according to imitation dynamics. Therefore, we define a steady state vector $\mathbf{u}$ according to the method Couto uses in [2].

Introspective Imitation Dynamics

Note that Introspective Imitation Dynamics takes into account both selection intensity and choice intensity. This means that both the level to which a player's fitness is considered when their strategy is being chosen for consideration, and also how much it is considered when choosing whether to accept the new strategy, is present in the dynamic.

Let $\eta_{i,j}(\mathbf{x},\mathbf{y}) = 1 + e^{\beta(f_i(\mathbf{x}) - f_j(\mathbf{y}))}$ be the denominator of the Fermi function. For the two-player state space $S = \{\mathbf{a},\mathbf{b},\mathbf{c},\mathbf{d}\}$, the explicit transition matrix is given in equation (16) in the supplementary material. Applying the absorption matrix calculation, we obtain:

$$\begin{bmatrix} \frac{\eta_{1,1}(\mathbf{b,d})f_1(b)}{\eta_{1,1}(\mathbf{b,d})f_1(b)+\eta_{2,2}(\mathbf{b,a})f_2(b)} & \frac{\eta_{2,2}(\mathbf{b,a})f_2(b)}{\eta_{1,1}(\mathbf{b,d})f_1(b)+\eta_{2,2}(\mathbf{b,a})f_2(b)} \\ \frac{\eta_{2,2}(\mathbf{c,d})f_2(c)}{\eta_{1,1}(\mathbf{c,a})f_1(c)+\eta_{2,2}(\mathbf{c,d})f_2(c)} & \frac{\eta_{1,1}(\mathbf{c,a})f_1(c)}{\eta_{1,1}(\mathbf{c,a})f_1(c)+\eta_{2,2}(\mathbf{c,d})f_2(c)} \end{bmatrix} \tag{15}$$

Unlike other imitation dynamics, we can see that the absorption probability in an introspective imitation process relies on the fitness that at least one individual will possess in the absorbing state. This means that players will enter absorbing states with a higher probability if it improves their own fitness, despite any heterogeneity between payoffs for the same action played by different players. This solves the problem of players making worsening moves based on the fitness of another individual in the population, while still retaining the property of imitation dynamics that new strategies cannot be introduced to the system.

Let $\eta_{i,j}(\mathbf{x},\mathbf{y}) = 1 + e^{\beta(f_i(\mathbf{x}) - f_j(\mathbf{y}))}$.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{f_1(b)}{2\eta_{2,2}(\mathbf{b,a})(f_1(b)+f_2(b))} & 1 - \frac{f_2(b)}{\eta_{1,1}(\mathbf{b,d})(2f_1(b)+2f_2(b))} - \frac{f_1(b)}{\eta_{2,2}(\mathbf{b,a})(2f_1(b)+2f_2(b))} & 0 & \frac{f_2(b)}{2\eta_{1,1}(\mathbf{b,d})(f_1(b)+f_2(b))} \\ \frac{f_2(c)}{2\eta_{1,1}(\mathbf{c,a})(f_1(c)+f_2(c))} & 0 & 1 - \frac{f_1(c)}{\eta_{2,2}(\mathbf{c,d})(2f_1(c)+2f_2(c))} - \frac{f_2(c)}{\eta_{1,1}(\mathbf{c,a})(2f_1(c)+2f_2(c))} & \frac{f_1(c)}{2\eta_{2,2}(\mathbf{c,d})(f_1(c)+f_2(c))} \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{16}$$

Moran process with mutation

$$\begin{bmatrix} -\frac{\mu_{12}}{2} - \frac{\mu_{22}}{2} + 1 & \frac{\mu_{22}}{2} & \frac{\mu_{12}}{2} & 0 \\ \frac{\mu_{21}}{2} + \frac{(\epsilon f_1(b)+1)(-\mu_{21}-\mu_{22}+1)}{2(\epsilon f_1(b)+\epsilon f_2(b)+2)} & -\frac{\mu_{12}}{2} - \frac{\mu_{21}}{2} - \frac{(\epsilon f_1(b)+1)(-\mu_{21}-\mu_{22}+1)}{2(\epsilon f_1(b)+\epsilon f_2(b)+2)} - \frac{(\epsilon f_2(b)+1)(-\mu_{11}-\mu_{12}+1)}{2(\epsilon f_1(b)+\epsilon f_2(b)+2)} + 1 & 0 & \frac{\mu_{12}}{2} + \frac{(\epsilon f_2(b)+1)(-\mu_{11}-\mu_{12}+1)}{2(\epsilon f_1(b)+\epsilon f_2(b)+2)} \\ \frac{\mu_{11}}{2} + \frac{(\epsilon f_2(c)+1)(-\mu_{11}-\mu_{12}+1)}{2(\epsilon f_1(c)+\epsilon f_2(c)+2)} & 0 & -\frac{\mu_{11}}{2} - \frac{\mu_{22}}{2} - \frac{(\epsilon f_1(c)+1)(-\mu_{21}-\mu_{22}+1)}{2(\epsilon f_1(c)+\epsilon f_2(c)+2)} - \frac{(\epsilon f_2(c)+1)(-\mu_{11}-\mu_{12}+1)}{2(\epsilon f_1(c)+\epsilon f_2(c)+2)} + 1 & \frac{\mu_{22}}{2} + \frac{(\epsilon f_1(c)+1)(-\mu_{21}-\mu_{22}+1)}{2(\epsilon f_1(c)+\epsilon f_2(c)+2)} \\ 0 & \frac{\mu_{11}}{2} & \frac{\mu_{21}}{2} & -\frac{\mu_{11}}{2} - \frac{\mu_{21}}{2} + 1 \end{bmatrix} \tag{17}$$

Fermi imitation dynamics with mutation

$$\begin{bmatrix} -\frac{\mu_{12}}{2} - \frac{\mu_{22}}{2} + 1 & \frac{\mu_{22}}{2} & \frac{\mu_{12}}{2} & 0 \\ \frac{\mu_{21}}{2} + \frac{-\mu_{21}-\mu_{22}+1}{2(e^{\beta(-f_1(b)+f_2(b))}+1)} & -\frac{\mu_{12}}{2} - \frac{\mu_{21}}{2} + 1 - \frac{-\mu_{11}-\mu_{12}+1}{2(e^{\beta(f_1(b)-f_2(b))}+1)} - \frac{-\mu_{21}-\mu_{22}+1}{2(e^{\beta(-f_1(b)+f_2(b))}+1)} & 0 & \frac{\mu_{12}}{2} + \frac{-\mu_{11}-\mu_{12}+1}{2(e^{\beta(f_1(b)-f_2(b))}+1)} \\ \frac{\mu_{11}}{2} + \frac{-\mu_{11}-\mu_{12}+1}{2(e^{\beta(f_1(c)-f_2(c))}+1)} & 0 & -\frac{\mu_{11}}{2} - \frac{\mu_{22}}{2} + 1 - \frac{-\mu_{11}-\mu_{12}+1}{2(e^{\beta(f_1(c)-f_2(c))}+1)} - \frac{-\mu_{21}-\mu_{22}+1}{2(e^{\beta(-f_1(c)+f_2(c))}+1)} & \frac{\mu_{22}}{2} + \frac{-\mu_{21}-\mu_{22}+1}{2(e^{\beta(-f_1(c)+f_2(c))}+1)} \\ 0 & \frac{\mu_{11}}{2} & \frac{\mu_{21}}{2} & -\frac{\mu_{11}}{2} - \frac{\mu_{21}}{2} + 1 \end{bmatrix} \tag{18}$$

Aspiration dynamics with mutation

$$\begin{bmatrix} 1 - \frac{0.5}{1+e^{-\beta(A_2-f_2(a))}} - \frac{0.5}{1+e^{-\beta(A_1-f_1(a))}} & \frac{0.5e^{A_2\beta}}{e^{A_2\beta}+e^{\beta f_2(a)}} & \frac{0.5e^{A_1\beta}}{e^{A_1\beta}+e^{\beta f_1(a)}} & 0 \\ \frac{0.5e^{A_2\beta}}{e^{A_2\beta}+e^{\beta f_2(b)}} & 1 - \frac{0.5}{1+e^{-\beta(A_2-f_2(b))}} - \frac{0.5}{1+e^{-\beta(A_1-f_1(b))}} & 0 & \frac{0.5e^{A_1\beta}}{e^{A_1\beta}+e^{\beta f_1(b)}} \\ \frac{0.5e^{A_1\beta}}{e^{A_1\beta}+e^{\beta f_1(c)}} & 0 & 1 - \frac{0.5}{1+e^{-\beta(A_2-f_2(c))}} - \frac{0.5}{1+e^{-\beta(A_1-f_1(c))}} & \frac{0.5e^{A_2\beta}}{e^{A_2\beta}+e^{\beta f_2(c)}} \\ 0 & \frac{0.5e^{A_1\beta}}{e^{A_1\beta}+e^{\beta f_1(d)}} & \frac{0.5e^{A_2\beta}}{e^{A_2\beta}+e^{\beta f_2(d)}} & 1 - \frac{0.5}{1+e^{-\beta(A_2-f_2(d))}} - \frac{0.5}{1+e^{-\beta(A_1-f_1(d))}} \end{bmatrix} \tag{19}$$

Introspection dynamics with mutation

$$\begin{bmatrix} -\frac{\mu_{12}}{2} - \frac{\mu_{22}}{2} + 1 - \frac{-\mu_{21}-\mu_{22}+1}{2(e^{\beta(f_2(a)-f_2(b))}+1)} - \frac{-\mu_{11}-\mu_{12}+1}{2(e^{\beta(f_1(a)-f_1(c))}+1)} & \frac{\mu_{22}}{2} + \frac{-\mu_{21}-\mu_{22}+1}{2(e^{\beta(f_2(a)-f_2(b))}+1)} & \frac{\mu_{12}}{2} + \frac{-\mu_{11}-\mu_{12}+1}{2(e^{\beta(f_1(a)-f_1(c))}+1)} & 0 \\ \frac{\mu_{21}}{2} + \frac{-\mu_{21}-\mu_{22}+1}{2(e^{\beta(-f_2(a)+f_2(b))}+1)} & -\frac{\mu_{12}}{2} - \frac{\mu_{21}}{2} + 1 - \frac{-\mu_{21}-\mu_{22}+1}{2(e^{\beta(-f_2(a)+f_2(b))}+1)} - \frac{-\mu_{11}-\mu_{12}+1}{2(e^{\beta(f_1(b)-f_1(d))}+1)} & 0 & \frac{\mu_{12}}{2} + \frac{-\mu_{11}-\mu_{12}+1}{2(e^{\beta(f_1(b)-f_1(d))}+1)} \\ \frac{\mu_{11}}{2} + \frac{-\mu_{11}-\mu_{12}+1}{2(e^{\beta(-f_1(a)+f_1(c))}+1)} & 0 & -\frac{\mu_{11}}{2} - \frac{\mu_{22}}{2} + 1 - \frac{-\mu_{21}-\mu_{22}+1}{2(e^{\beta(f_2(c)-f_2(d))}+1)} - \frac{-\mu_{11}-\mu_{12}+1}{2(e^{\beta(-f_1(a)+f_1(c))}+1)} & \frac{\mu_{22}}{2} + \frac{-\mu_{21}-\mu_{22}+1}{2(e^{\beta(f_2(c)-f_2(d))}+1)} \\ 0 & \frac{\mu_{11}}{2} + \frac{-\mu_{11}-\mu_{12}+1}{2(e^{\beta(-f_1(b)+f_1(d))}+1)} & \frac{\mu_{21}}{2} + \frac{-\mu_{21}-\mu_{22}+1}{2(e^{\beta(-f_2(c)+f_2(d))}+1)} & -\frac{\mu_{11}}{2} - \frac{\mu_{21}}{2} + 1 - \frac{-\mu_{21}-\mu_{22}+1}{2(e^{\beta(-f_2(c)+f_2(d))}+1)} - \frac{-\mu_{11}-\mu_{12}+1}{2(e^{\beta(-f_1(b)+f_1(d))}+1)} \end{bmatrix} \tag{20}$$

Introspective Imitation Dynamics with mutation

Let $\eta_{i,j}(\mathbf{x},\mathbf{y}) = 1 + e^{\beta(f_i(\mathbf{x}) - f_j(\mathbf{y}))}$.

$$\begin{bmatrix} -\frac{\mu_{12}}{2} - \frac{\mu_{22}}{2} + 1 & \frac{\mu_{22}}{2} & \frac{\mu_{12}}{2} & 0 \\ \frac{\mu_{21}}{2} + \frac{(\epsilon f_1(b)+1)(-\mu_{21}-\mu_{22}+1)}{2\eta_{2,2}(\mathbf{b},\mathbf{a})(\epsilon f_1(b)+\epsilon f_2(b)+2)} & -\frac{\mu_{12}}{2} - \frac{\mu_{21}}{2} - \frac{(\epsilon f_1(b)+1)(-\mu_{21}-\mu_{22}+1)}{2\eta_{2,2}(\mathbf{b},\mathbf{a})(\epsilon f_1(b)+\epsilon f_2(b)+2)} - \frac{(\epsilon f_2(b)+1)(-\mu_{11}-\mu_{12}+1)}{2\eta_{1,1}(\mathbf{b},\mathbf{d})(\epsilon f_1(b)+\epsilon f_2(b)+2)} + 1 & 0 & \frac{\mu_{12}}{2} + \frac{(\epsilon f_2(b)+1)(-\mu_{11}-\mu_{12}+1)}{2\eta_{1,1}(\mathbf{b},\mathbf{d})(\epsilon f_1(b)+\epsilon f_2(b)+2)} \\ \frac{\mu_{11}}{2} + \frac{(\epsilon f_2(c)+1)(-\mu_{11}-\mu_{12}+1)}{2\eta_{1,1}(\mathbf{c},\mathbf{a})(\epsilon f_1(c)+\epsilon f_2(c)+2)} & 0 & -\frac{\mu_{11}}{2} - \frac{\mu_{22}}{2} - \frac{(\epsilon f_1(c)+1)(-\mu_{21}-\mu_{22}+1)}{2\eta_{2,2}(\mathbf{c},\mathbf{d})(\epsilon f_1(c)+\epsilon f_2(c)+2)} - \frac{(\epsilon f_2(c)+1)(-\mu_{11}-\mu_{12}+1)}{2\eta_{1,1}(\mathbf{c},\mathbf{a})(\epsilon f_1(c)+\epsilon f_2(c)+2)} + 1 & \frac{\mu_{22}}{2} + \frac{(\epsilon f_1(c)+1)(-\mu_{21}-\mu_{22}+1)}{2\eta_{2,2}(\mathbf{c},\mathbf{d})(\epsilon f_1(c)+\epsilon f_2(c)+2)} \\ 0 & \frac{\mu_{11}}{2} & \frac{\mu_{21}}{2} & -\frac{\mu_{11}}{2} - \frac{\mu_{21}}{2} + 1 \end{bmatrix} \tag{21}$$

5.2 Mutation

5.2.1 The Moran Process

In the case of the Moran process, our transition matrix becomes:

$$T_{\mathbf{ab}}^{(m)} = \begin{cases} \left(\frac{\sum_{a_i = b_{I(\mathbf{a},\mathbf{b})}} f(a_i)}{N \sum_{a_j} f(a_j)}\right)(1 - \sum_k \mu_{I(\mathbf{a},\mathbf{b}),k}) + \left(\frac{\mu_{i,\mathbf{b}_I(\mathbf{a},\mathbf{b})}}{N}\right) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \text{, differing at position } I(\mathbf{a},\mathbf{b}) \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b} \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \tag{22}$$

The explicit transition matrix for the two-player case is given in equation (17) in the supplementary material. This yields an ergodic Markov chain whose steady state is:

5.2.2 Fermi Imitation Dynamics

When we introduce mutation to Fermi Imitation Dynamics, we obtain the following transition matrix:

$$T_{\mathbf{ab}}^{(m)} = \begin{cases} \left(\frac{1}{N(N-1)} \sum_{a_j = b_{I(\mathbf{a},\mathbf{b})}} \phi(\pi_{\text{I}(\mathbf{a},\,\mathbf{b})}(a) - \pi_j(\mathbf{a}))\right)(1 - \sum_k \mu_{I(\mathbf{a},\mathbf{b}),k}) + \left(\frac{\mu_{i,\mathbf{b}_I(\mathbf{a},\mathbf{b})}}{N}\right) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b} \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \tag{23}$$

The explicit transition matrix for the two-player case is given in equation (18) in the supplementary material. The resulting steady state is:

5.2.3 Aspiration Dynamics

As aspiration and introspection dynamics do not form absorbing Markov chains, we do not need mutation to be applied in order to form an ergodic chain. However, we will consider these cases for completeness, and thus we obtain the following transition matrix:

$$T_{\mathbf{ab}}^{(m)} = \begin{cases} \left(\frac{1}{N} \phi(\pi_{I(\mathbf{a},\mathbf{b})} - A_i)\right)(1 - \sum_k \mu_{I(\mathbf{a},\mathbf{b}),k}) + \left(\frac{\mu_{i,\mathbf{b}_I(\mathbf{a},\mathbf{b})}}{N}\right) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \text{ and } j = I(\mathbf{a},\mathbf{b}), \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases}$$

The explicit transition matrix for the two-player case is given in equation (19) in the supplementary material.

5.2.4 Introspection Dynamics

As introspection dynamics does not form an absorbing Markov chain, we do not need mutation to form an ergodic chain. However, we will consider the following transition matrix for the case with mutation:

$$T^{(m)}_{\mathbf{ab}} = \begin{cases} (\dfrac{1}{N(m_j-1)}\,\phi(\Delta\pi_{I(\mathbf{a},\mathbf{b})}))(1-\sum_k \mu_{I(\mathbf{a},\mathbf{b}),k}) + (\frac{\mu_{i,\mathbf{b}_{I(\mathbf{a},\mathbf{b})}}}{N}) & \text{if } \mathbf{b}\in \text{Neb}(\mathbf{a}) \text{ and } j = I(\mathbf{a},\mathbf{b}),\\ 0 & \text{if } \mathbf{b}\notin \text{Neb}(\mathbf{a})\text{and } \mathbf{a}\neq\mathbf{b},\\ 1-\sum_{\mathbf{b}\in S\setminus\{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a}=\mathbf{b} \end{cases}$$

The explicit transition matrix for the two-player case is given in equation (20) in the supplementary material.

5.2.5 Introspective Imitation Dynamics

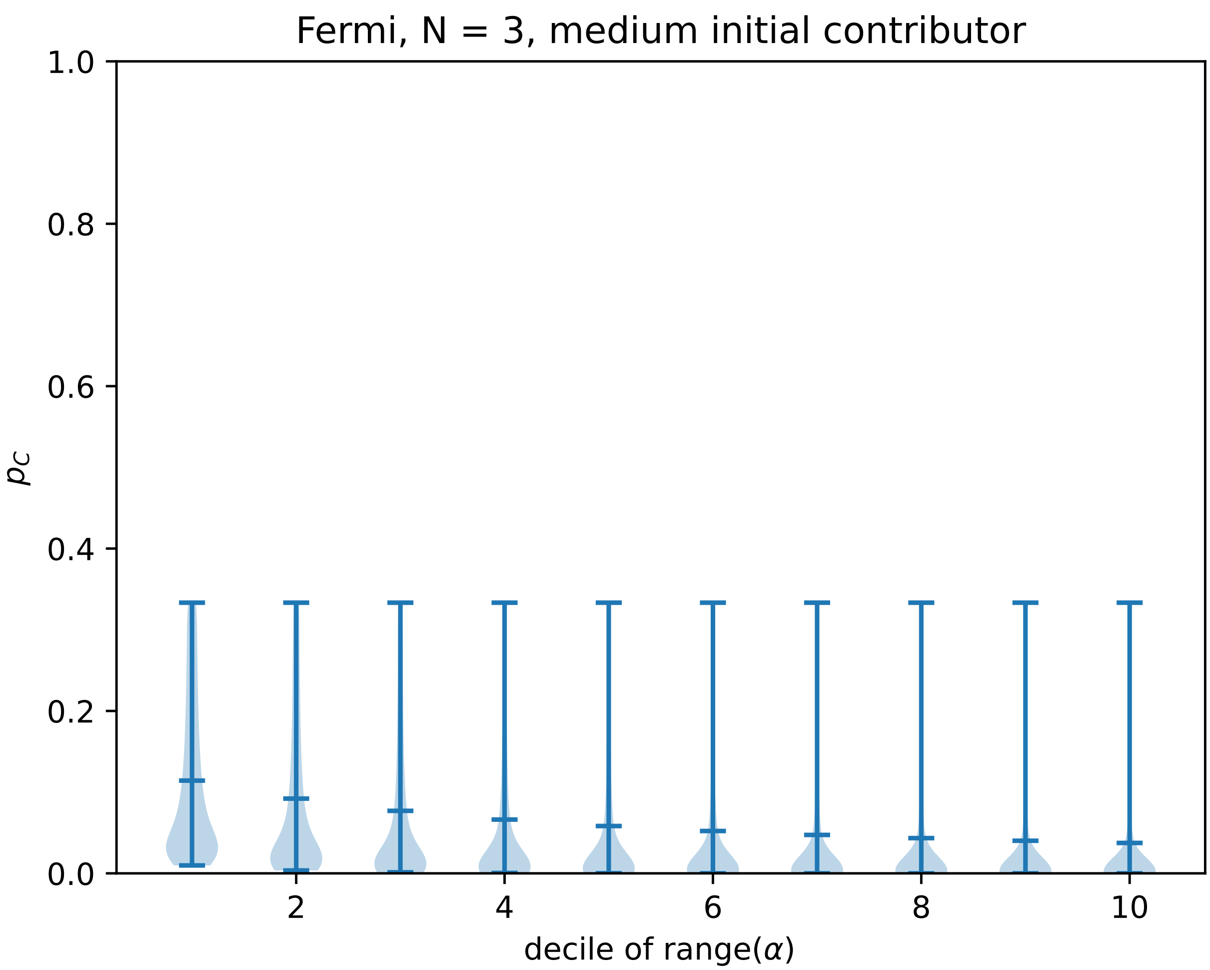
Finally, for the case of introspective imitation dynamics, we get the following transition matrix with mutation:

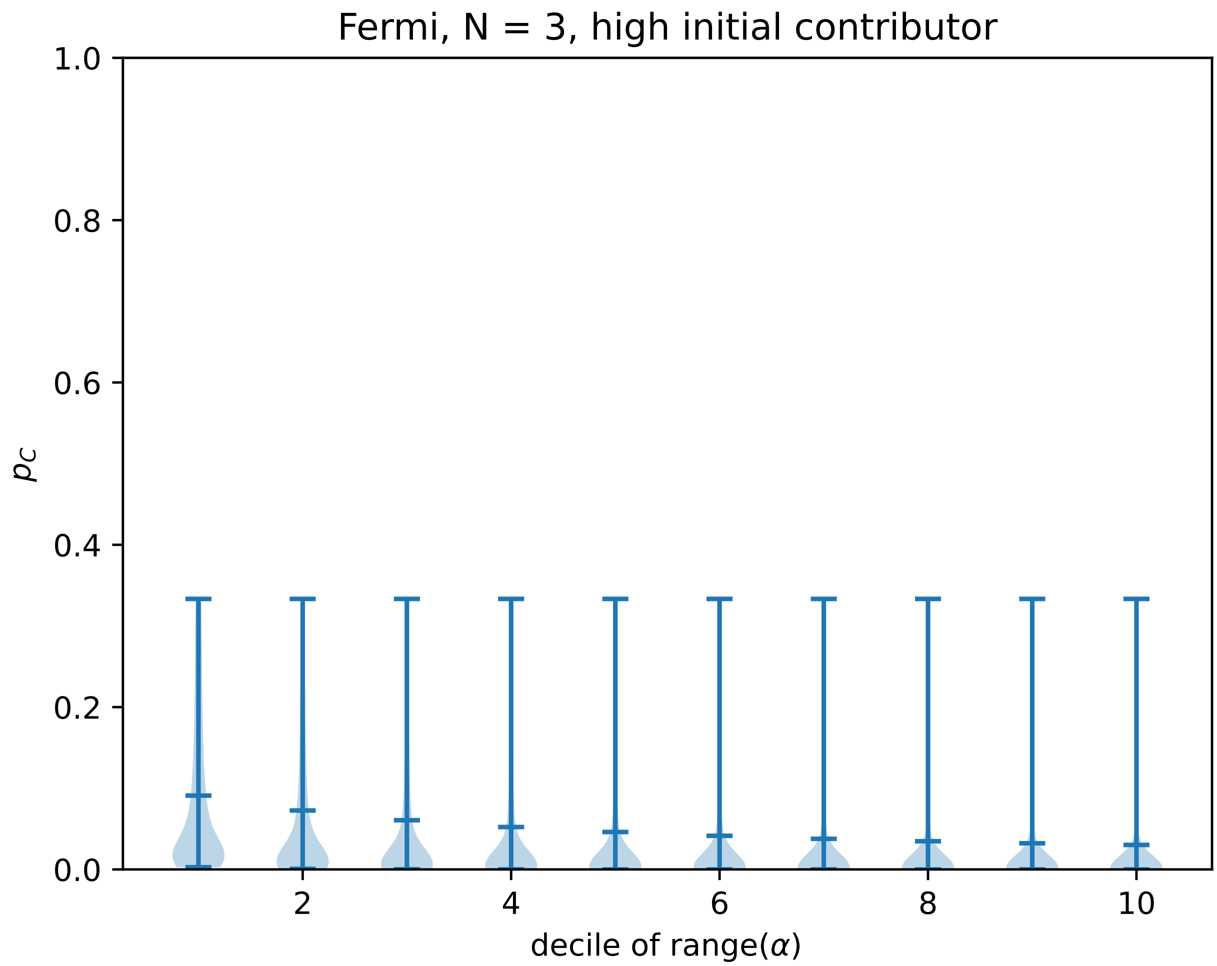
$$T^{(m)}_{\mathbf{ab}} = \begin{cases} (\frac{1}{N}\frac{\sum_{a_j=b_{I(\mathbf{a},\mathbf{b})}} f_j(\mathbf{a})}{\sum_k f_k(\mathbf{a})}\phi(\Delta(f_{I(\mathbf{a},\mathbf{b})})))(1-\sum_k \mu_{I(\mathbf{a},\mathbf{b}),k}) + (\frac{\mu_{i,\mathbf{b}_{I(\mathbf{a},\mathbf{b})}}}{N}) & \text{if } \mathbf{b}\in \text{Neb}(\mathbf{a})\\ 0 & \text{if } \mathbf{b}\notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a}\neq\mathbf{b}\\ 1-\sum_{\mathbf{b}\in S\setminus\{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a}=\mathbf{b} \end{cases} \tag{24}$$

The explicit transition matrix for the two-player case is given in equation (21) in the supplementary material. The resulting steady state is:

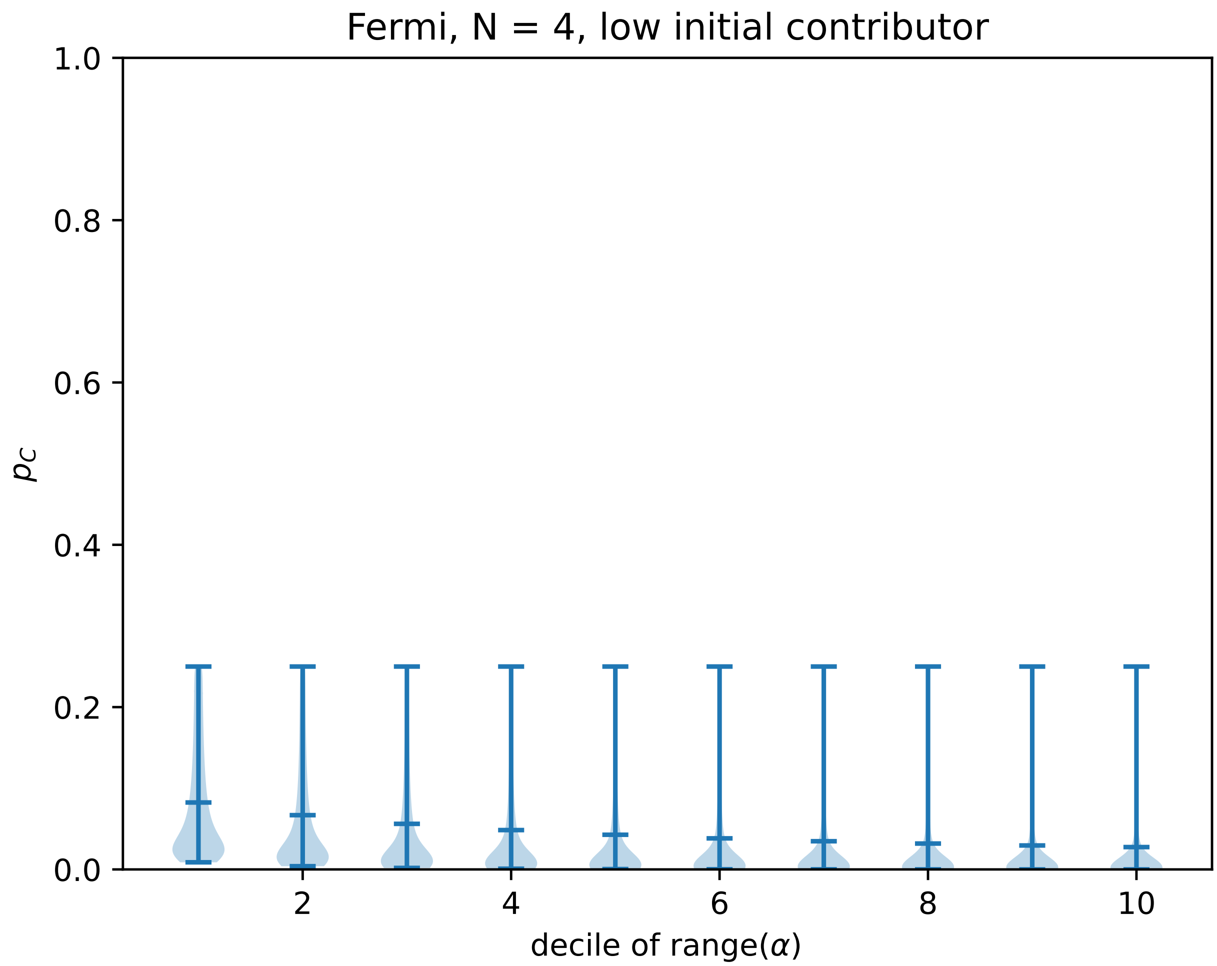
5.3 Extended parameter sweeps

Fermi imitation dynamics: linear population, $N = 3$–8

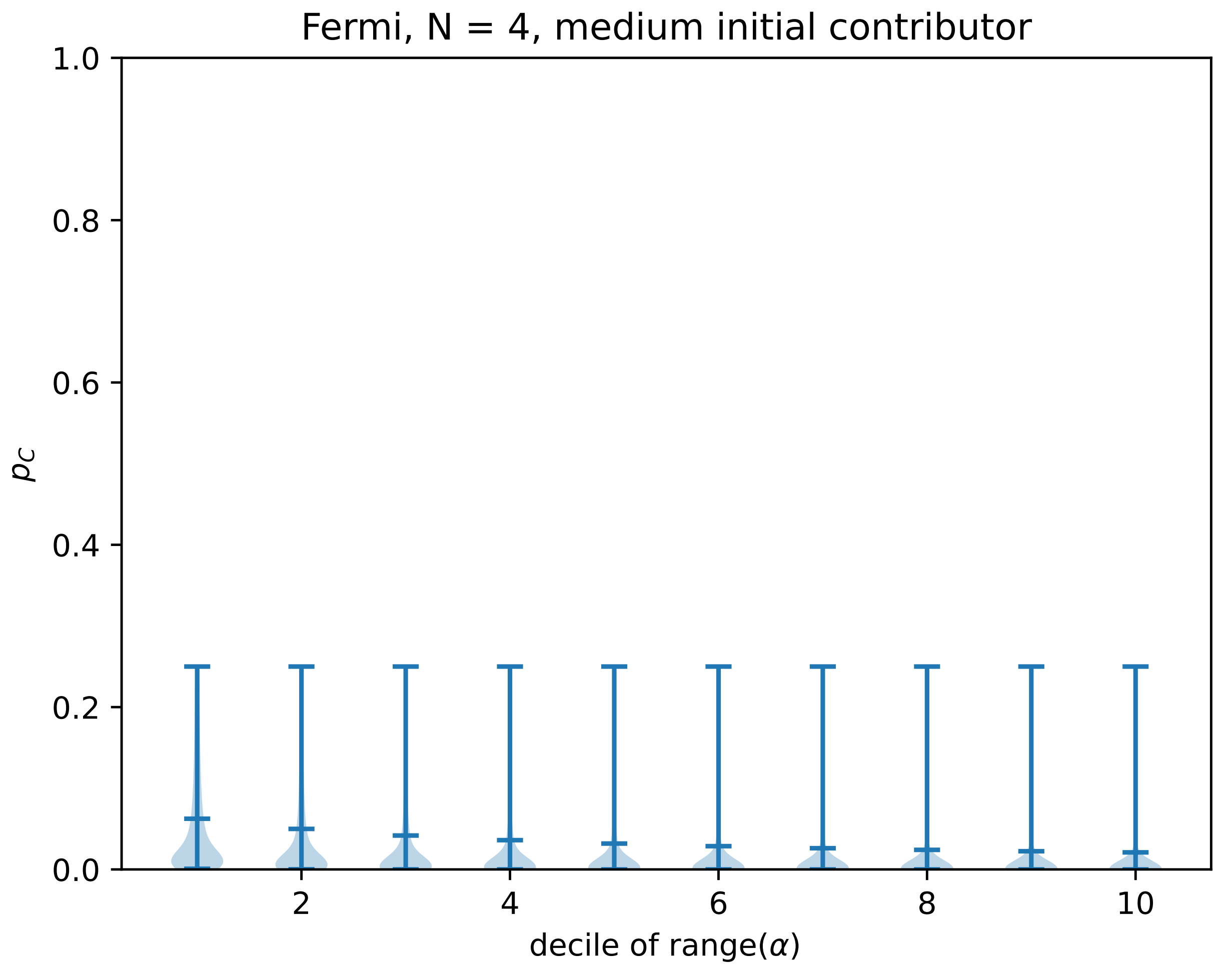


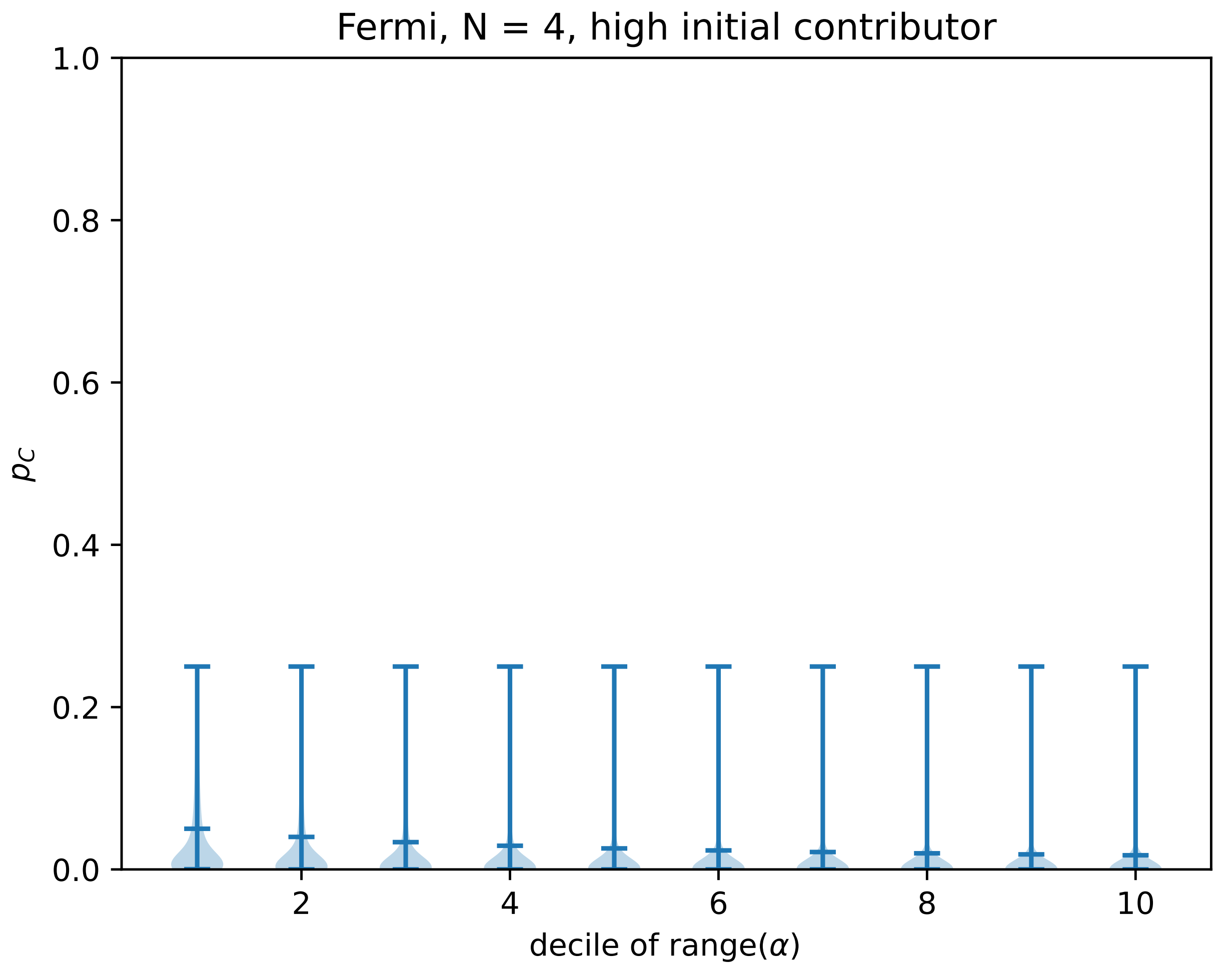


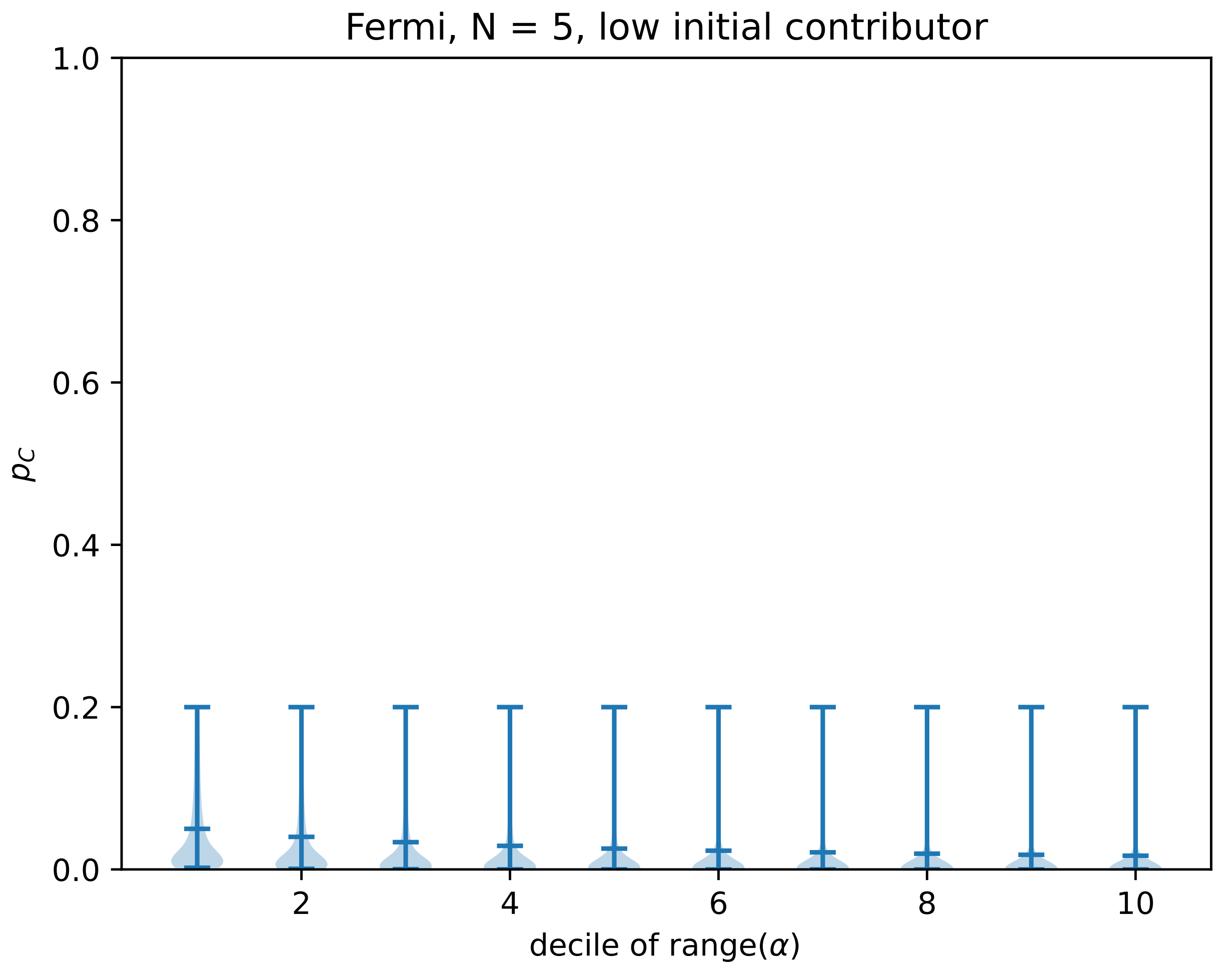
Fermi, N = 3, high initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)



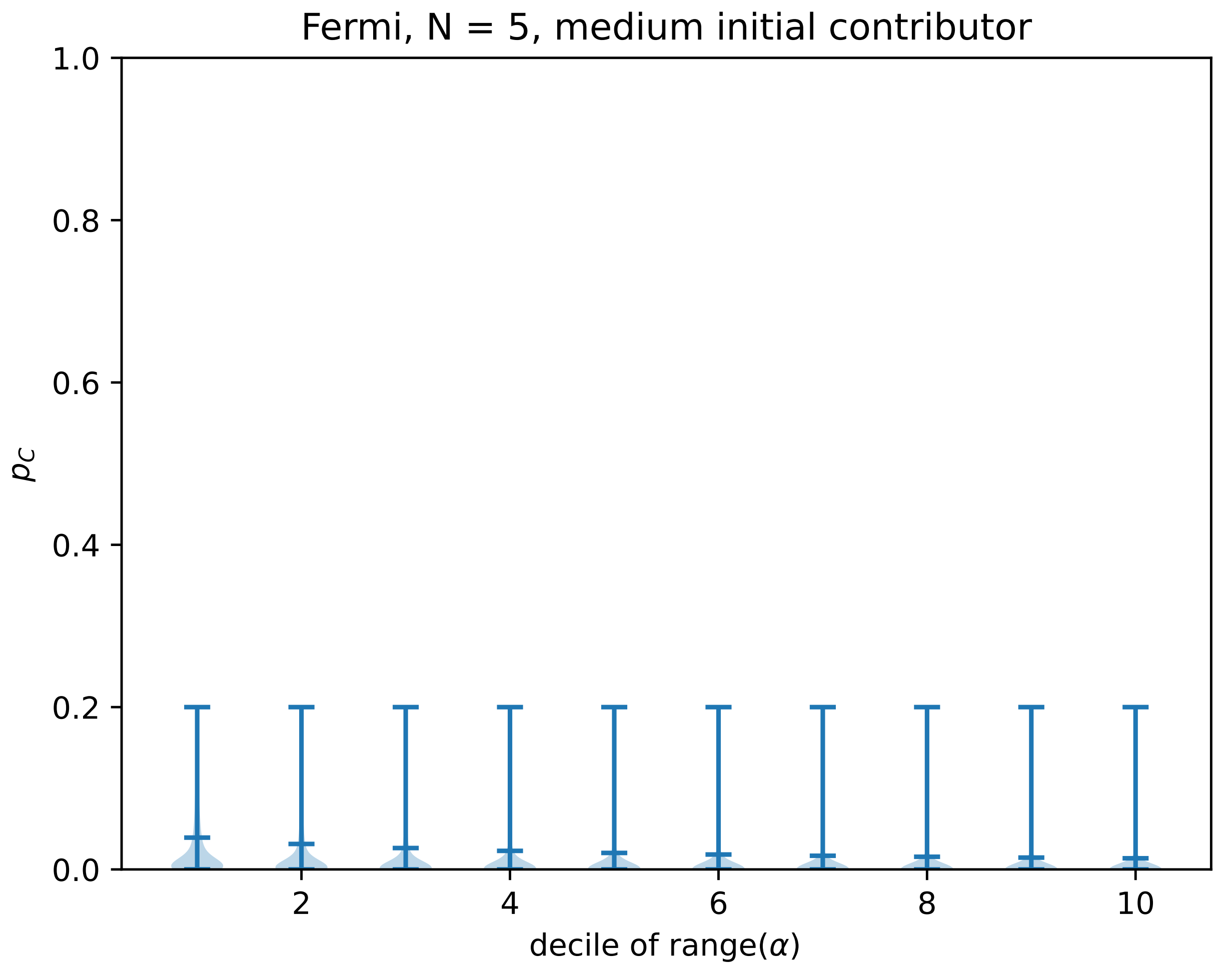
Fermi, N = 4, low initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

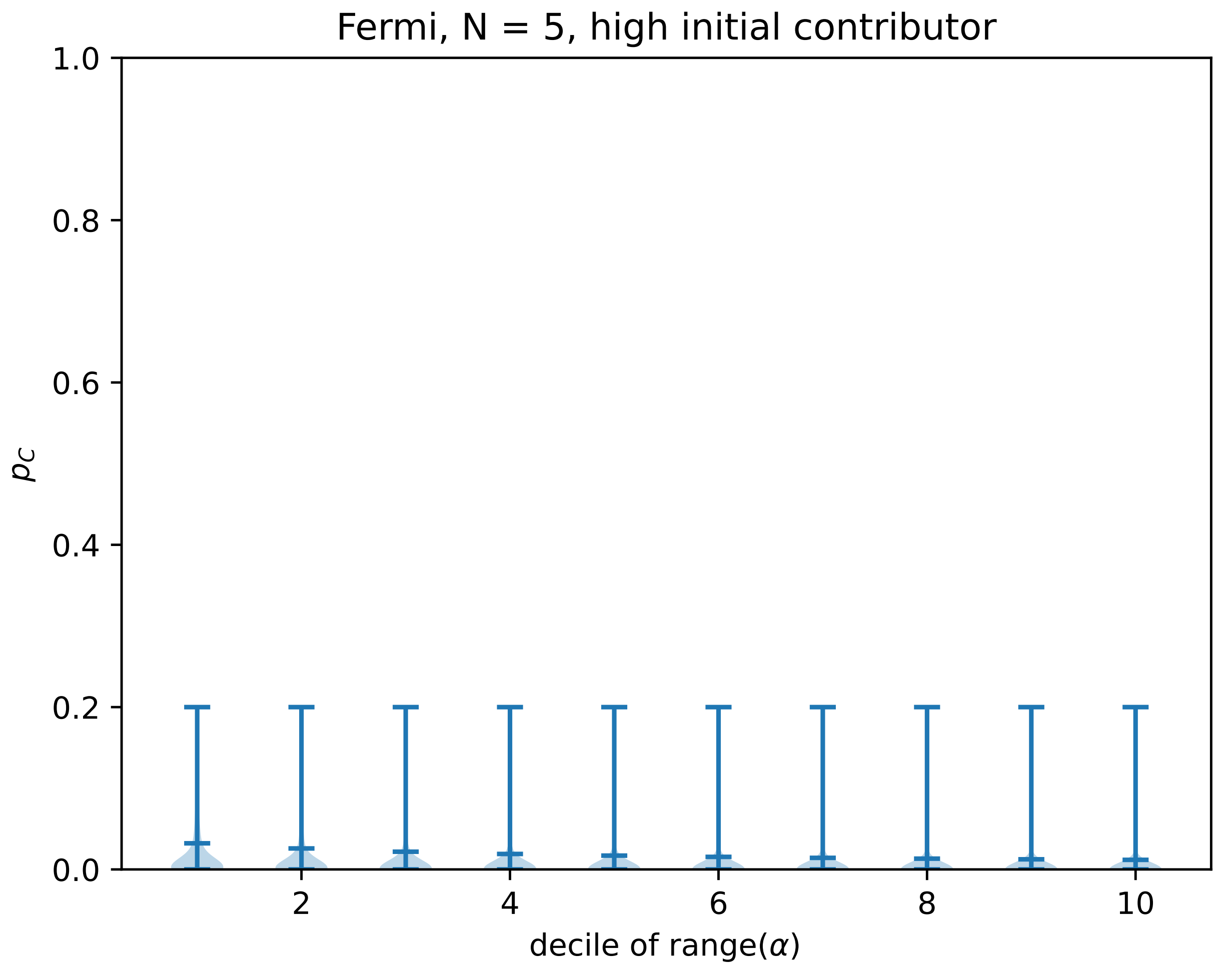

Fermi, N = 4, medium initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

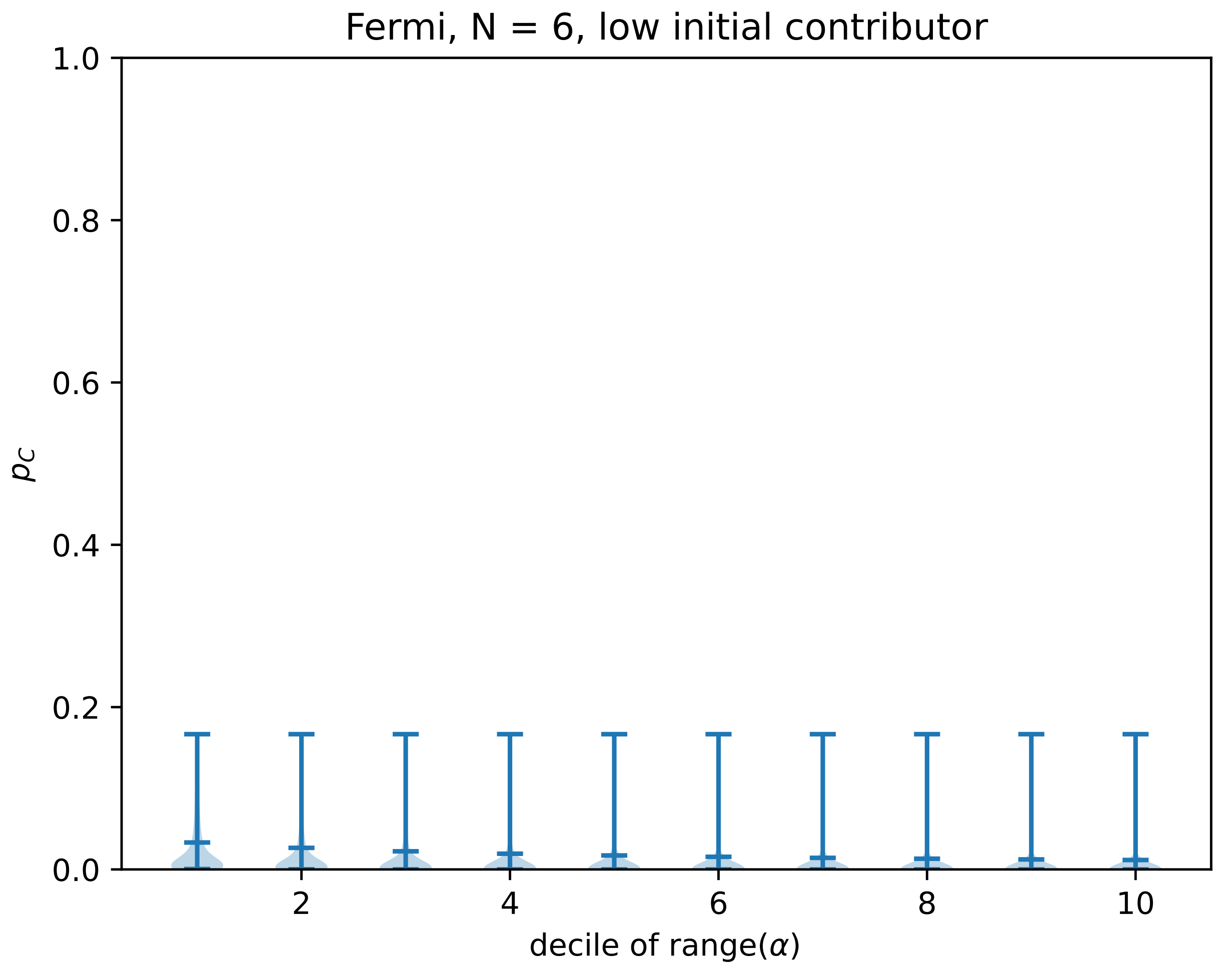

Fermi, N = 4, high initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

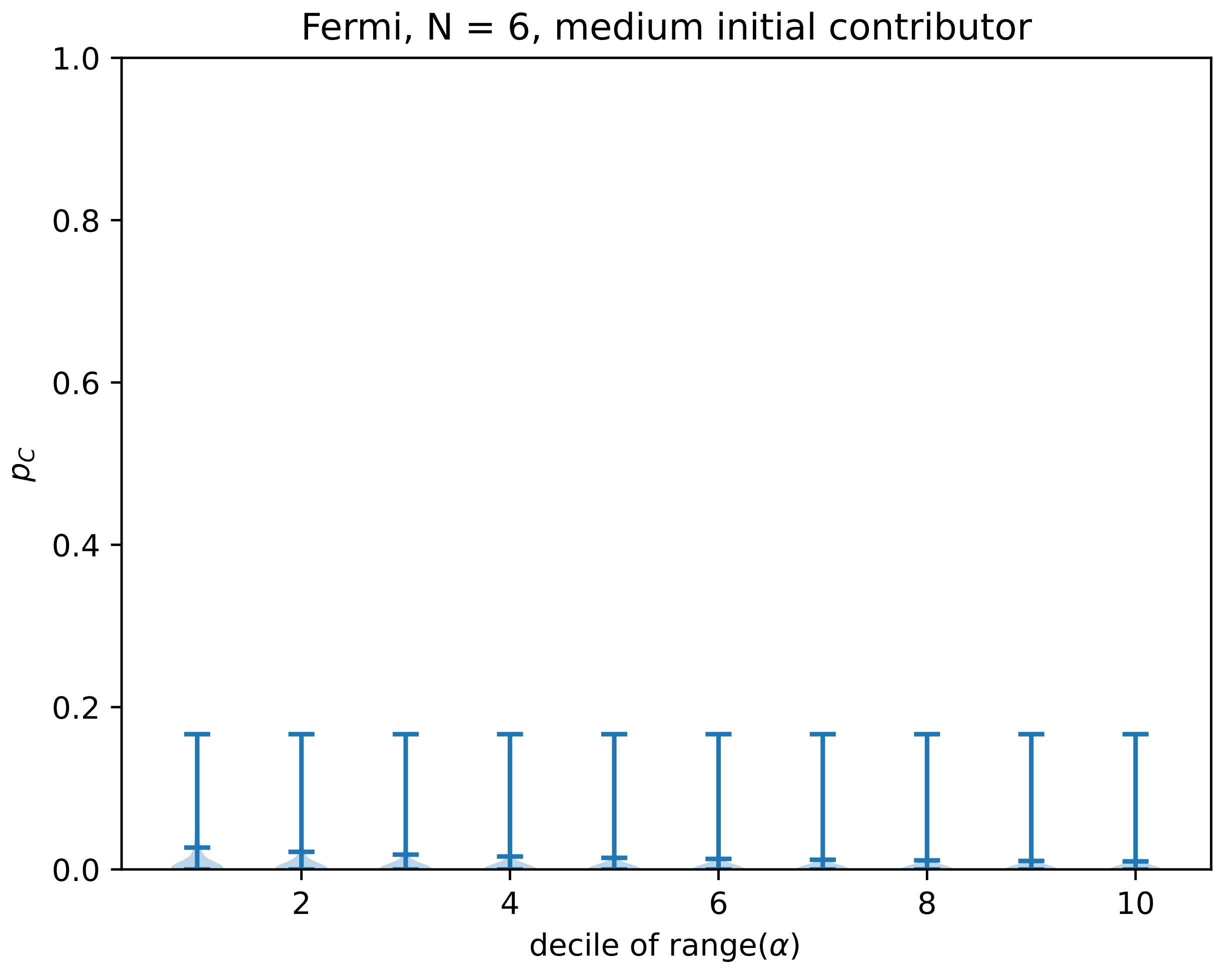


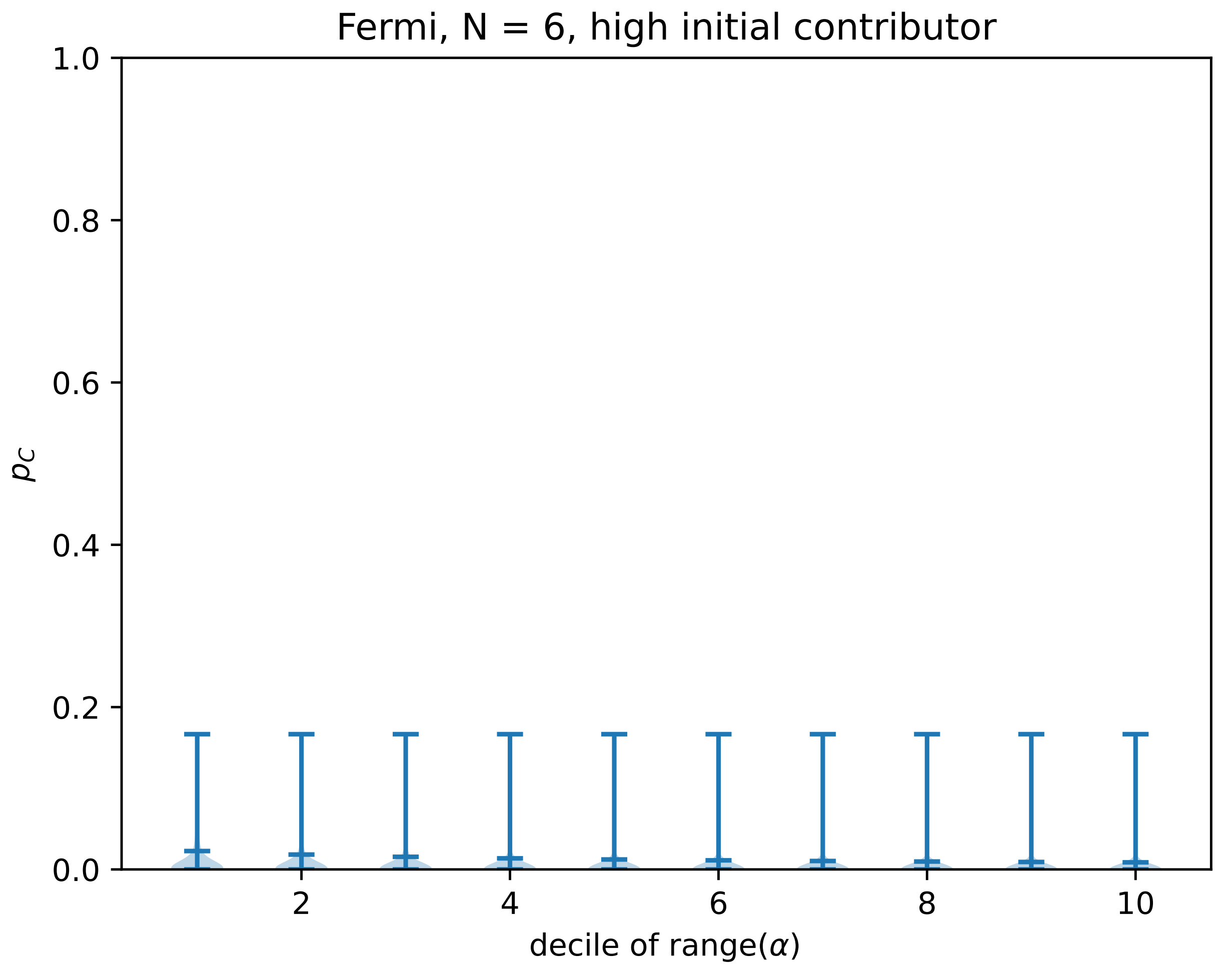
Fermi, N = 5, low initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

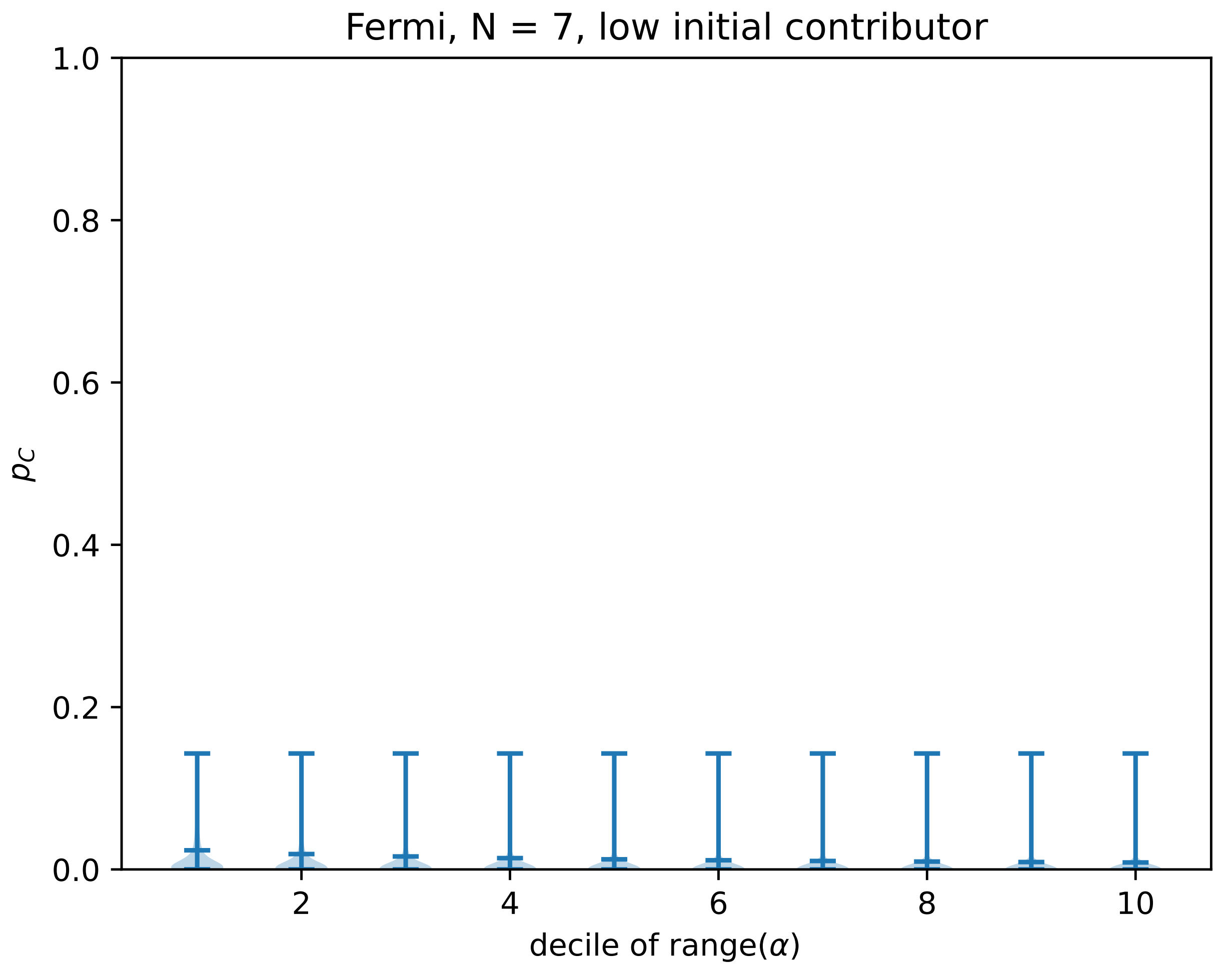

Fermi, N = 5, medium initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

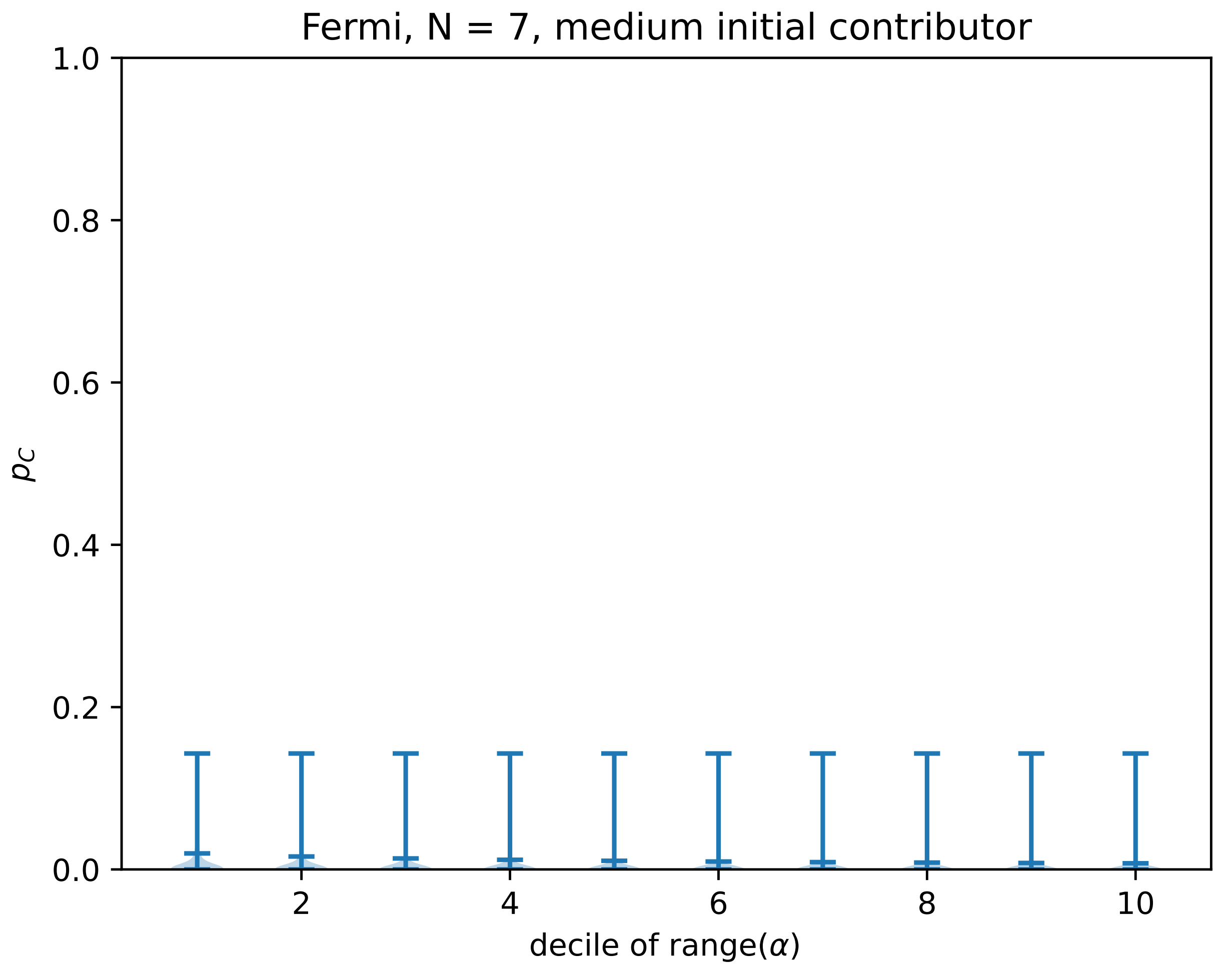

Fermi, N = 5, high initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

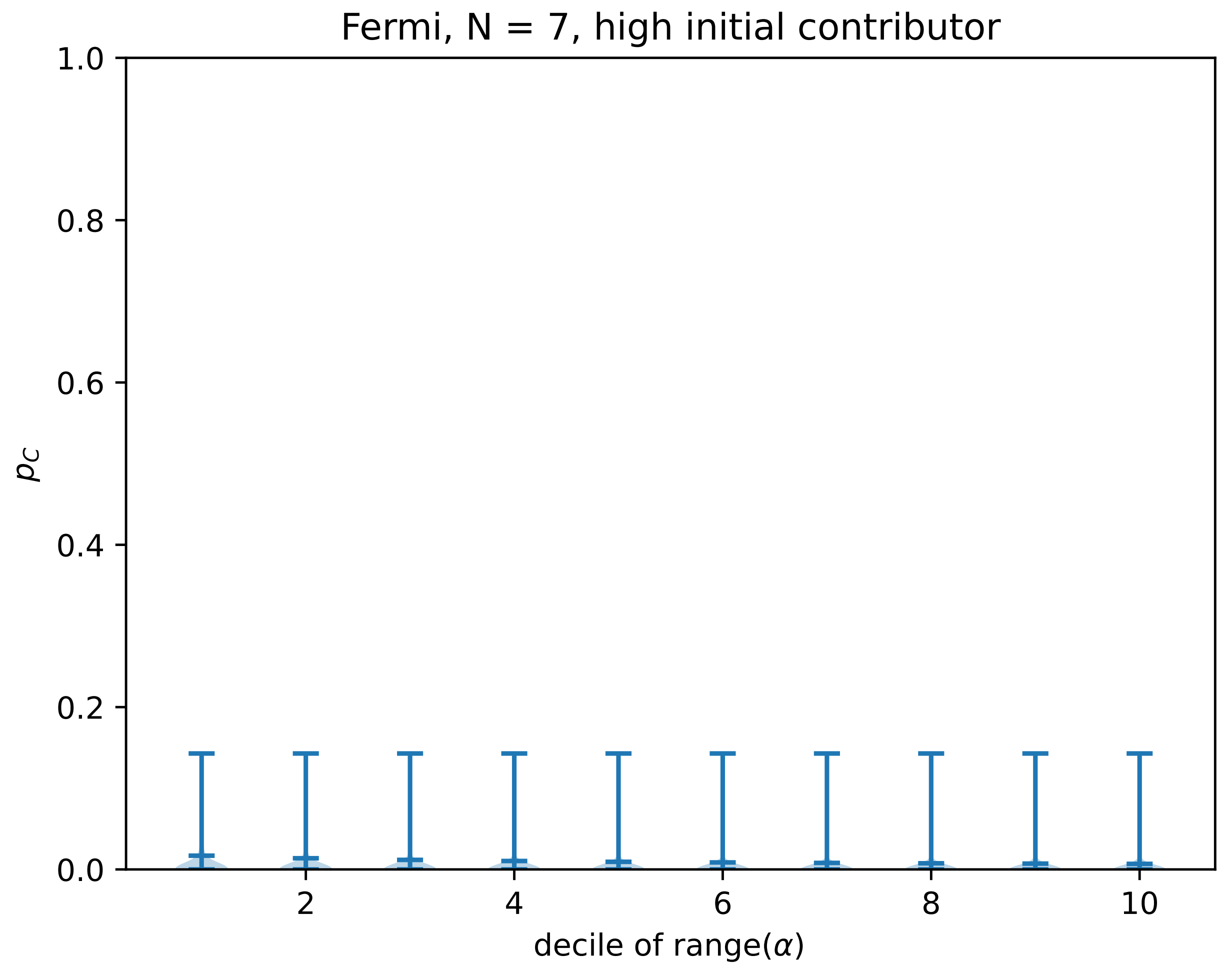

Fermi, N = 6, low initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

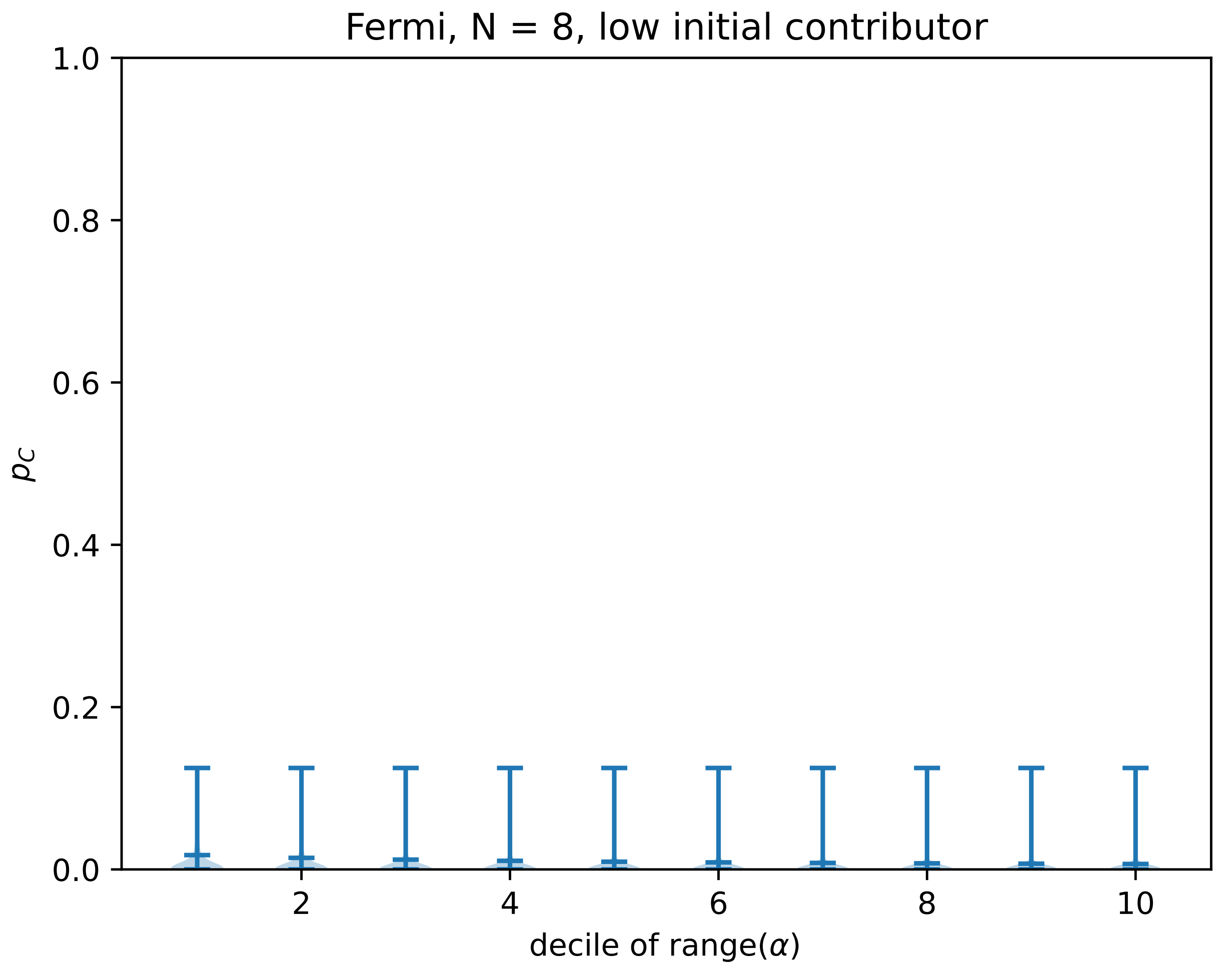

Fermi, N = 6, medium initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

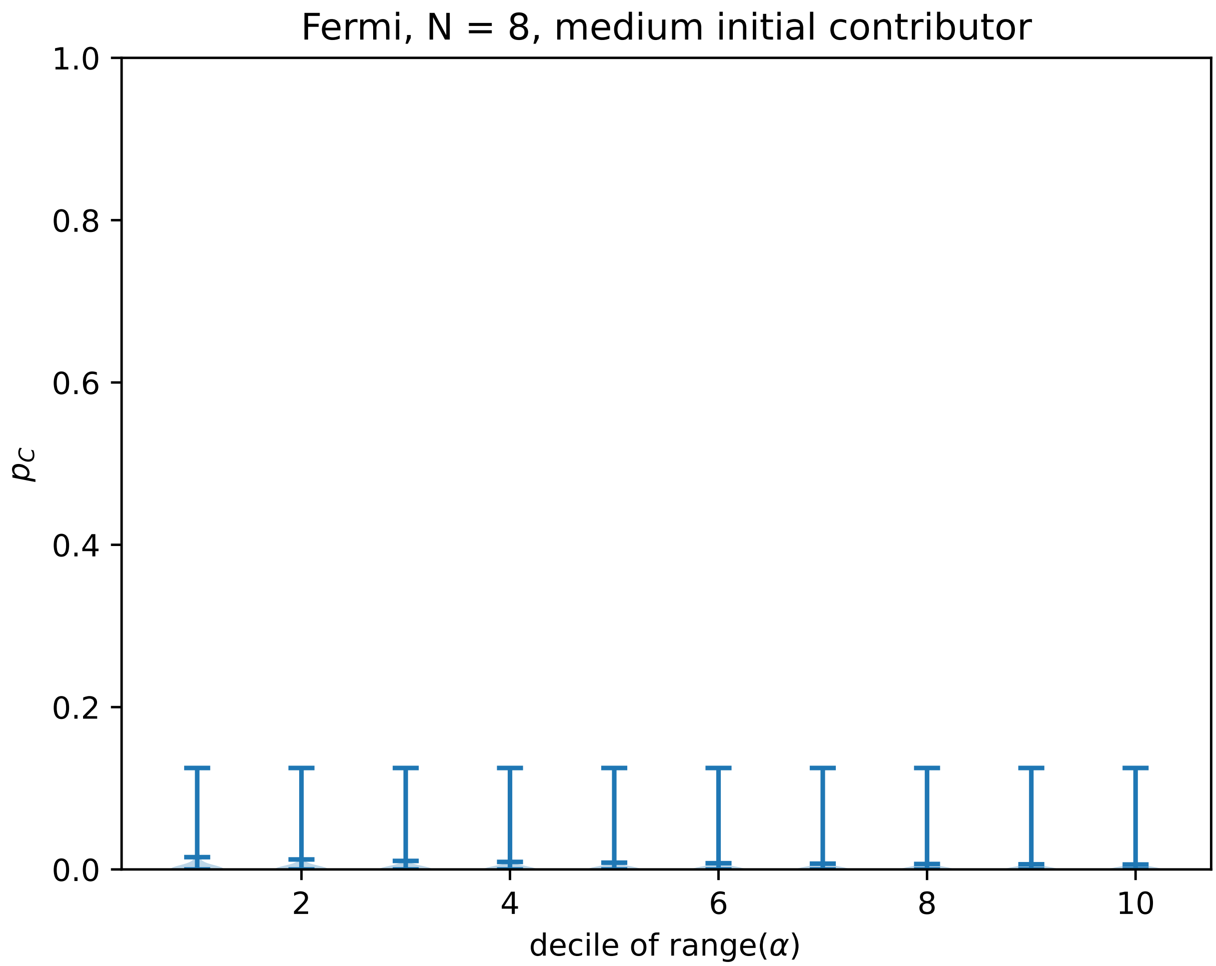

Fermi, N = 6, high initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)


Fermi, N = 7, low initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

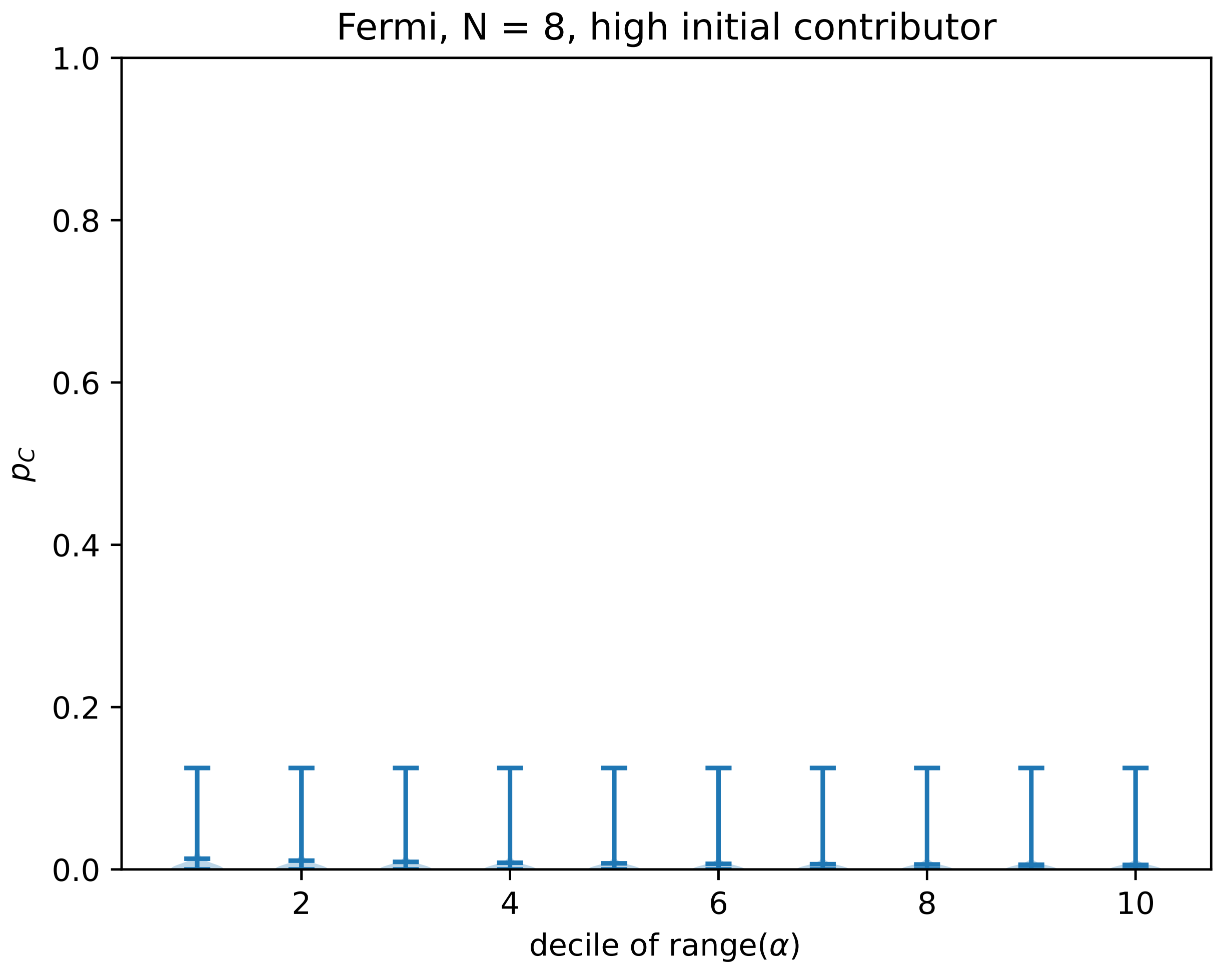

Fermi, N = 7, medium initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)


Fermi, N = 7, high initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

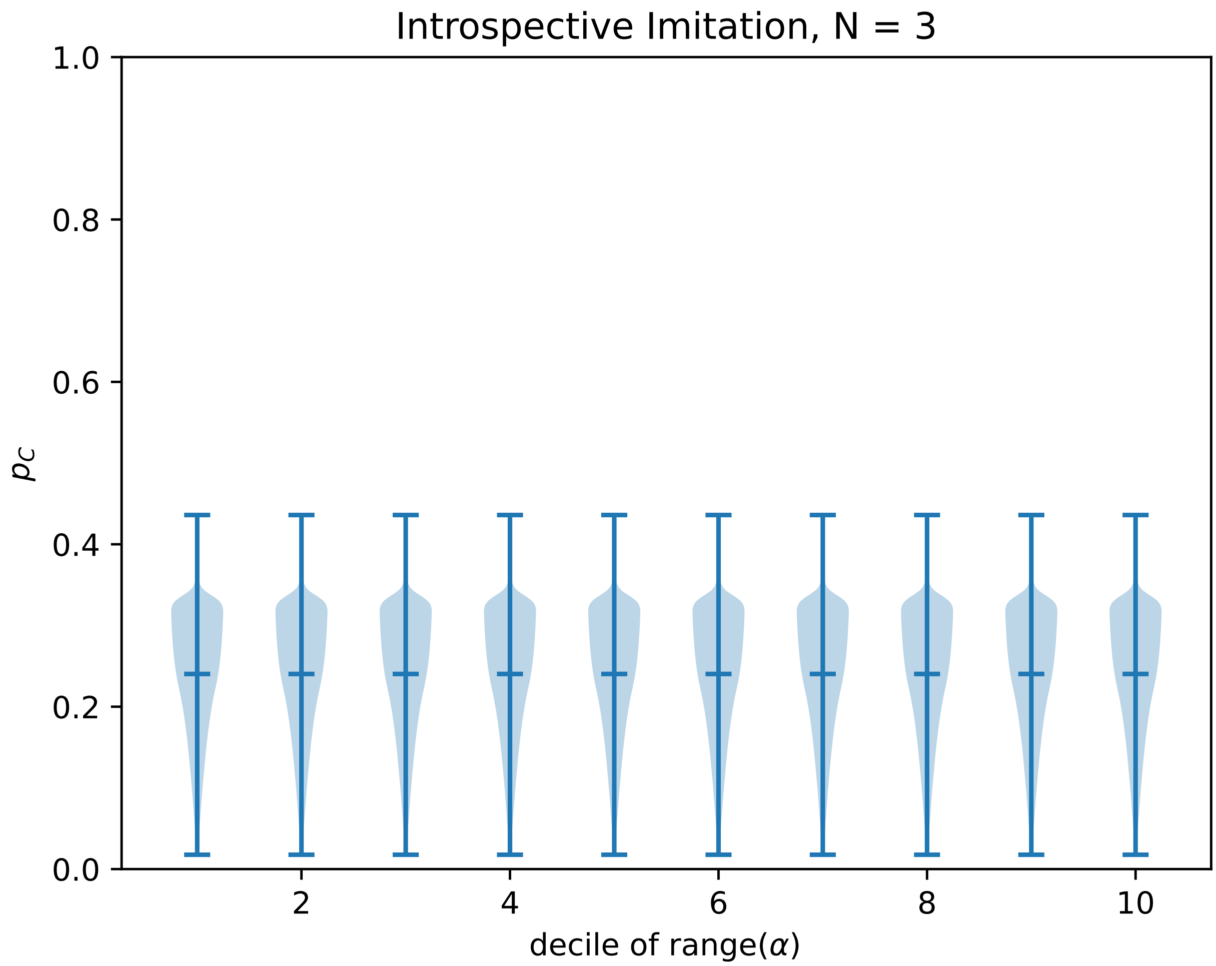

Fermi, N = 8, low initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

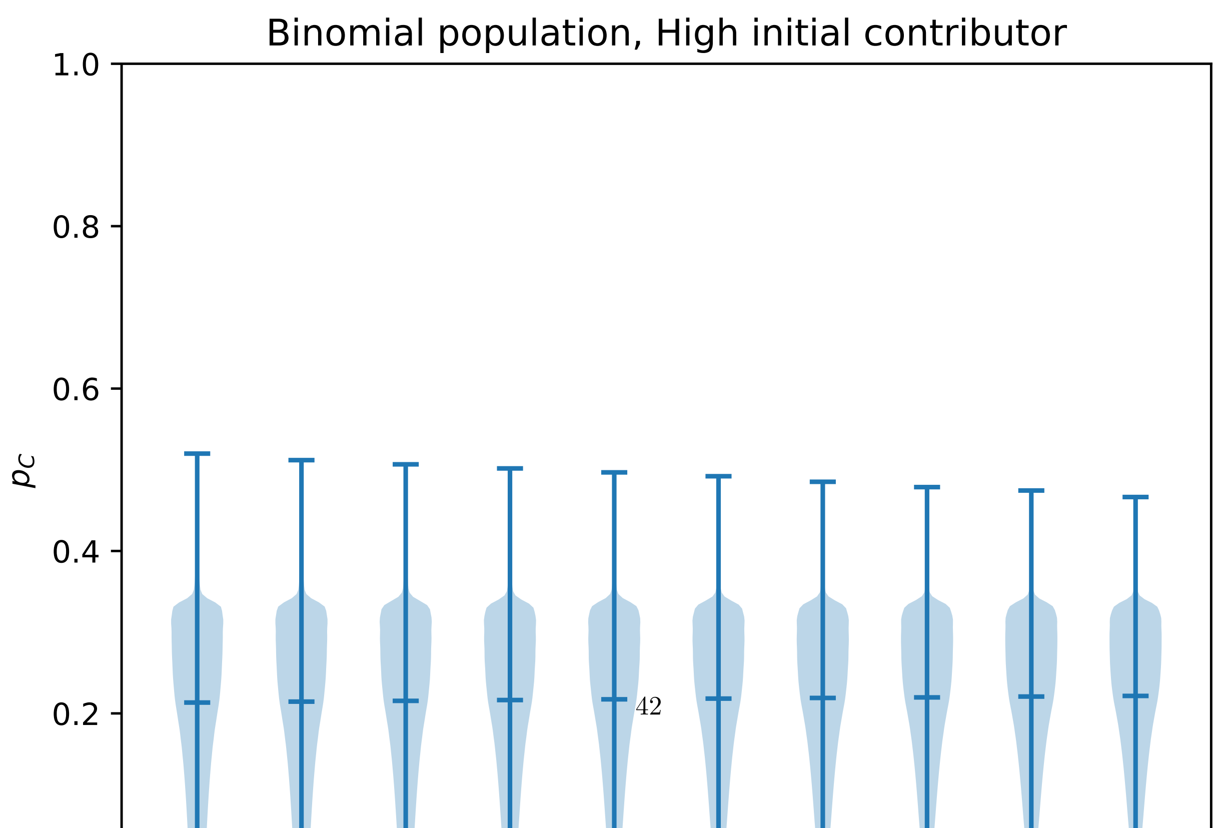


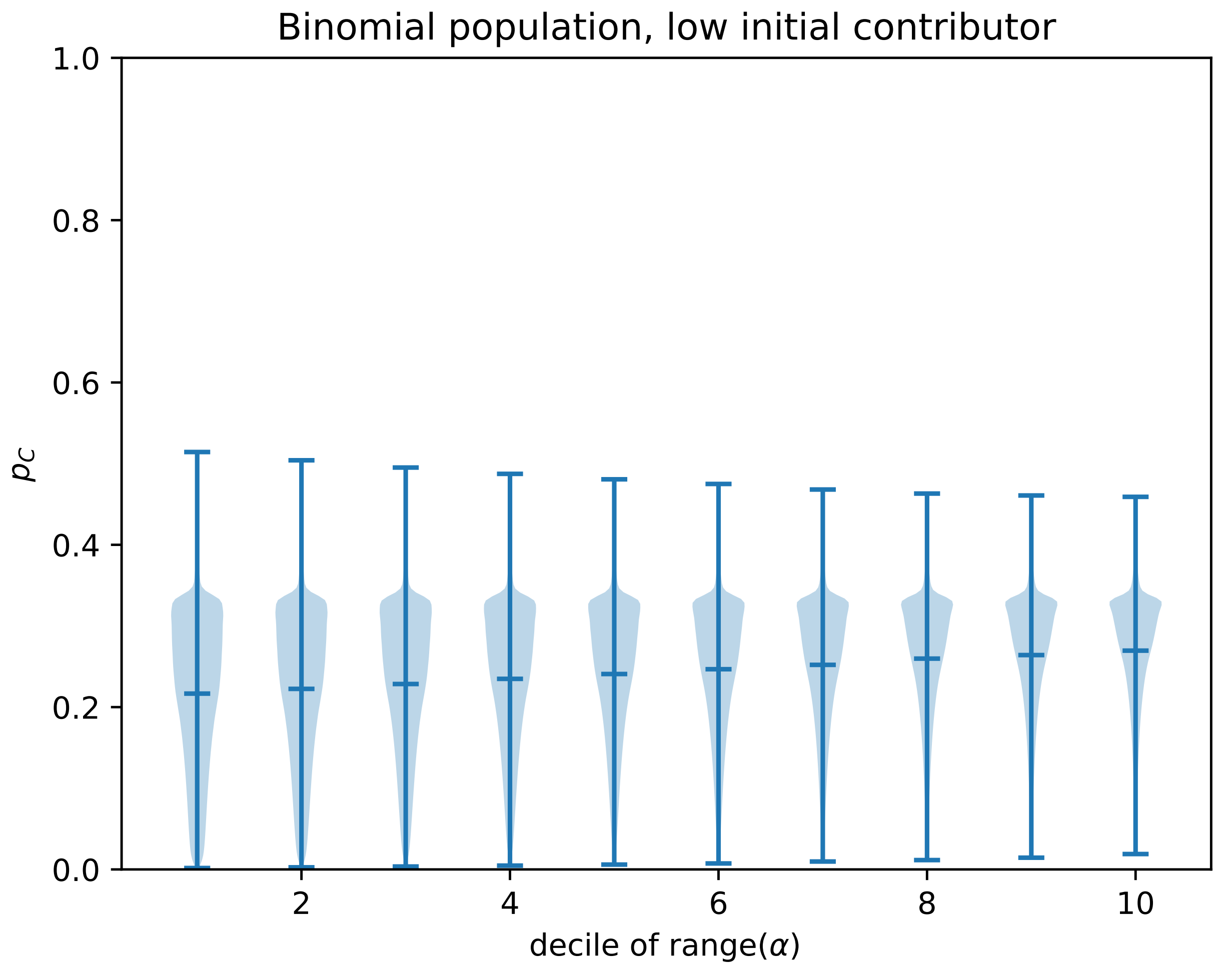
Fermi, N = 8, medium initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

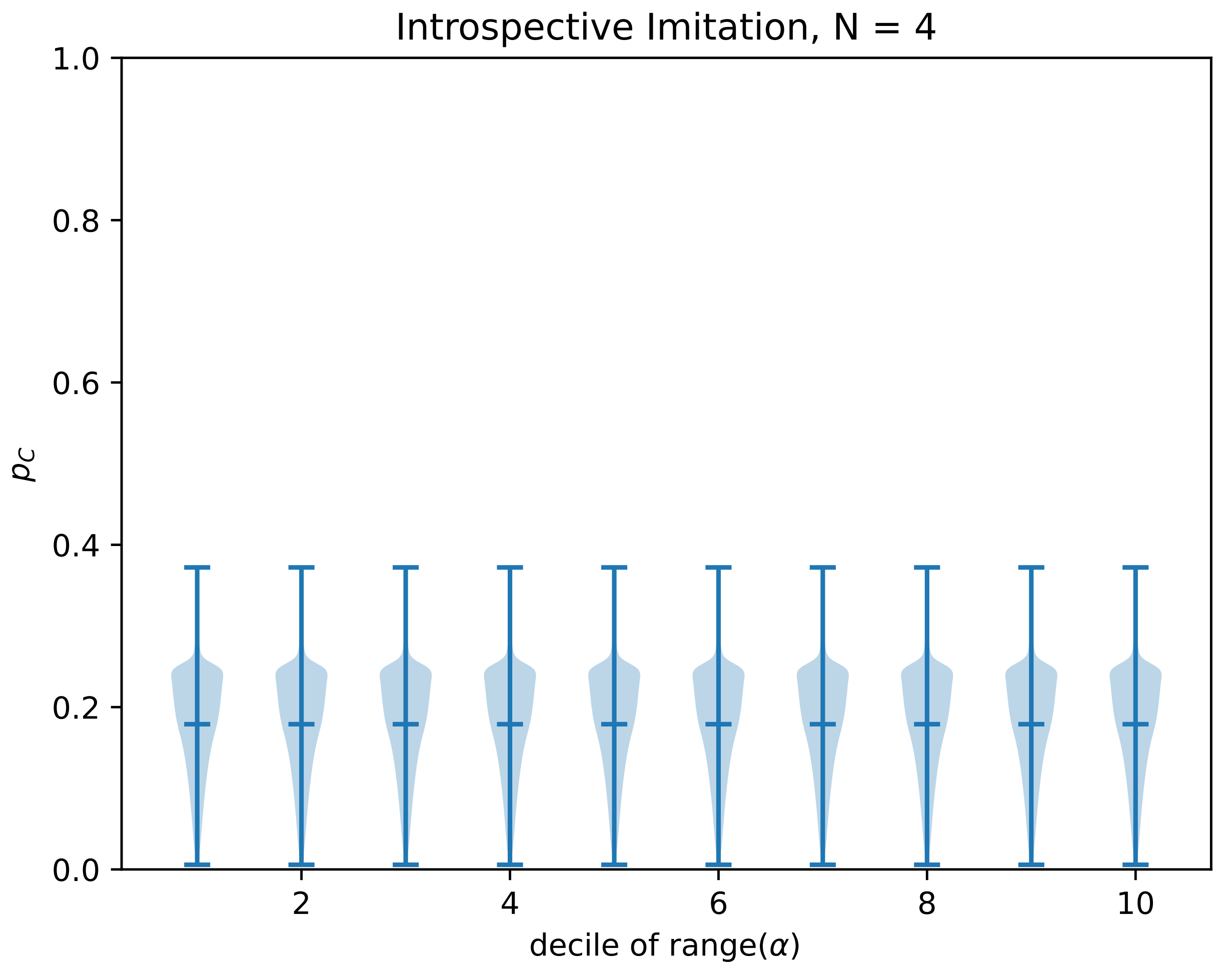


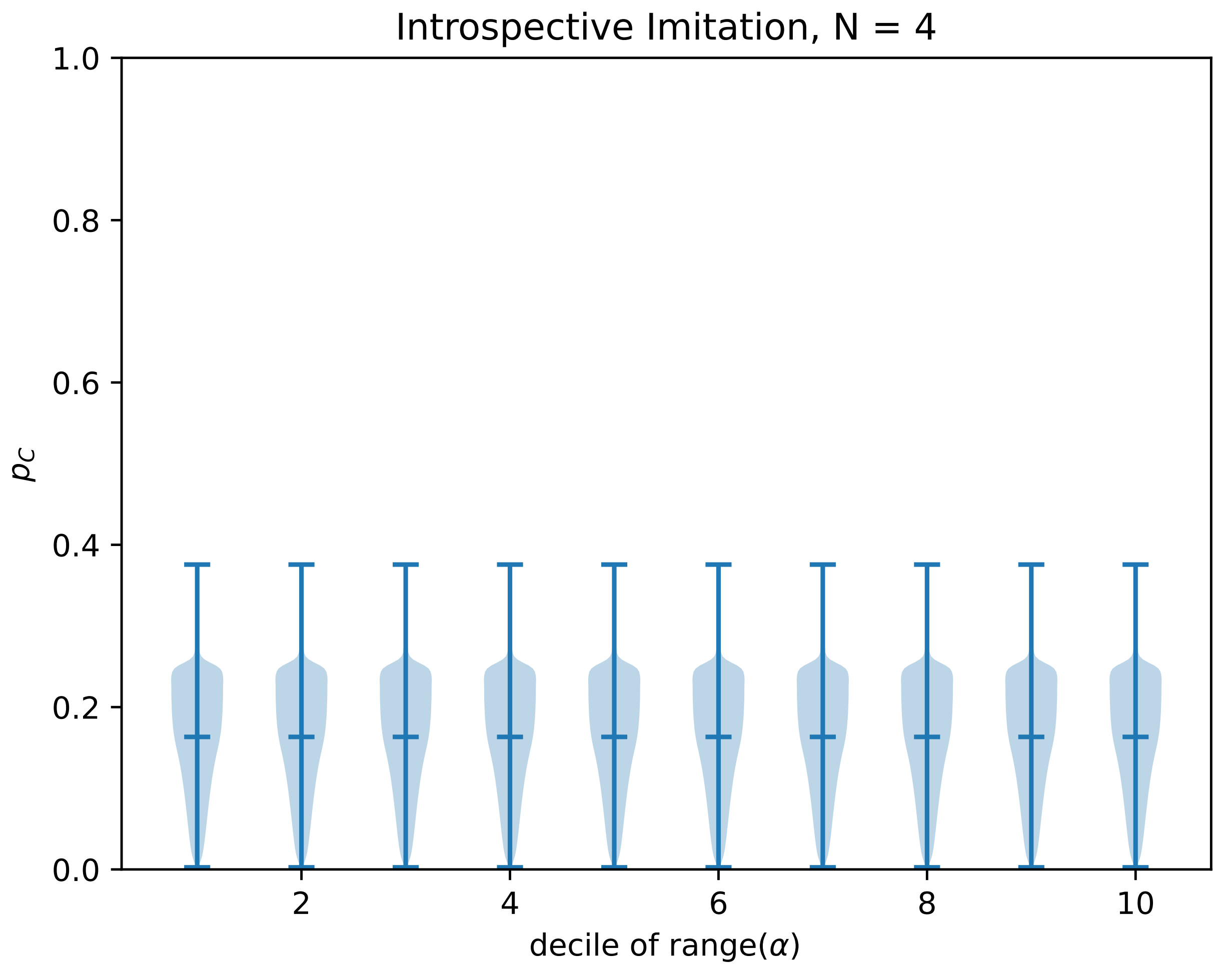
Fermi, N = 8, high initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

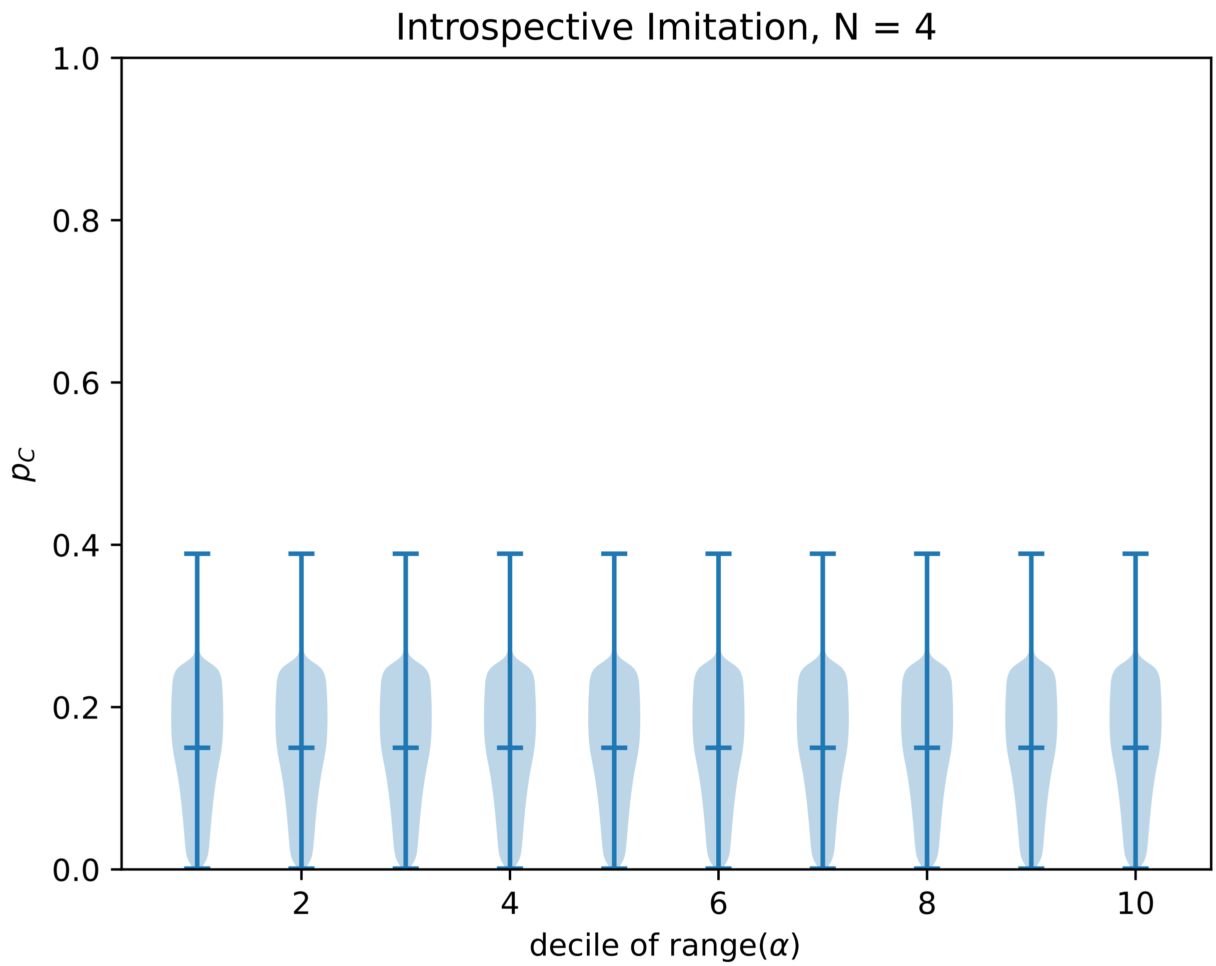

Introspective Imitation, N = 3
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)


Binomial population, High initial contributor
1.0
0.8
0.6
0.4
0.2
p_C

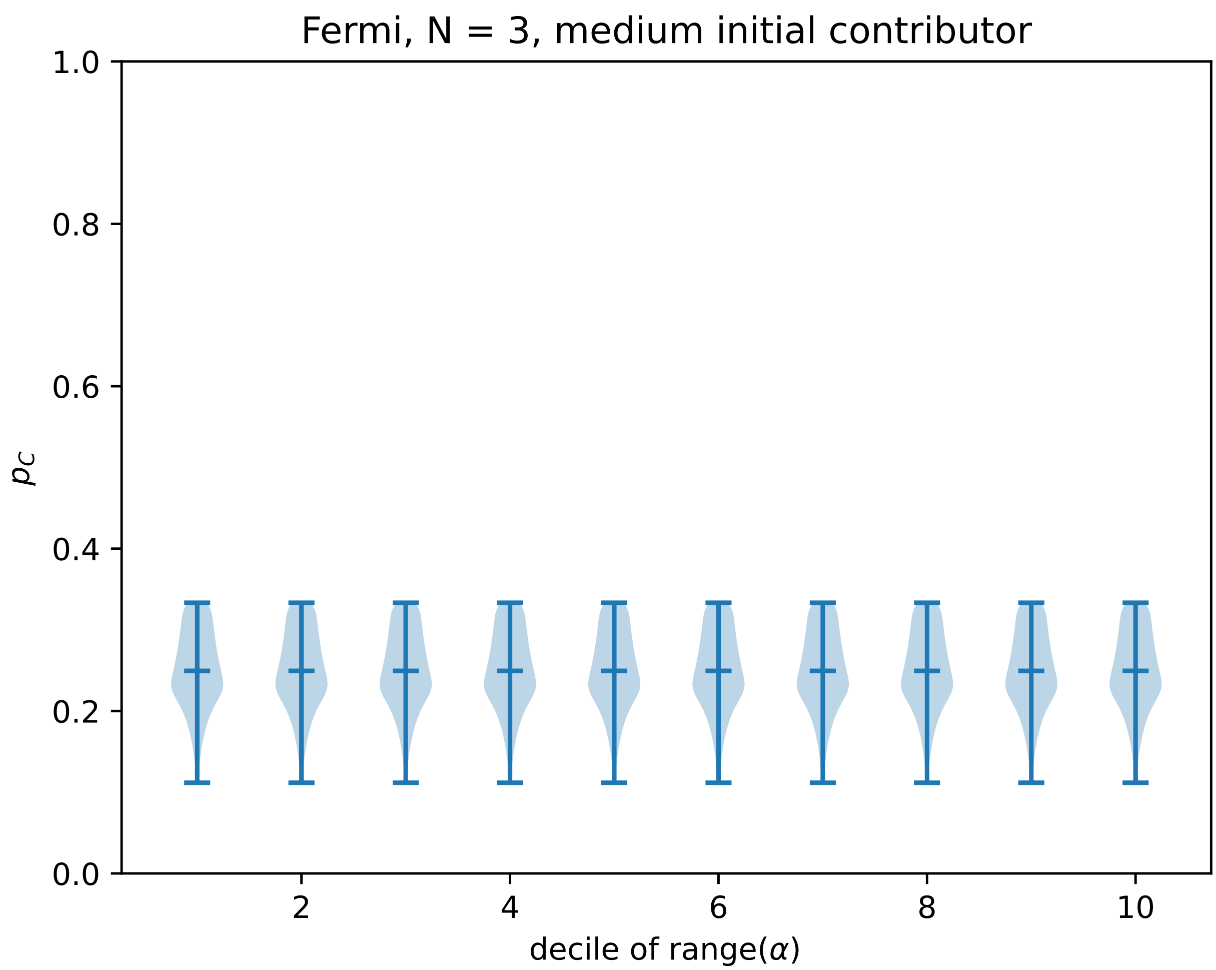

Binomial population, low initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

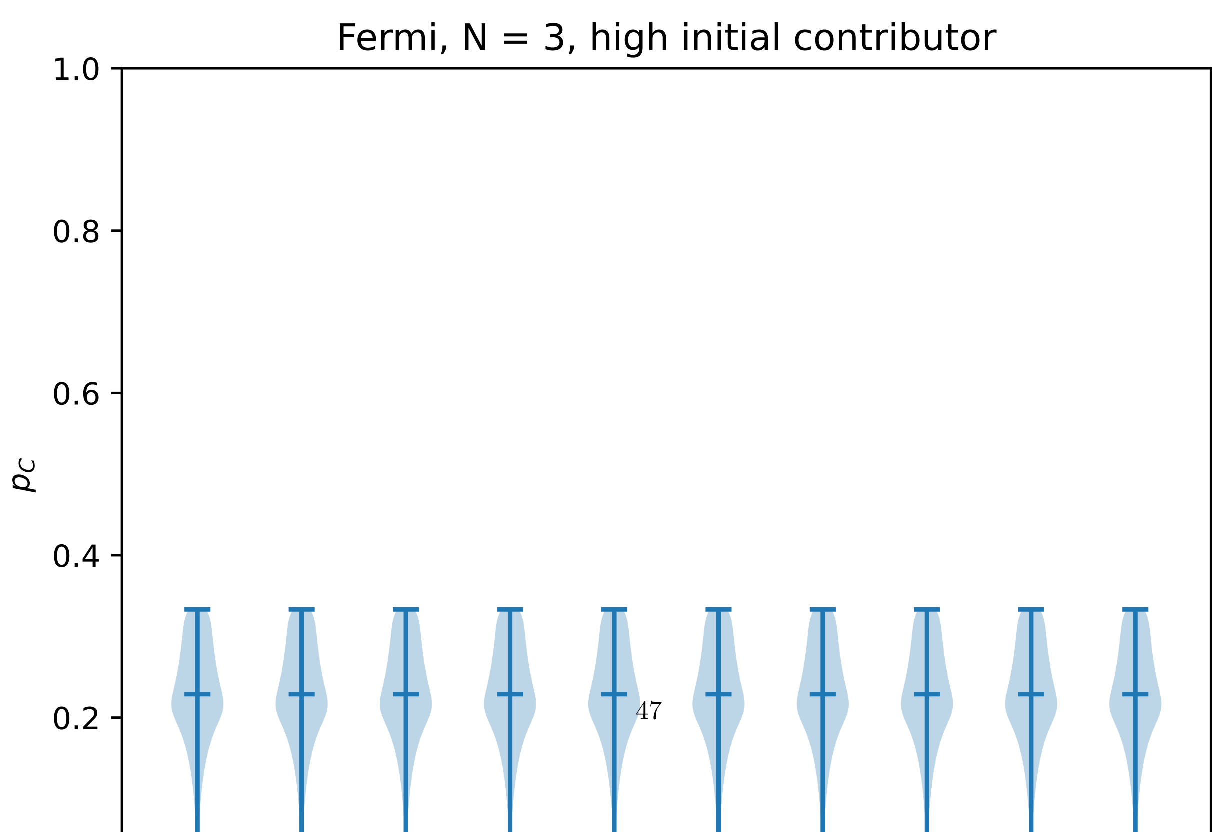

Introspective Imitation, N = 4
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

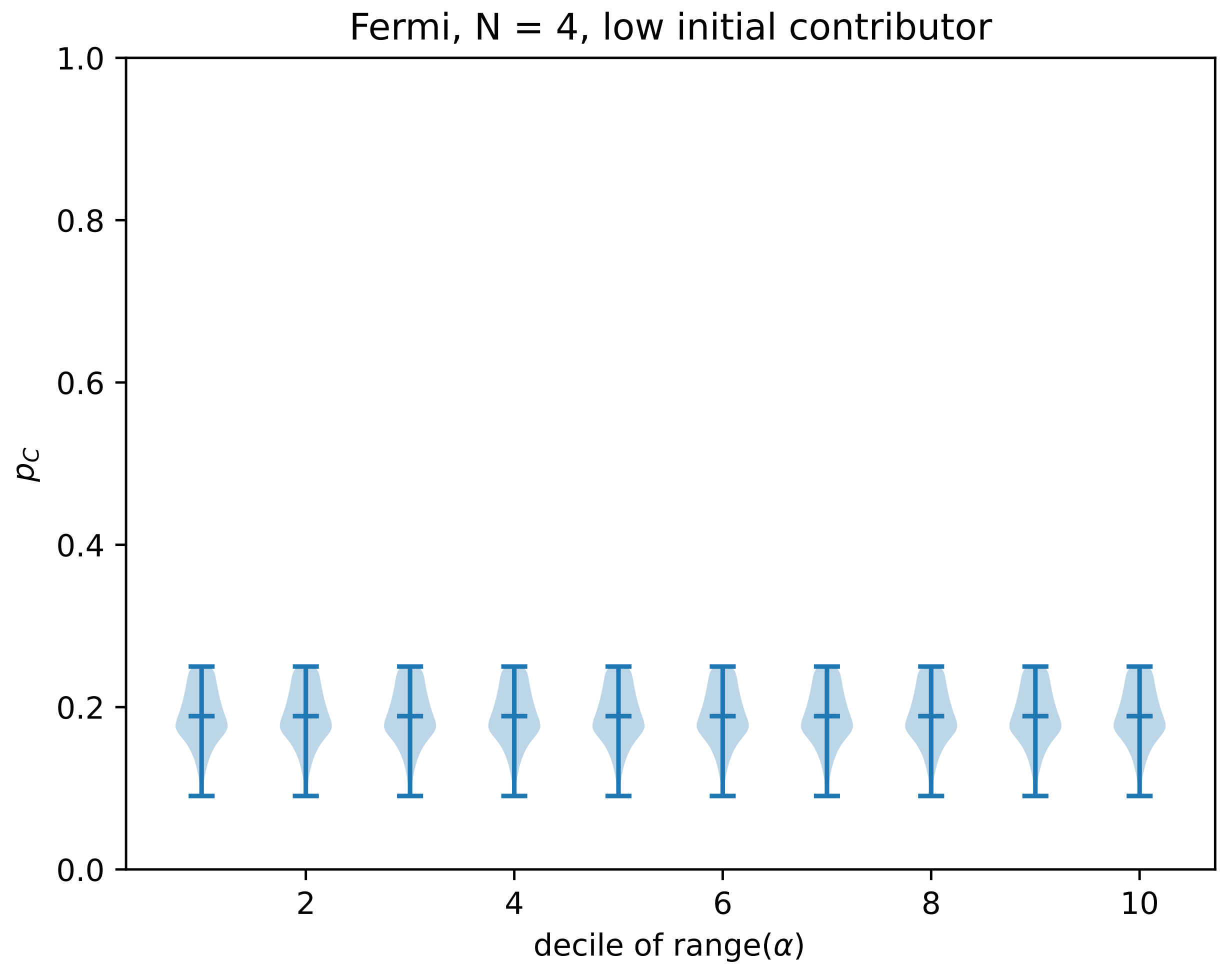

Introspective Imitation, N = 4
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

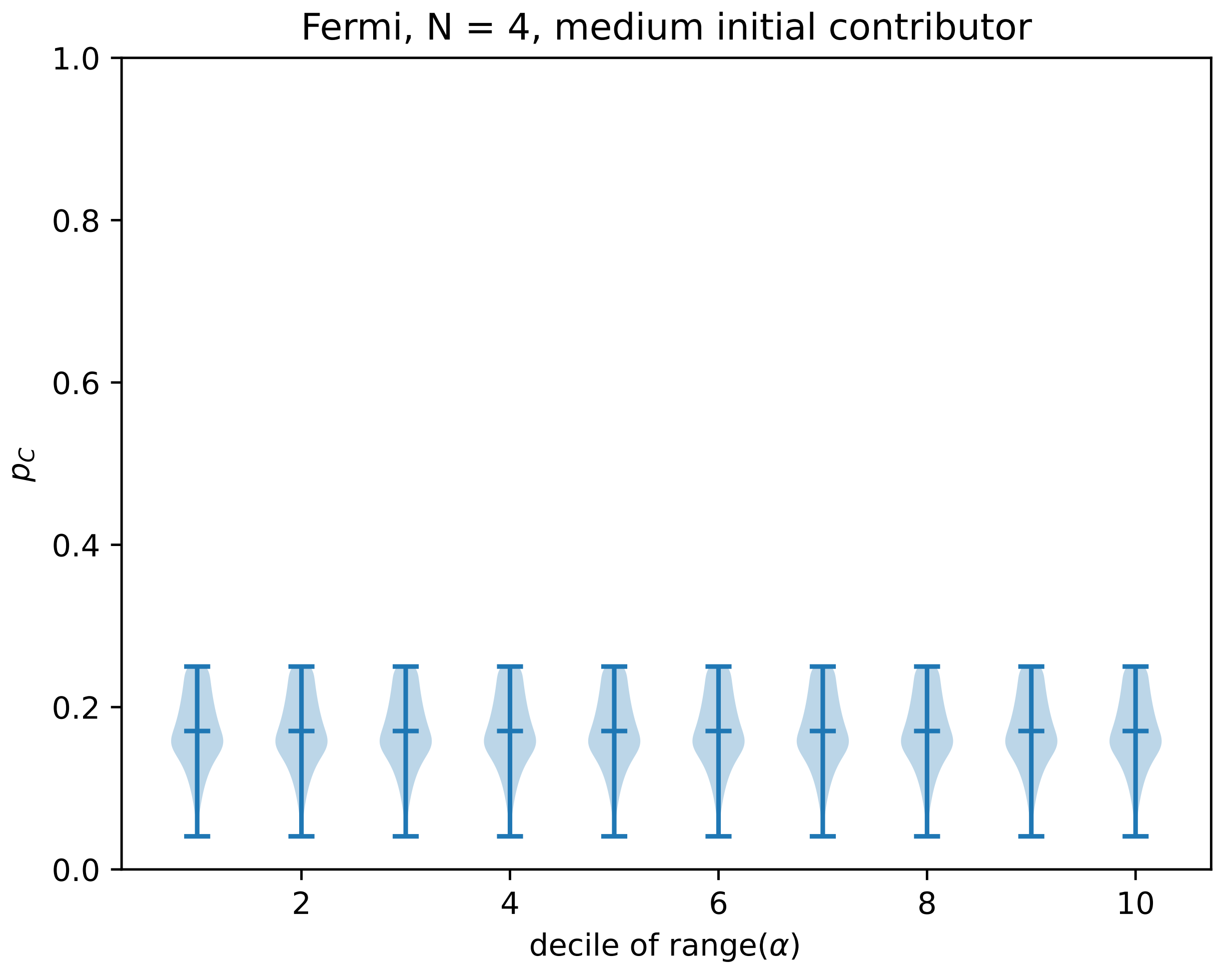

Introspective Imitation, N = 4
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

Moran process: linear population, $N = 3$–8

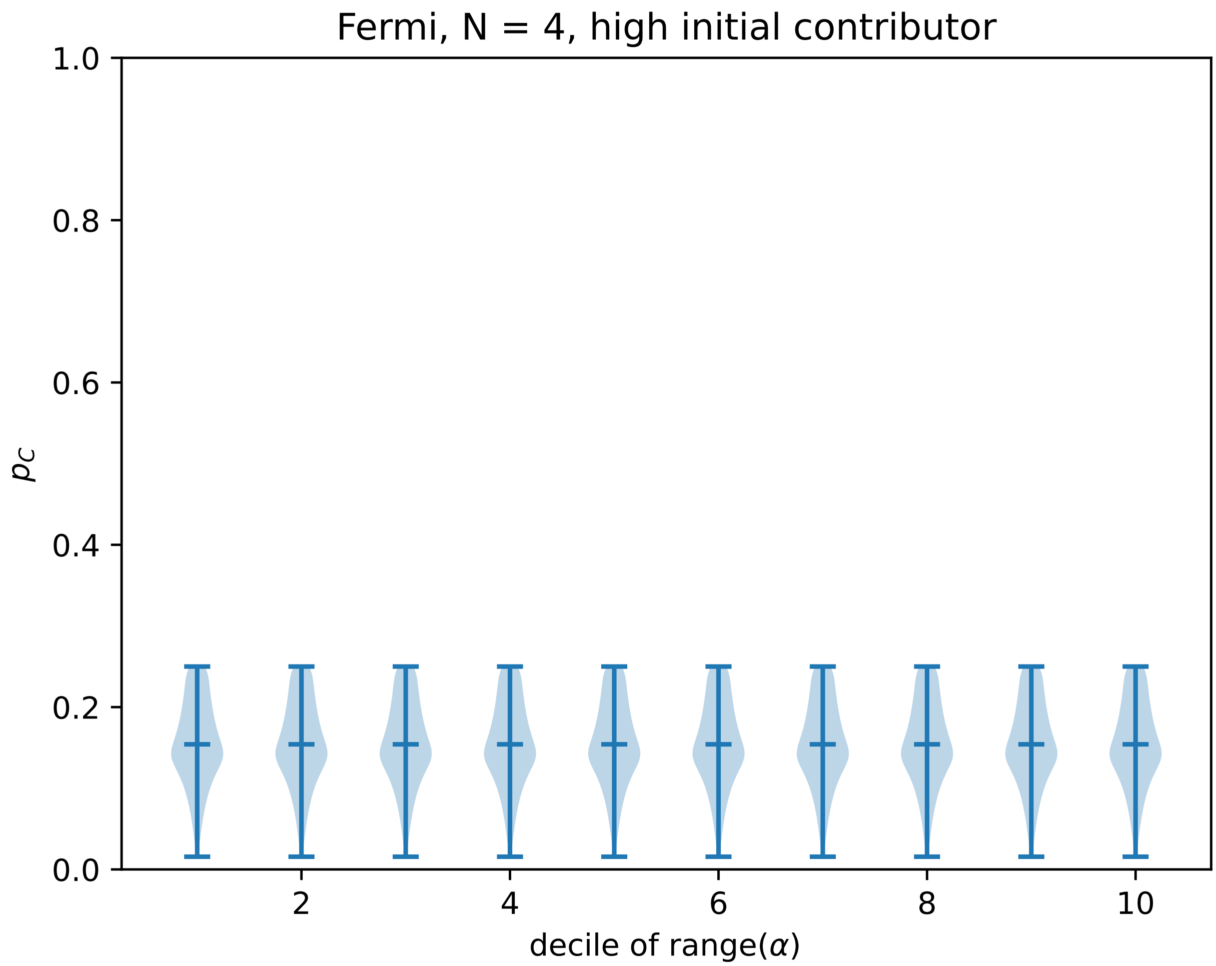




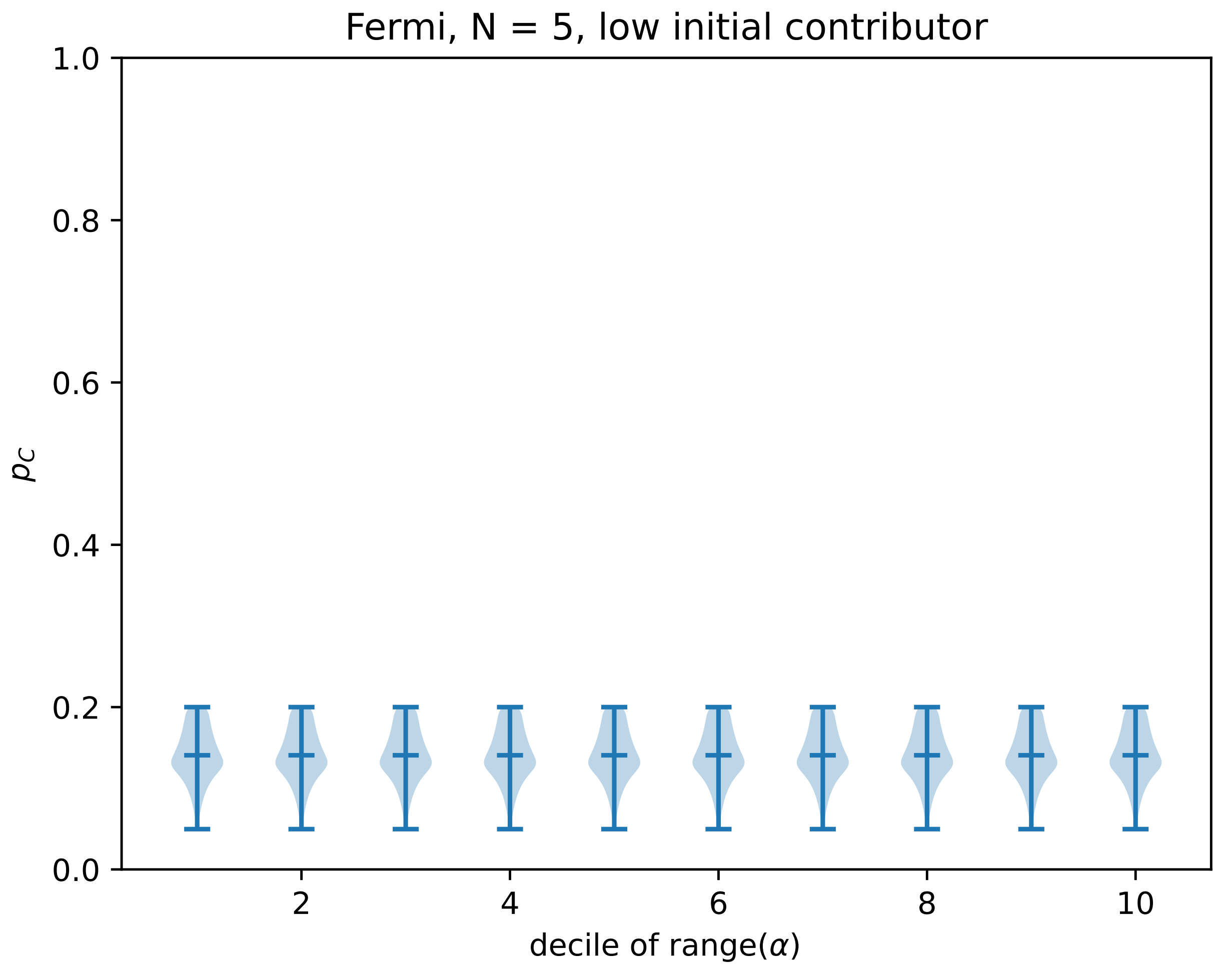

Fermi, N = 4, low initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)



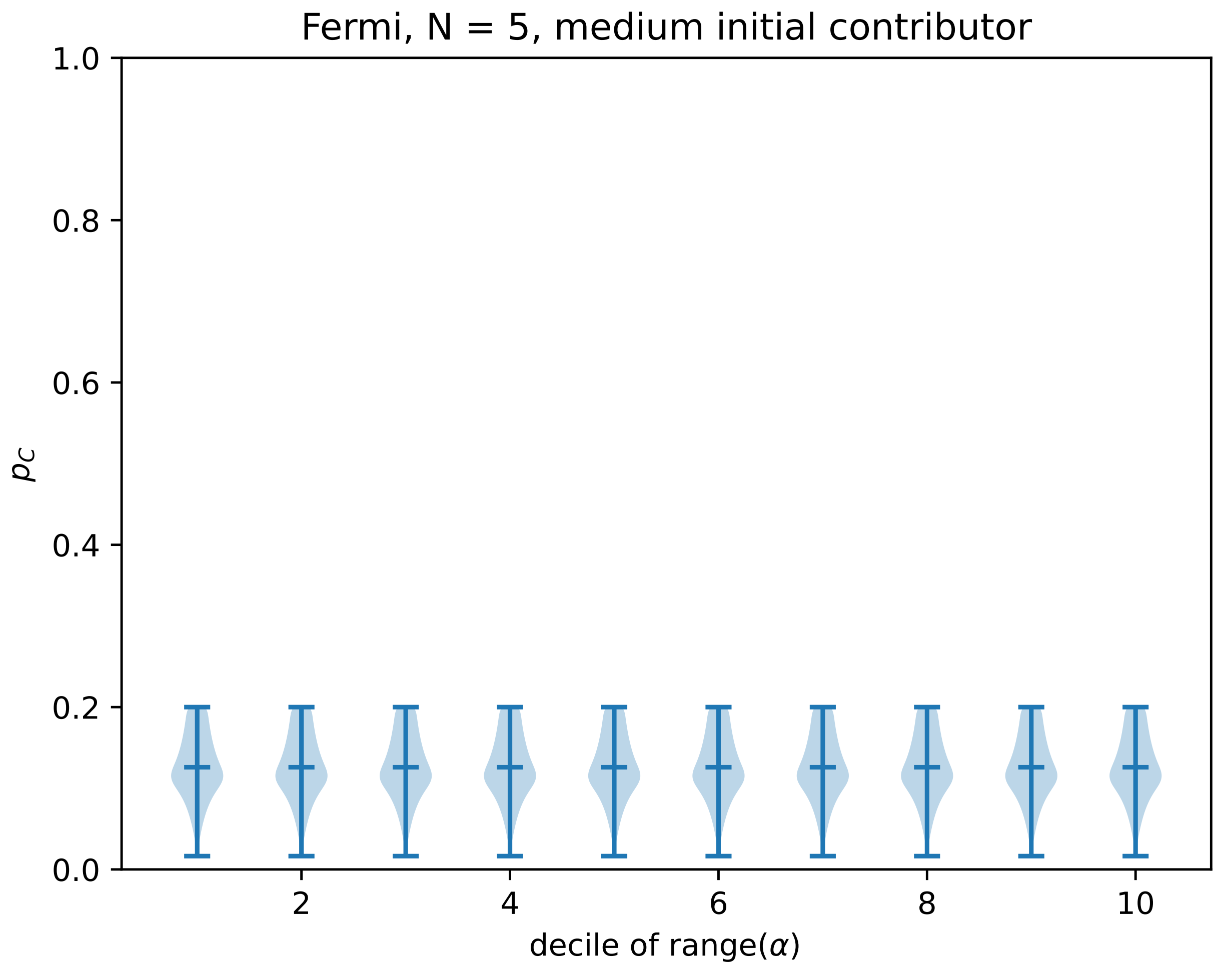
Fermi, N = 4, medium initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

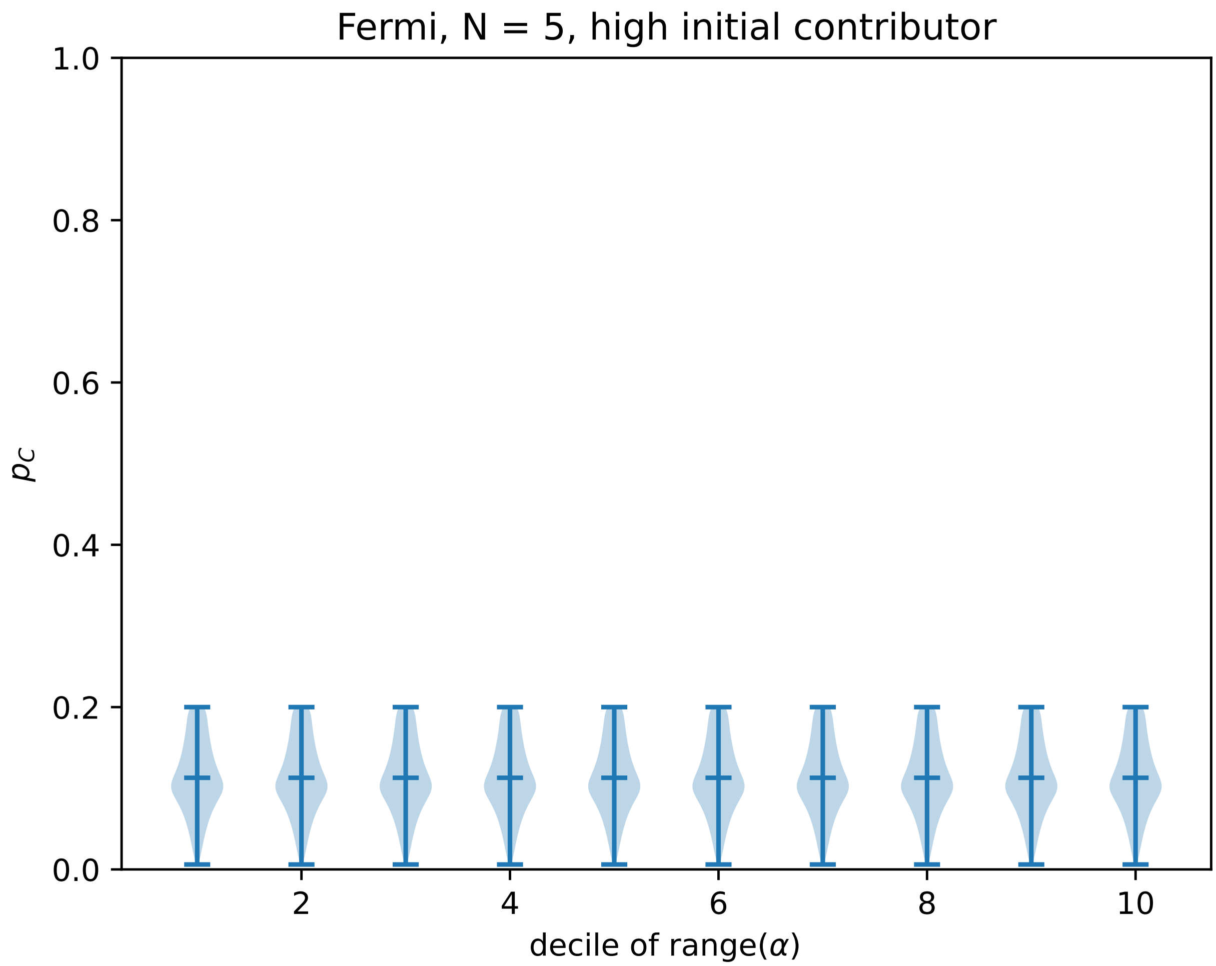


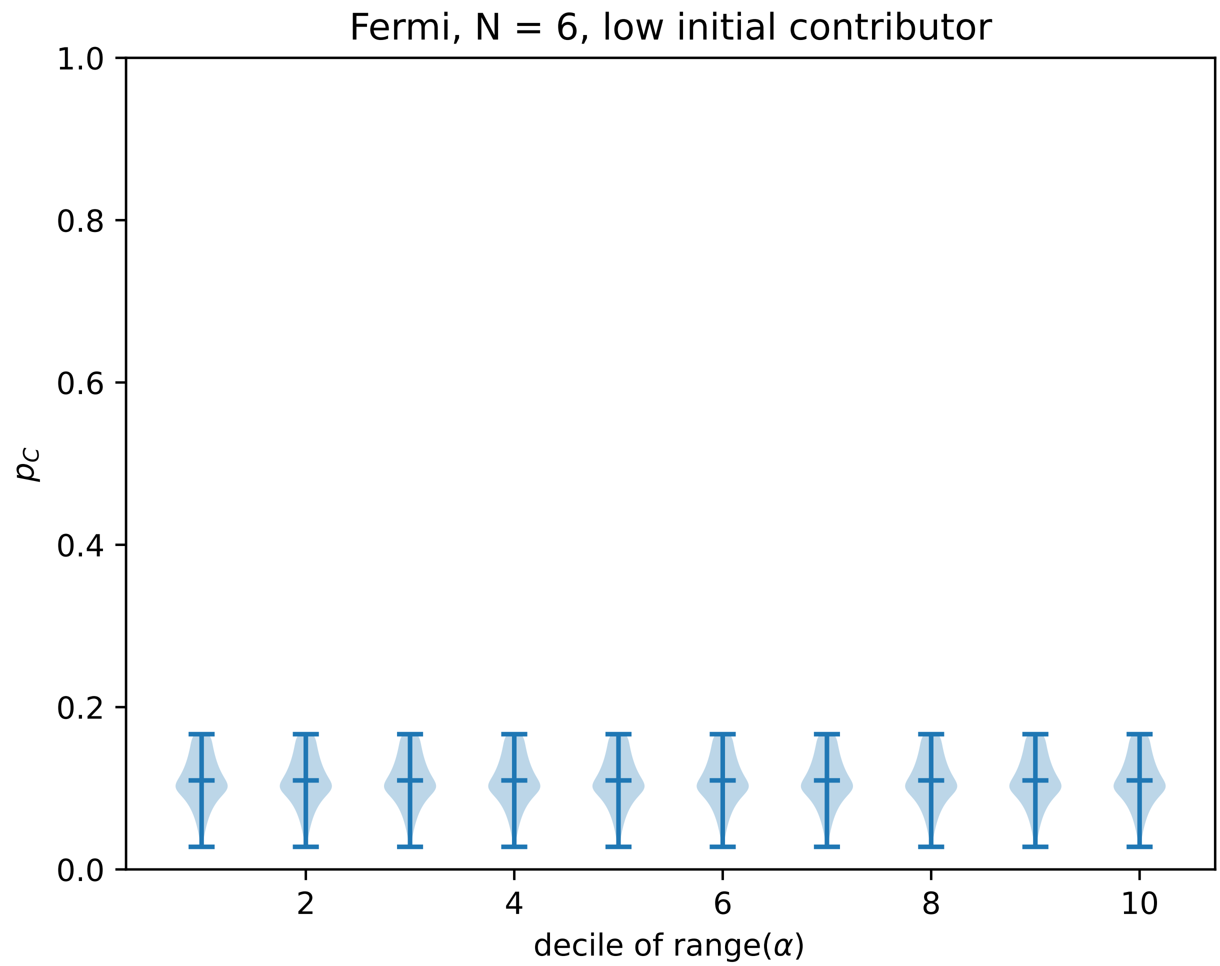
Fermi, N = 4, high initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)



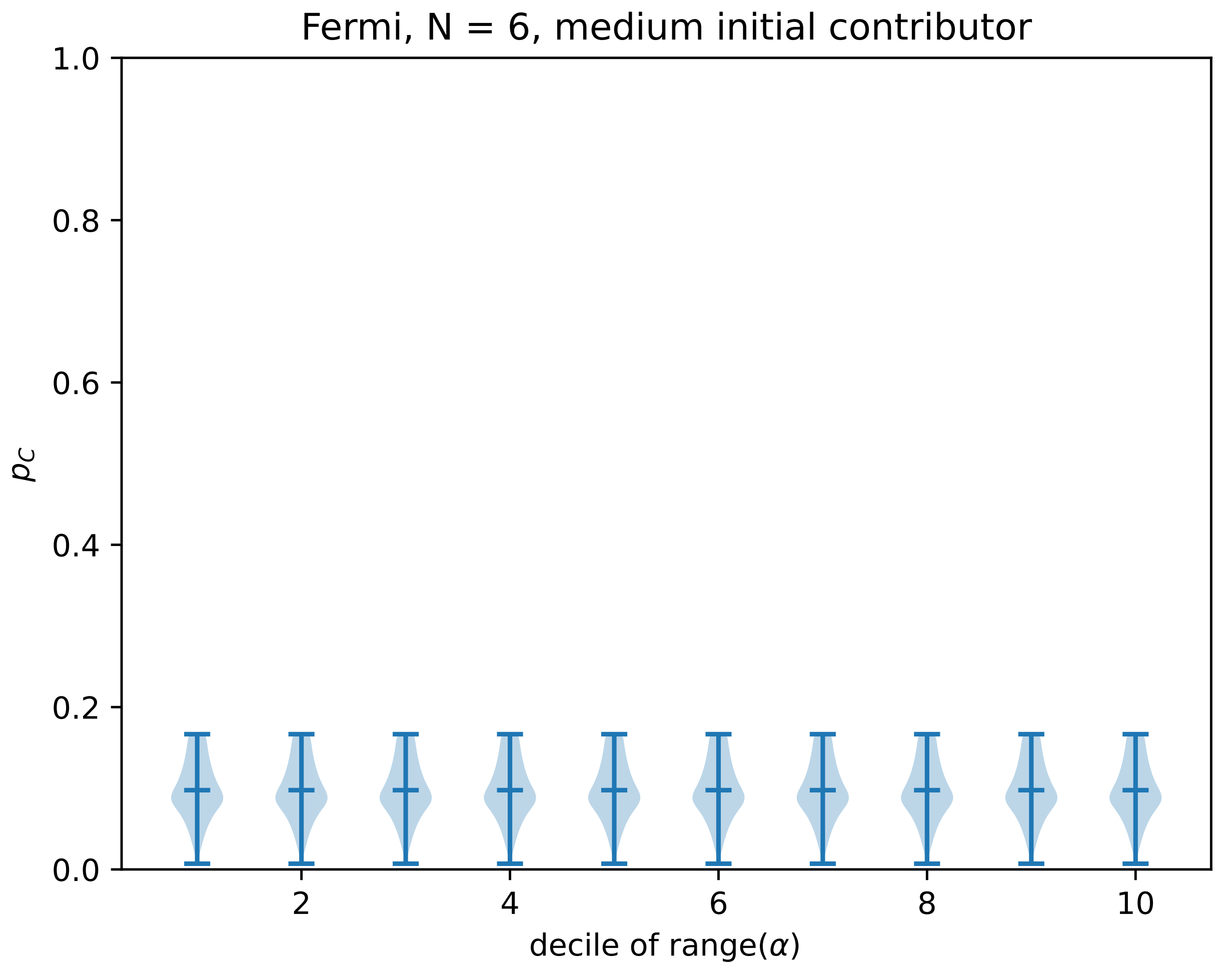
Fermi, N = 5, low initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

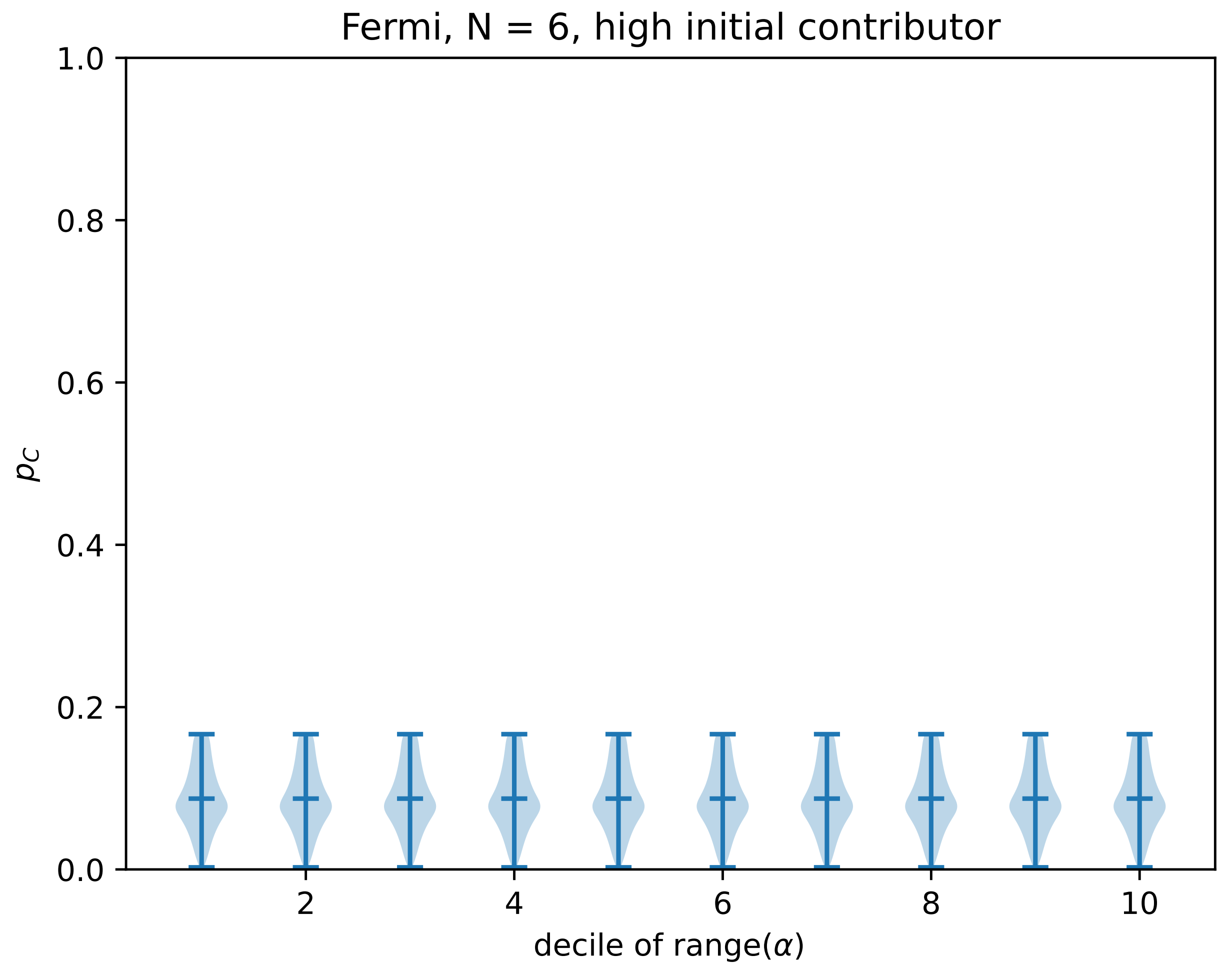

Fermi, N = 5, medium initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

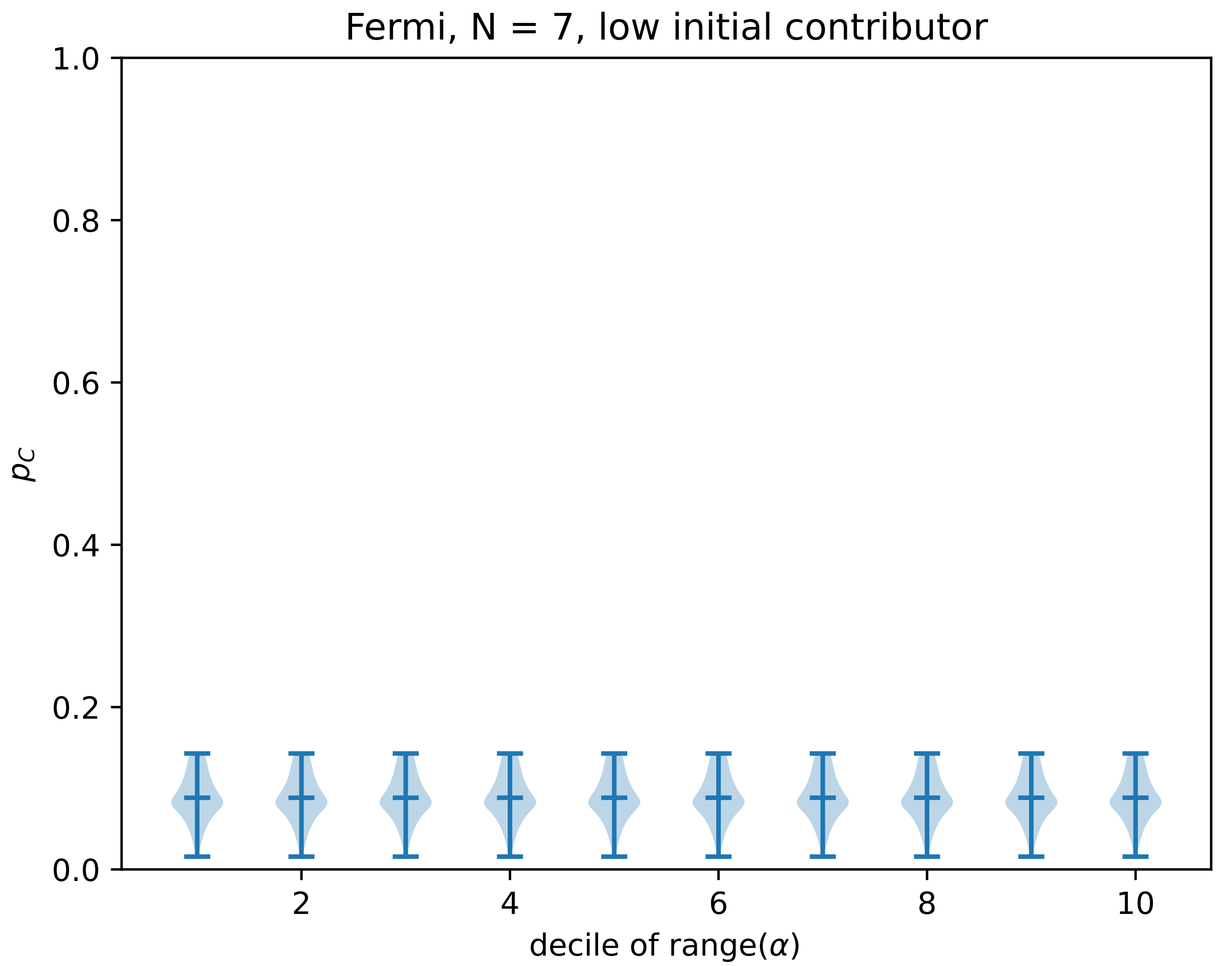

Fermi, N = 5, high initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

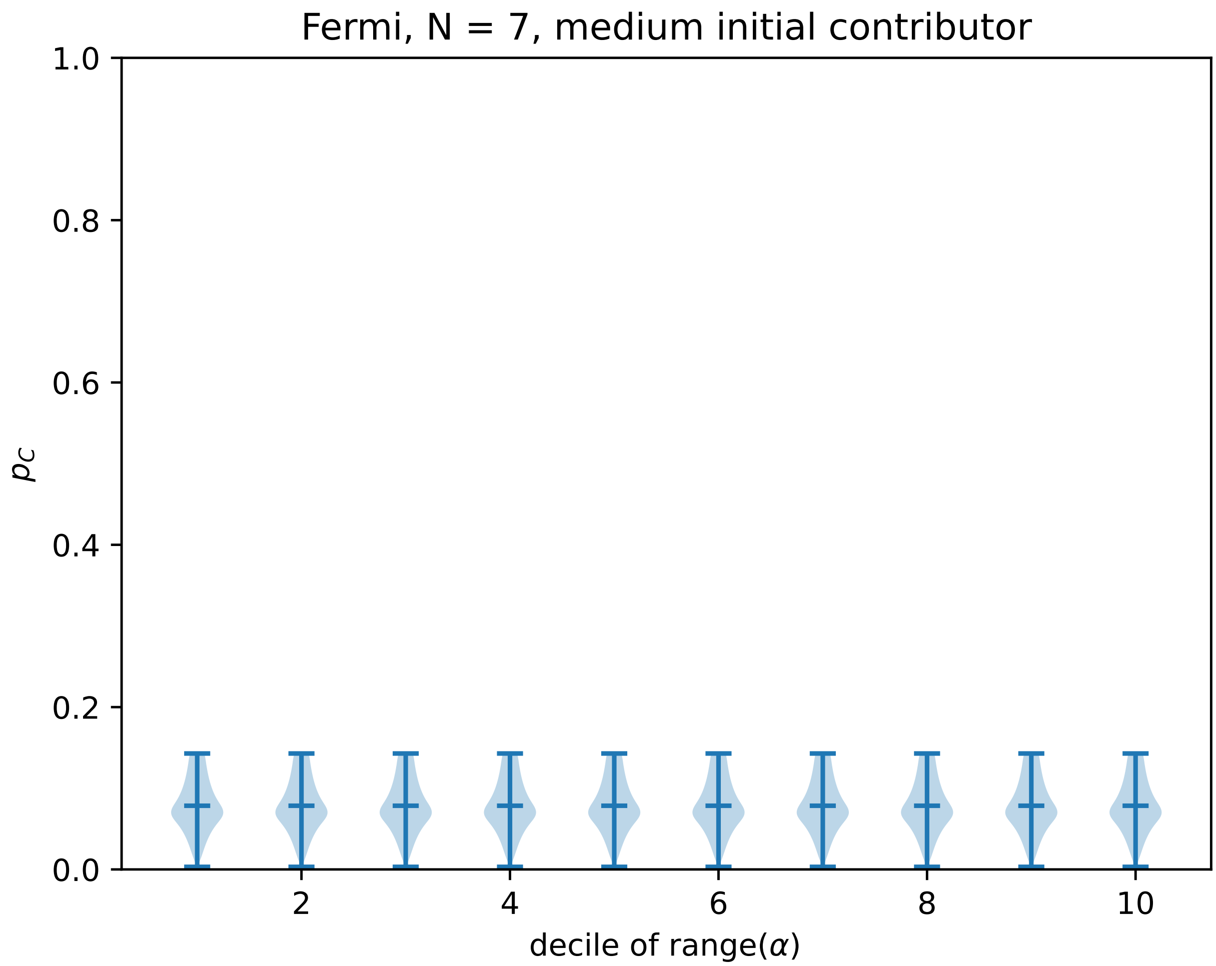


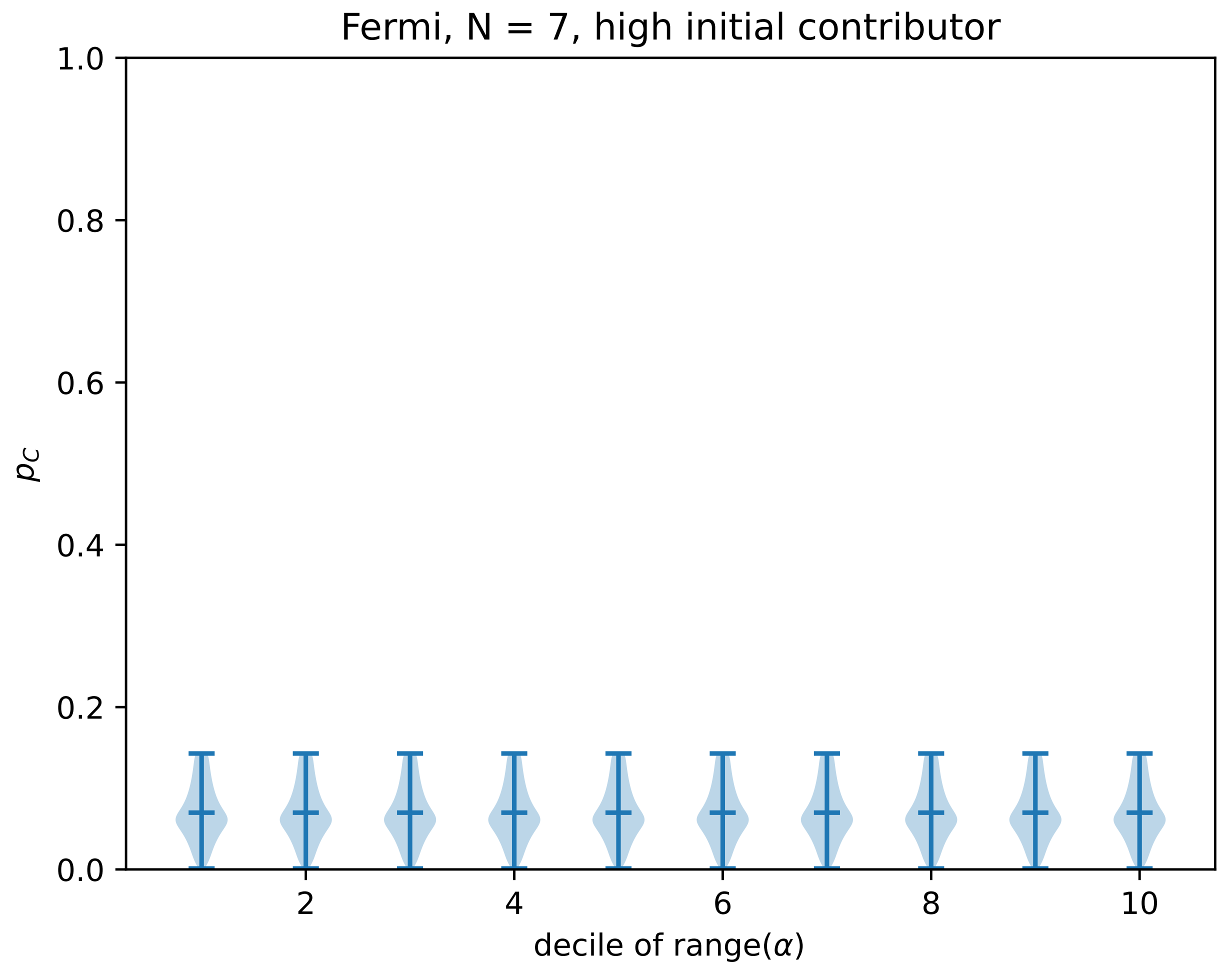
Fermi, N = 6, low initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

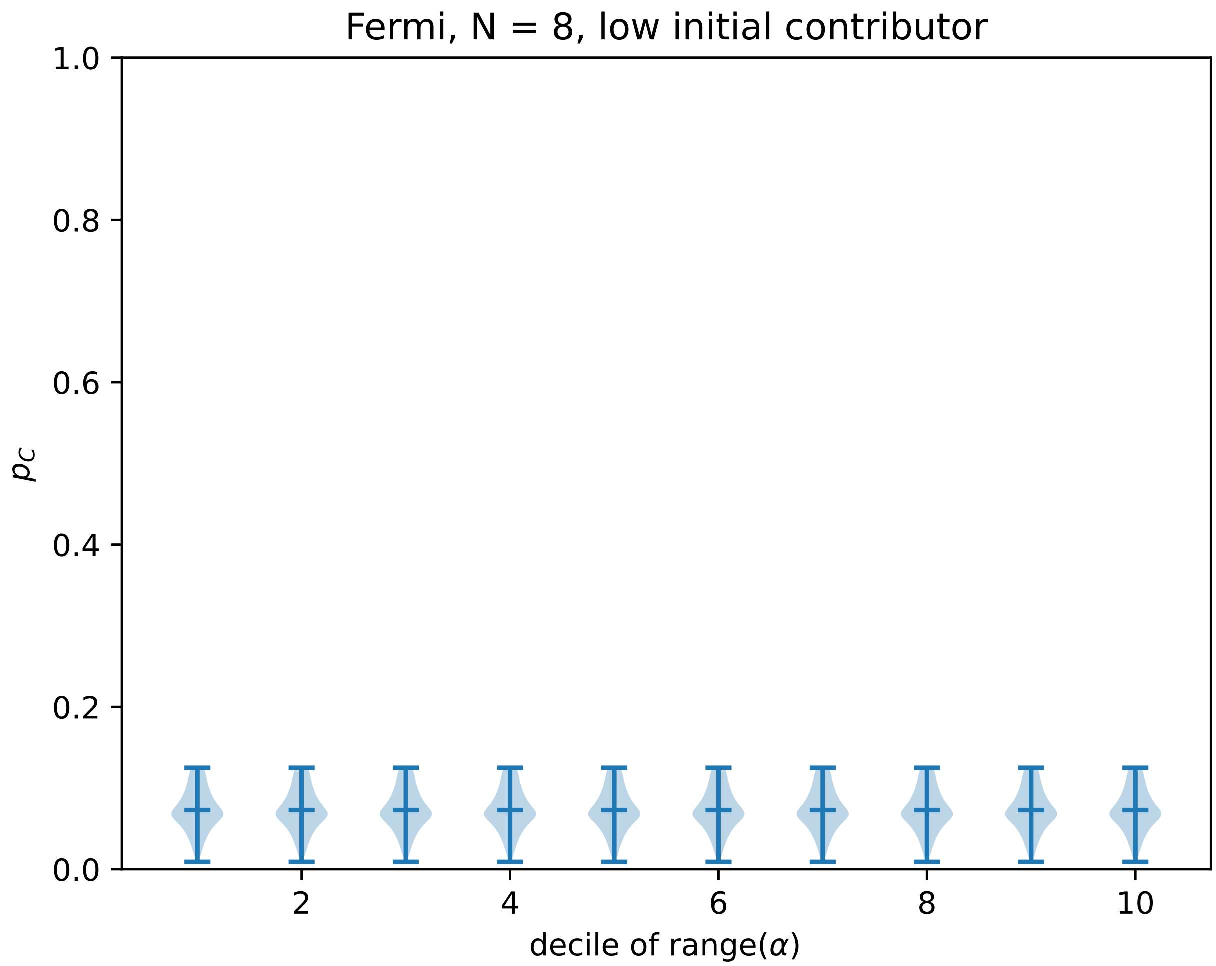

Fermi, N = 6, medium initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

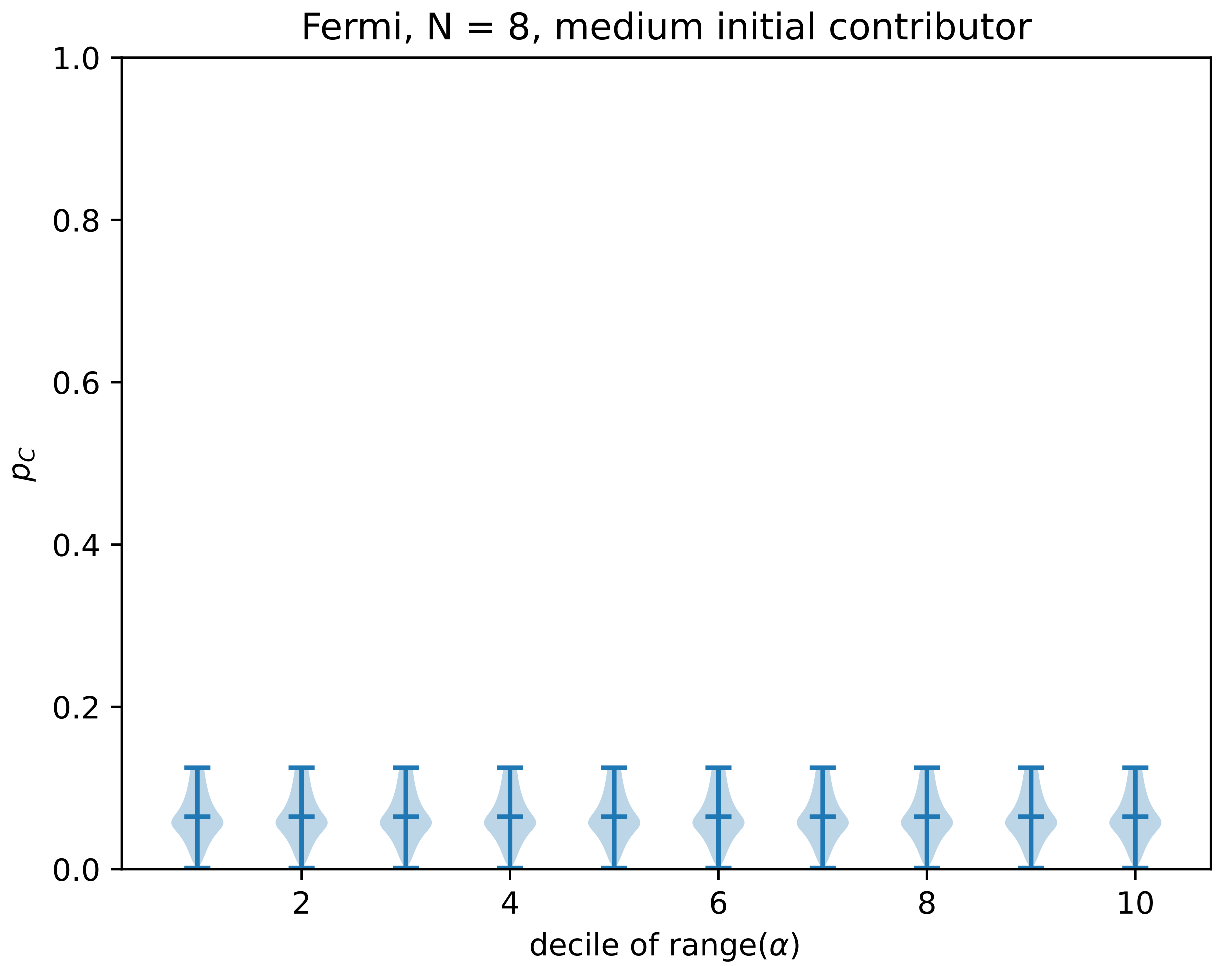

Fermi, N = 6, high initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

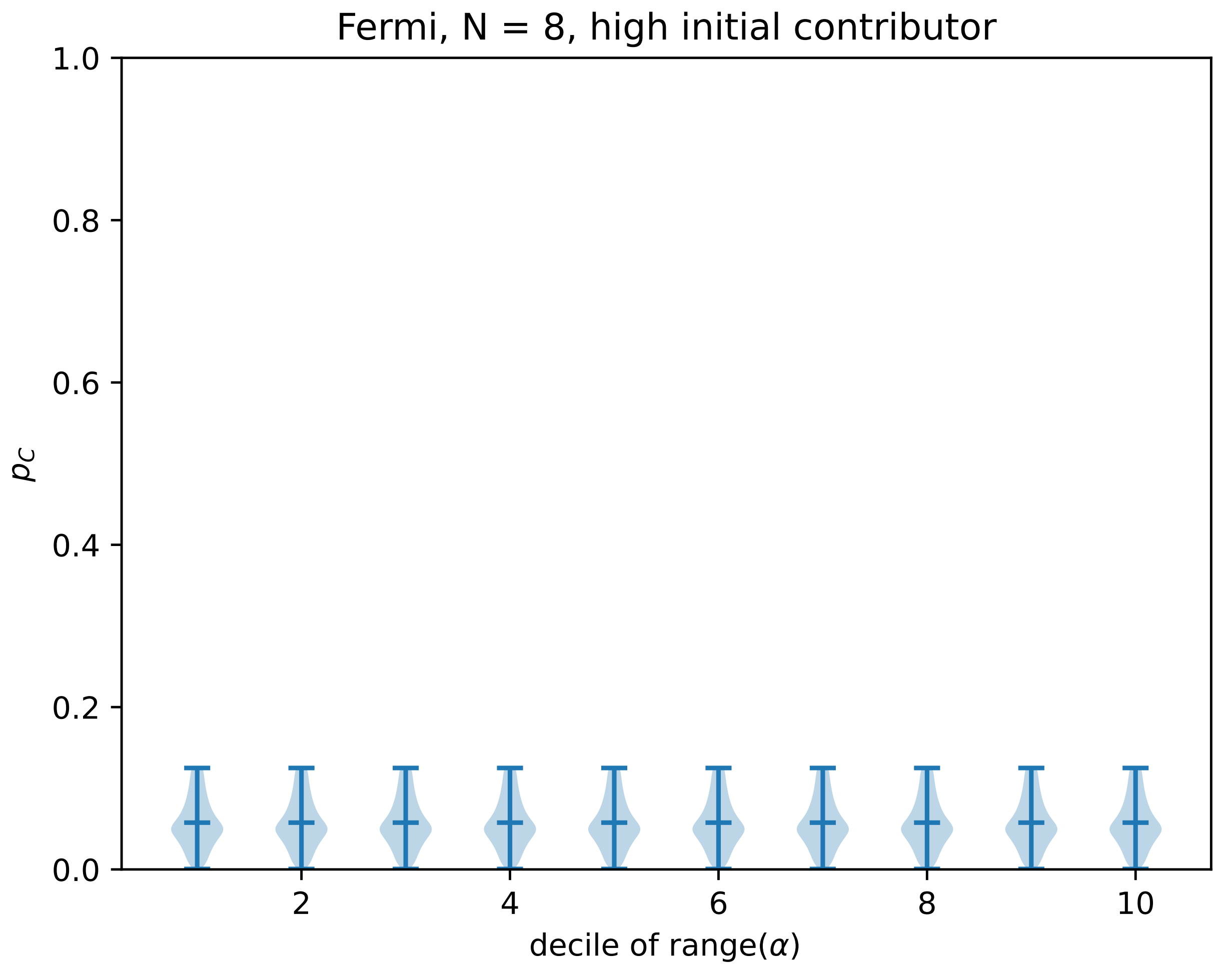

Fermi, N = 7, low initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

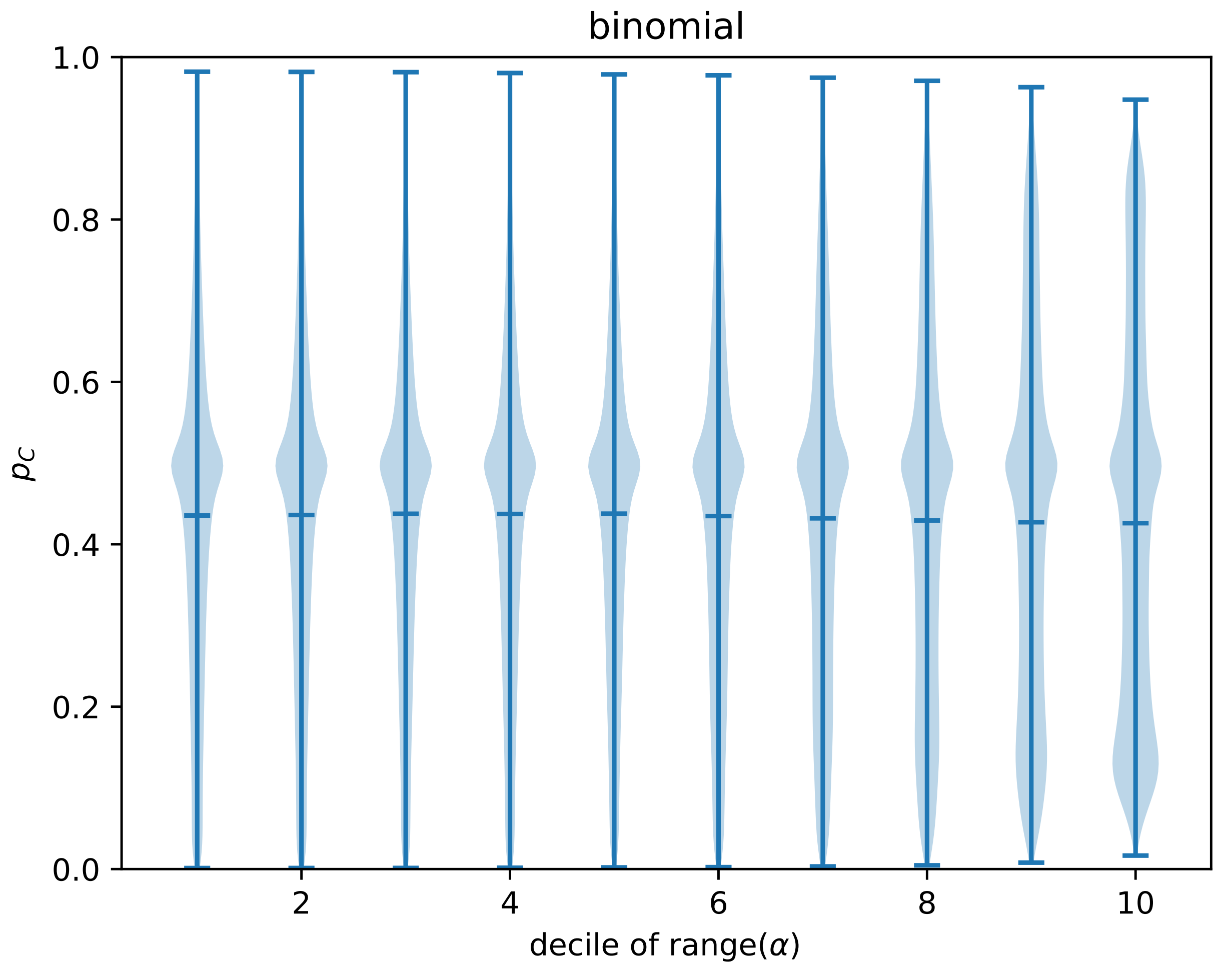

Fermi, N = 7, medium initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

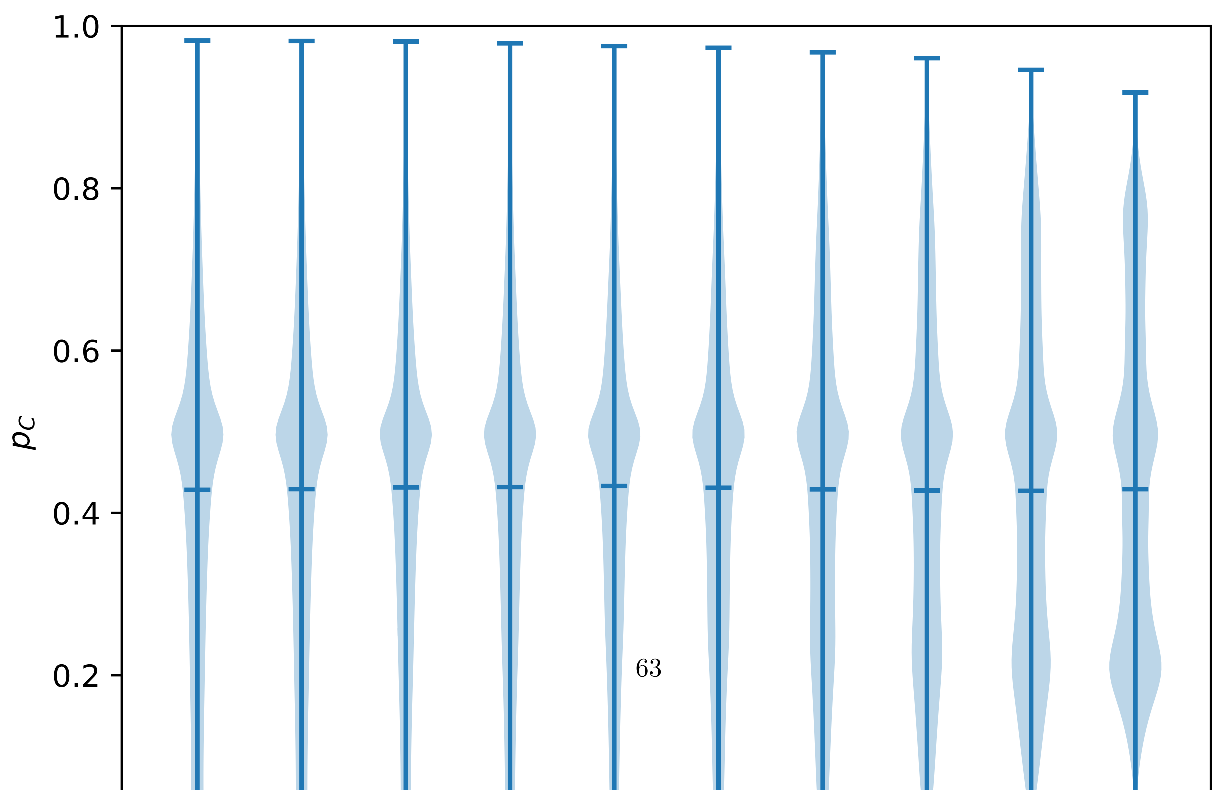

Fermi, N = 7, high initial contributor
p_C
1.0
0.8
0.6
0.4
0.2
0.0
2
4
6
8
10
decile of range(α)

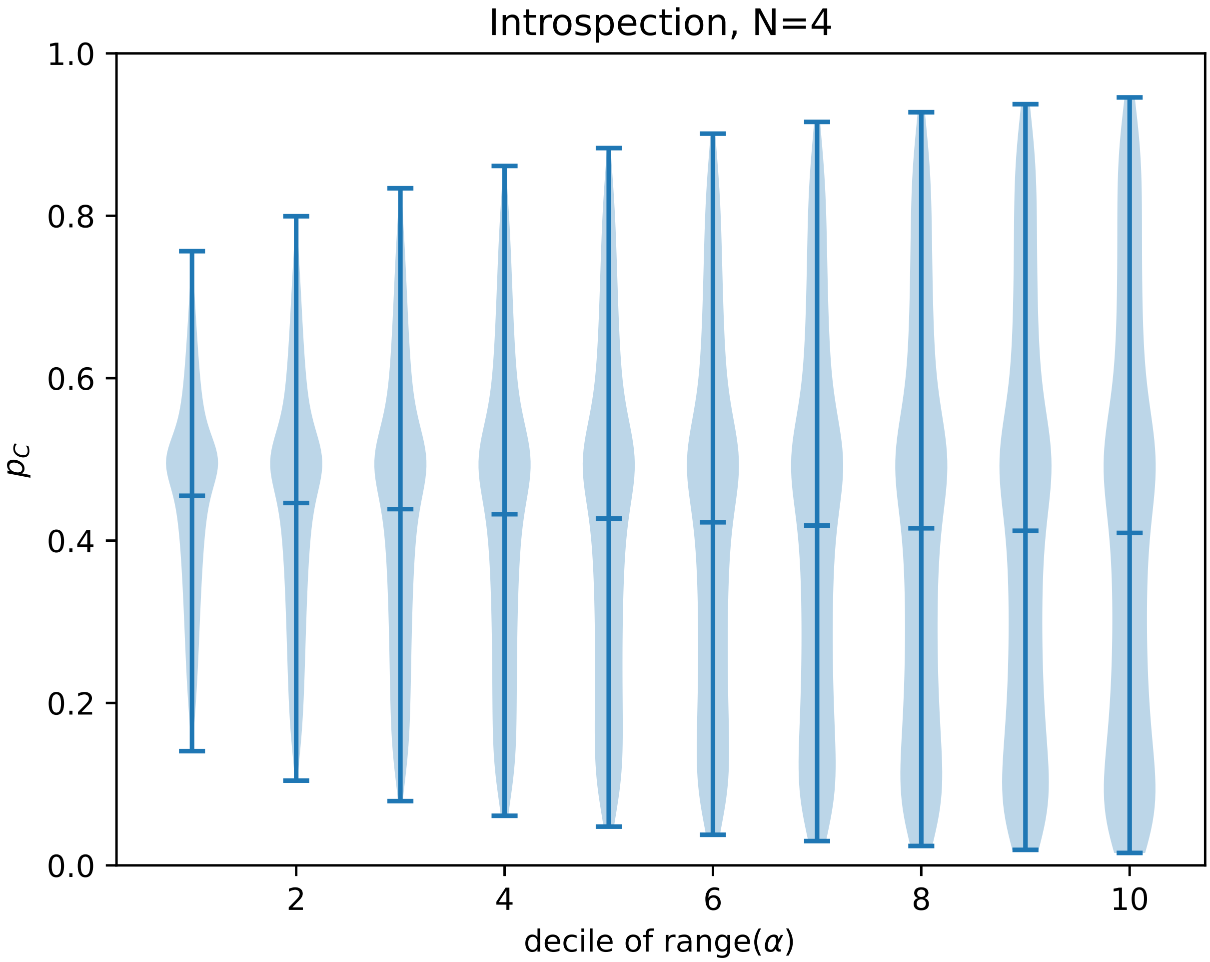

Fermi, N = 8, low initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)


Fermi, N = 8, medium initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)


Fermi, N = 8, high initial contributor
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

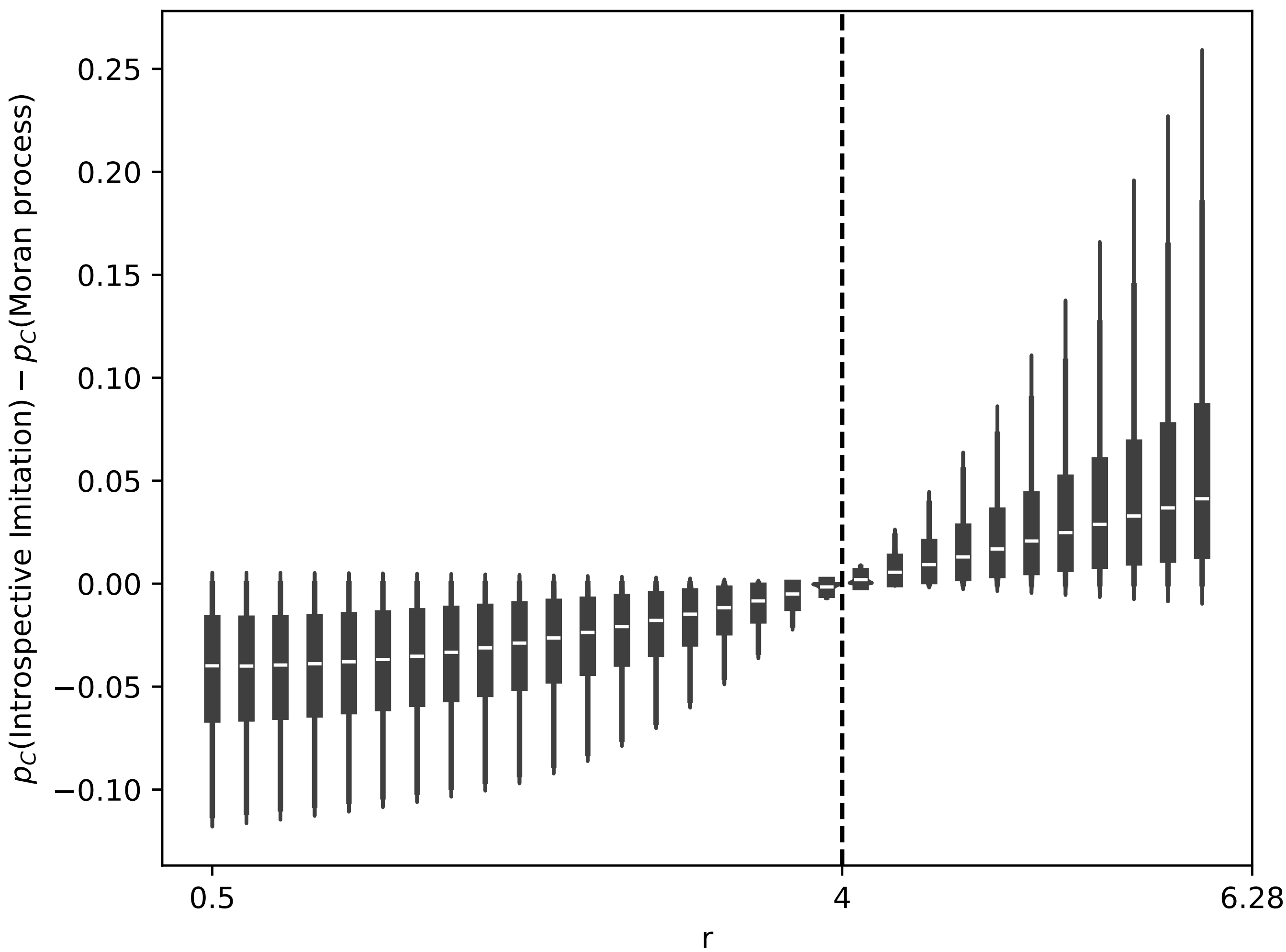

binomial
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

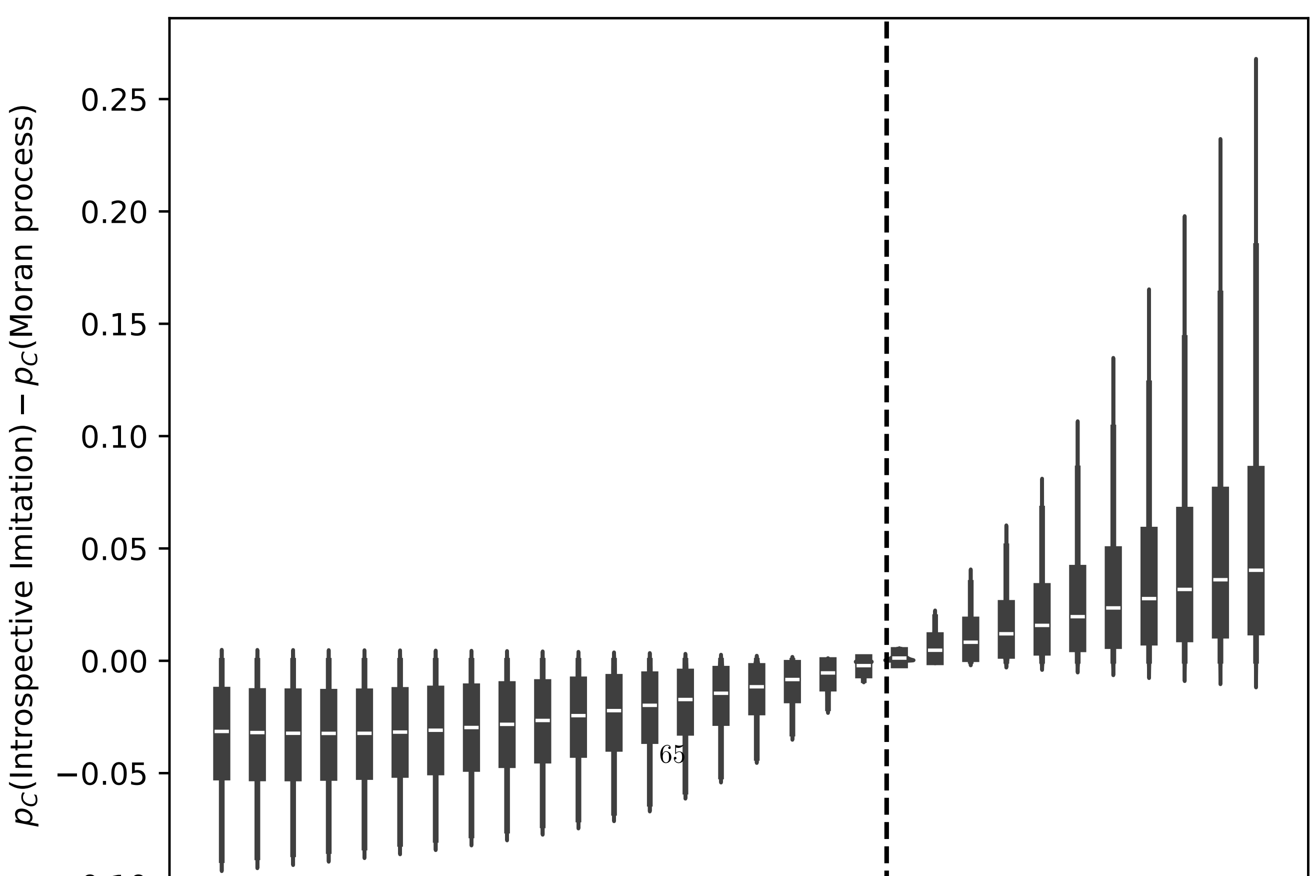

1.0
0.8
0.6
0.4
0.2
p_C



Introspection, N=4
1.0
0.8
0.6
0.4
0.2
0.0
p_C
2
4
6
8
10
decile of range(α)

Moran process vs Introspective Imitation Dynamics: $N = 4, 5$





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